



Silverstone System and VA Test Plan

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Draft

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2 Document Version History

Version	Date	Editor(s)/Reviewer(s)	Comments & Changes
Draft	12 July 2022	Goy Teck Lim	Prepared for review

3 References & Applicable Documents

Reference	Document

4 Glossary

Ant	Antenna
CW	Continuous Wave
DC	Direct Current
EMC	Electromagnetic Compatibility
EV	Electric Vehicle
GA	Ground assembly
GUI	Graphical User Interface
HF	High Frequency
LSB	Lower Side Band
NVIS	Near Vertical Incidence Skywave
OTP	Over temperature protection
OATS	Open Area Test Site
RF	Radio Frequency
RX	Receiver
RMS	Root-mean-square
RSSI	Radio Signal Strength Indication
SDR	Software Defined Radio
TX	Transmitter
USB	Upper Side Band
VA	Vehicle assembly
WPT	Wireless Power Transfer

5 Introduction

This document defines the general functional, performance and safety test items, and their test conditions, methods and requirements for WiTricity wireless power transfer (WPT) overall system and vehicle assembly (VA) for the Silverstone program. This Silverstone program shall first develop the 800V configuration for VA (referred as “Silverstone 800”) to be implemented, and thereon extend to the 400V configuration (referred as “Silverstone 400”) as needed. To establish the overall system specification in relation to Silverstone 800 VA, the Silverstone program shall consider the ground assembly and wall box of the R1 version as a reference.

6 System Performance Test

6.1 Input voltage range test

Test condition:

1. System input: 220 Vac/50 Hz.
2. The system works in constant voltage output mode with $V_{out} = 350$ V and rated power output $P_{out} = 6.6$ kW.
3. Nominal ground clearance: $Z = 175$ mm.
4. The inlet water temperature of VA is 45 °C and the flow rate is 8 L/min.

Test method:

1. Under the condition of rated input and rated power output, adjust the input voltage upward to 115% rated input voltage.
2. Under the condition of rated input and rated power output, adjust the input voltage downward to 85% rated input voltage.

Reference Requirement:

Input voltage is in the range of 85 % – 115 % rated voltage, and the system is full power output.

Test Item	Input voltage range test		
Pass Requirement	Input Voltage (V)	115% rated input voltage	85% rated input voltage
Test Result	Output Power (kW)		

6.2 Input frequency range test

Test condition:

1. System input: 220 Vac/50 Hz.

2. The system works in constant voltage output mode with $V_{out} = 350$ V and rated power output $P_{out} = 6.6$ kW.
3. Nominal ground clearance: $Z = 175$ mm.
4. The inlet water temperature of VA is 45°C and the flow rate is 8 L/min.

Test method:

1. Under the condition of rated input and rated power output, adjust the input frequency upward to 53 Hz.
2. Under the condition of rated input and rated power output, adjust the input frequency downward to 47 Hz.

Reference Requirement:

System can generate full power output in the range of input frequency.

Test Item	Input frequency range test		
Pass Requirement	Input frequency (Hz)	47 Hz	53 Hz
Test Result	Output power (kW)		

6.3 Input harmonic current limit test

Test condition:

1. System input: 220 Vac/50Hz.
2. The system works in constant voltage output mode with $V_{out} = 350$ V and rated power output $P_{out} = 6.6$ kW.
3. Nominal ground clearance: $Z = 175$ mm.
4. The inlet water temperature of VA is 45°C and the flow rate is 8L/min.

Test method:

Record the total current harmonic distortion iTHD using power analyzer under the condition of rated input and nominal ground clearance without offset.

Reference Requirement:

iTHD<10%.

Test Item	Total current harmonic distortion		
Test Requirement	Output Power (kW)	3.3 kW	6.6 kW
Test Result	iTHD		

6.4 Input surge current test

Test condition:

1. System input: 220 Vac/50 Hz.
2. The system works in constant voltage output mode with $V_{out} = 350$ V and rated power output $P_{out} = 6.6$ kW.
3. Nominal ground clearance: $Z = 175$ mm.
4. The inlet water temperature of VA is 45 °C and the flow rate is 8 L/min.

Test method:

1. Digital oscilloscope is used to monitor the input current waveform under the rated input and output conditions of the system.
2. The system starts up with rated power and observes and records the change of input current waveform from the start-up to the output of rated power.

Reference Requirement:

The input impulse current does not exceed 110% of the rated input current.

Test Item	Input surge current test		
Test Result	Rated input current (A)	Max input current (A)	Max input current /Rated input current

6.5 Source effect test

Test condition:

1. System input: 220 Vac/50 Hz.
2. The system works in constant voltage output mode with $V_{out} = 350$ V and rated power output $P_{out} = 6.6$ kW.
3. Nominal ground clearance: $Z = 175$ mm.
4. The inlet water temperature of VA is 45 °C and the flow rate is 8 L/min.

Test method:

1. Record output voltage U_0 under rated input and rated output conditions.
2. Record the output voltage U_1 and U_2 respectively when adjusting the input voltage to the lower and upper limits.
3. Calculate the source effect according to the following formula: source effect $\delta U = \Delta U / U_0 \times 100\%$ (where ΔU is the maximum difference between U_1 , U_2 and U_0).

Reference Requirement:

The source effect should not more than $\pm 1\%$ of the output voltage setting value.

Test Item	Source effect test				
Test Result	U_0 (V)	U_1 (V)	U_2 (V)	ΔU	δU

6.6 Load effect test

Test condition:

1. System input: 220 Vac/50 Hz.
2. The system works in constant voltage output mode with $V_{out} = 350$ V and rated power output $P_{out} = 6.6$ kW.
3. Nominal ground clearance: $Z = 175$ mm.
4. The inlet water temperature of VA is 45 °C and the flow rate is 8 L/min.

Test method:

1. Record output voltage U_0 under rated input and 50 % output load conditions.
2. Adjust the output current to the minimum value and full load value and record the output voltage U_1 and U_2 respectively.
3. Load effect is calculated according to the following formula: load effect $\delta i = \Delta U / U_0 \times 100\%$ (where ΔU is the maximum difference between U_1 , U_2 and U_0), and the maximum value of δi is taken as the load effect of the system.

Reference Requirement:

Load effect is less than +1 % of output voltage setting value.

Test Item	Load effect test				
Test Result	U_0 (V)	U_1 (V)	U_2 (V)	ΔU	δi

6.7 Output voltage regulation accuracy test

Test condition:

1. System input: 220 Vac/50 Hz.
2. The system works in constant voltage output mode with $V_{out} = 350$ V and rated power output $P_{out} = 6.6$ kW.
3. Nominal ground clearance: $Z = 175$ mm.
4. The inlet water temperature of VA is 45 °C and the flow rate is 8 L/min.
5. Test $V_{out} = 350$ V in constant voltage mode.

Test method:

1. Record output voltage U_0 under rated input and 50 % output load conditions.
2. Adjust the input voltage to 187 Vac, adjust the output current to minimum and full load respectively, record the output voltage U_1 and U_2 .

- Adjust the input voltage to 253 Vac, adjust the output current to minimum and full load respectively, record the output voltage U3 and U4.
- Use the following formula to calculate the voltage stabilization accuracy: the voltage stabilization accuracy $\Delta = (\Delta U / U_0) \times 100\%$ (where ΔU is the maximum difference between U1, U2, U3, U4 and U0).

Reference Requirement:

The precision of output voltage stabilization should not more than $\pm 1\%$ of the output voltage setting value.

Test Item	Output voltage regulation accuracy test						
Test Result	U0 (V)	U1 (V)	U2 (V)	U3 (V)	U4 (V)	ΔU	Δ

6.8 Output voltage ripple test

Test condition:

- System input: 220 Vac/50 Hz.
- The system works in constant voltage output mode with $V_{out} = 350\text{ V}$ and rated power output $P_{out} = 6.6\text{ kW}$.
- Nominal ground clearance: $Z = 175\text{ mm}$.
- The inlet water temperature of VA is $45\text{ }^{\circ}\text{C}$ and the flow rate is 8 L/min .

Test method:

- Under the rated input and output conditions, the peak-peak and effective values of AC component of DC output voltage are measured by adjusting the rated output current from 0 % to 100 %.
- Adjust the input voltage to the upper and lower limits and change the output voltage setting value in the upper and lower limits of the output voltage.

Repeat step 1 measurement.

The oscilloscope for ripple coefficient test requires a bandwidth of 20 MHz and a horizontal scanning speed of 0.5 s/DIV .

The ripple coefficient is calculated according to the following formula:

$$X_{rms} = \frac{U_{rms}}{U_{DC}} \times 100\%$$

$$X_{pp} = \frac{U_{pp}}{U_{DC}} \times 100\%$$

式中:

X_{rms} ——纹波有效值系数;

X_{pp} ——纹波峰值系数;

U_{rms} ——输出电压交流分量有效值;

U_{pp} ——输出电压交流分量峰-峰值;

U_{DC} ——直流输出电压平均值。

Reference Requirement:

The peak value coefficient of voltage ripple should not exceed $\pm 3\%$ and the RMS coefficient of voltage ripple should not exceed 1% .

Test Item	Output voltage ripple test				
Test Result	U_{rms} (V)	U_{pp} (V)	U_{DC} (V)	X_{rms}	X_{pp}

6.9 Efficiency and power factor test (for different x, y and z offset)

6.9.1 System efficiency test

Test condition:

1. System input: 220 Vac/50 Hz.
2. Constant voltage output: $V_{out} = 350$ V.
3. The inlet water temperature of VA is 45°C and the flow rate is 8 L/min.
4. Test the system efficiency under different ground clearance, primary and secondary aligned and offset conditions.

Test method:

1. Under the aligned position and nominal ground clearance ($Z = 175$ mm) condition, measure the efficiency of the system at 50 %, 75 % and 100 % of the rated output power.
2. Adjust to minimum Z height ($Z = 140$ mm) and repeat the test in step 1.
3. Adjust to maximum Z height ($Z = 210$ mm) and repeat the test in step 1.
4. The system efficiency at rated input and output of the system are tested under different ground clearance and x, y, z offset conditions.

Reference Requirements:

At rated operating point, the system efficiency should not be less than 88%. When the system outputs at rated power, the system efficiency should not be less than 85% under all permissible offsets positions.

6.9.2 Power factor test

Test conditions:

1. System input: 220 Vac/50 Hz;
2. Test the power factor under different ground clearance, primary and secondary aligned and offset conditions.
3. The inlet water temperature of VA is 45 °C and the flow rate is 8 L/min.

Test methods:

1. Under the aligned position and nominal ground clearance (Z=175mm) condition, measure the power factor of the system at 50%, 75% and 100% of the rated output power.
2. Adjust to minimum Z height (Z = 140mm) and repeat the test in step 1.
3. Adjust to maximum Z height (Z = 210mm) and repeat the test in step 1.
4. The power factor at rated input and output of the system are tested under different ground clearance and x, y offset conditions.

Reference Requirements:

Under the condition of rated input and output, the power factor of the system should not be less than 0.99.

Test item	System efficiency and power factor at different x, y and z offset					
Test Result	Output Power /KW	Z/mm	X-Offset /mm	Y-Offset /mm	System efficiency /%	Power Factor
	3.3 kW	140	0	0		
	5.0 kW	140	0	0		
	6.6 kW	140	0	0		
	3.3 kW	175	0	0		
	5.0 kW	175	0	0		
	6.6 kW	175	0	0		
	3.3 kW	210	0	0		
	5.0 kW	210	0	0		
	6.6 kW	210	0	0		
	6.6 kW	140	-75	0		
	6.6 kW	140	-100	0		
	6.6 kW	140	0	-100		
	6.6 kW	140	-75	-100		
Test Result	6.6 kW	140	-100	-100		
	6.6 kW	140	-100	100		
	6.6 kW	140	-100	100		

	6.6 kW	140	-75	100		
	6.6 kW	140	0	100		
	6.6 kW	140	75	100		
	6.6 kW	140	100	100		
	6.6 kW	140	100	0		
	6.6 kW	140	100	-100		
	6.6 kW	140	75	-100		
	6.6 kW	140	75	0		
	6.6 kW	175	0	0		
	6.6 kW	175	-75	0		
	6.6 kW	175	-100	0		
	6.6 kW	175	0	-100		
	6.6 kW	175	-75	-100		
	6.6 kW	175	-100	-100		
	6.6 kW	175	-100	100		
	6.6 kW	175	-75	100		
	6.6 kW	175	0	100		
	6.6 kW	175	75	100		
	6.6 kW	175	100	100		
	6.6 kW	175	100	0		
	6.6 kW	175	100	-100		
	6.6 kW	175	75	-100		
	6.6 kW	175	75	0		
	6.6 kW	210	0	0		
	6.6 kW	210	-75	0		
	6.6 kW	210	-100	0		
Test Result	6.6 kW	210	0	-100		
	6.6 kW	210	-75	-100		
	6.6 kW	210	-100	-100		
	6.6 kW	210	-100	100		
	6.6 kW	210	-75	100		
	6.6 kW	210	0	100		
	6.6 kW	210	75	100		
	6.6 kW	210	100	100		
	6.6 kW	210	100	0		
	6.6 kW	210	100	-100		
	6.6 kW	210	75	-100		
	6.6 kW	210	75	0		

6.10 Efficiency and power factor test at different roll, pitch, and yaw

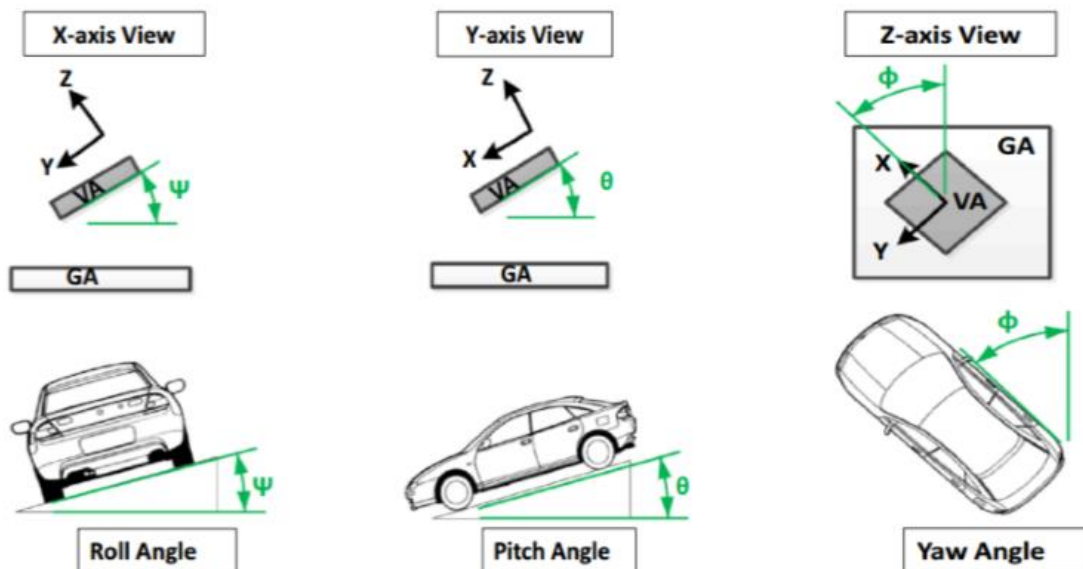
Test condition:

1. System input: 220 Vac/50 Hz;
2. At nominal ground clearance, $Z = 175$ mm, and primary and secondary sides are aligned
3. Power output: $P_{out} = 6.6$ kW.

4. Water temperature of VA inlet: 45 °C, flow rate: 8 L/min
5. Test the system efficiency under roll, pitch, and yaw conditions.

Test method:

1. The wireless charging system starts normally and works at rated power. After the system is stabilized, three different rotation angles are carried out: roll, pitch and yaw, as follows.



2. Adjust the test bench according to the recording table. After the system is stable, record the corresponding system efficiency and power factor.

Reference Requirement:

The system efficiency at each point should not less than 85%.

Roll, pitch and yaw test					
Output Power	Roll angle	Pitch angle	Yaw angle	System Efficiency %	Power Factor
6.6 kW	0	0	0		
6.6 kW	2	0	0		
6.6 kW	-2	0	0		
6.6 kW	0	2	0		
6.6 kW	0	-2	0		
6.6 kW	0	0	10		
6.6 kW	0	0	-10		
6.6 kW	2	2	0		
6.6 kW	2	-2	0		
6.6 kW	-2	2	0		
6.6 kW	-2	-2	0		
6.6 kW	2	2	10		
6.6 kW	2	-2	10		
6.6 kW	-2	2	10		

6.6 kW	-2	2	10		
6.6 kW	2	2	-10		
6.6 kW	2	-2	-10		
6.6 kW	-2	2	-10		
6.6 kW	-2	-2	-10		

6.11 Load dump test

Test condition:

1. System input 220 Vac/50 Hz;
2. At nominal ground clearance, $Z = 175$ mm. No offset.
3. Power output: $P_{out} = 6.6$ kW.
4. Water temperature of VA inlet: 45 °C, flow rate: 8 L/min

Test method:

The wireless charging system starts normally. After working at rated power, the load will be disconnected instantaneously (the current drops below 500 mA). The peak output voltage of the wireless charging system will be recorded by an oscilloscope, and the time of the drop of the output voltage will be recorded.

Reference Requirement:

The wireless charging system should be able to detect the output open circuit, turn off the output and alarm. The peak output voltage should not exceed the overvoltage protection value, and the output voltage should be reduced to less than 60 V within 1 s.

Test Item	Load dump test	
Test Result	Peak output voltage/V	Output voltage drop time /ms

7 System Functional Test

7.1 Input overvoltage protection test

Test condition:

1. System input: 220 Vac/50 Hz.
2. At nominal ground clearance, $Z = 175$ mm. No offset.
3. Power output: $P_{out} = 6.6$ kW.
4. Water temperature of VA inlet: 45 °C, flow rate: 8 L/min

Test method:

When the system operates at rated input and output conditions, adjust the input voltage upward until the system output shut down.

Reference Requirement:

The system should have the input overvoltage protection function.

Test item	Input overvoltage protection
Test Result	Input overvoltage protection value /V

7.2 Input undervoltage protection test

Test condition:

1. System input: 220 Vac/50 Hz;
2. At nominal ground clearance, Z = 175 mm. No offset.
3. Power output: $P_{out} = 6.6$ kW.
4. Water temperature of VA inlet: 45 °C, flow rate: 8 L/min

Test method:

When the system operates at rated input and output conditions, adjust the input voltage down until the system output shut down.

Reference Requirements:

The system should have the input undervoltage protection function.

Test item	Input undervoltage protection
Test Result	Input undervoltage protection threshold /V

7.3 Input grid frequency protection test

Test condition:

1. System input: 220 Vac/50 Hz.
2. At nominal ground clearance, Z = 175 mm. No offset.
3. Power output: $P_{out} = 6.6$ kW.
4. Water temperature of VA inlet: 45 °C, flow rate: 8 L/min.

Test methods:

1. The system works under the rated input and rated output conditions, and slowly adjusts the input frequency upwards until the system is turned off.
2. The system works under the rated input and rated output conditions, and slowly adjusts the input frequency downwards until the system is turned off.

Reference Requirements:

The system should have the input grid frequency protection function.

Test item	Input grid frequency protection test	
Test Result	Frequency protection lower limit/Hz	Frequency protection upper limit/Hz

7.4 Input overcurrent protection test

Test conditions:

1. System input: 220 Vac/50 Hz.
2. At nominal ground clearance, $Z = 175$ mm. No offset.
3. Power output: $P_{out} = 6.6$ kW.
4. Water temperature of VA inlet: 45 °C, flow rate: 8 L/min.

Test methods:

At rated input and output conditions, increase the output load slowly until the input current increases to the input overcurrent protection threshold, and the system is turned off.

Reference Requirements:

The system should have the input overcurrent protection function.

Test item	Input overcurrent protection test
Test Result	Input overcurrent protection threshold /V

7.5 Output overvoltage protection test

Test condition:

1. System input: 220 Vac/50 Hz.
2. At nominal ground clearance, $Z = 175$ mm. No offset.
3. Power output: $P_{out} = 6.6$ kW.
4. Water temperature of VA inlet: 45 °C, flow rate: 8 L/min.

Test method:

After the system works normally, the output voltage is pulled high to simulate output overvoltage condition. The system can protect and shutdown normally without damage.

Reference Requirement:

The system should have the output overvoltage protection function.

Test item	Output overvoltage protection test
Test Result	Output overvoltage protection threshold /V

7.6 Output undervoltage protection test

Test condition:

1. System input: 220 Vac/50 Hz.
2. At nominal ground clearance, $Z = 175$ mm. No offset.
3. Power output $P_{out} = 6.6$ kW.
4. Water temperature of VA inlet: 45 °C, flow rate: 8 L/min.

Test method:

After the system works normally, the output voltage is pulled low to simulate output undervoltage condition. The system can protect and shutdown normally without damage.

Reference Requirement:

The system should have the output undervoltage protection function.

Test item	Output undervoltage protection test
Test Result	Output undervoltage protection threshold /V

7.7 System coupling transient test

Test condition:

1. System input: 220 Vac/50 Hz;
2. At nominal ground clearance, $Z = 175$ mm. No offset.
3. Power output: $P_{out} = 6.6$ kW.
4. Water temperature of VA inlet: 45 °C, flow rate: 8 L/min.

Test method:

The wireless charging system starts normally, and after the rated power is applied, the vehicle side is instantly removed.

Reference Requirements:

The wireless charging system should be able to detect that the secondary device is removed, turn off the output and alarm.

7.8 System over temperature protection test**Test condition:**

1. System input: 220 Vac/50 Hz.
2. At nominal ground clearance, $Z = 175$ mm. No offset.
3. Power output: $P_{out} = 6.6$ kW.
4. Water temperature of VA inlet: 45 °C, flow rate: 8 L/min.
5. Oven environment.

Test method:

Set the ambient temperature between 55 °C and 60 °C. When the system works under rated input and output conditions, observe whether the system can be overheated and shut down.

Reference Requirement:

The system should have the over temperature protection (OTP) function.

Test item	System over temperature protection test		
Test Result	WB OTP value /°C	GA OTP value /°C	VA OTP value /°C

7.9 System alignment function (PD) test**Test condition:**

1. System input: 220 Vac/50 Hz.
2. At nominal ground clearance, $Z = 175$ mm.
3. Power output: $P_{out} = 6.6$ kW.
4. Water temperature of VA inlet: 45 °C, flow rate: 8 L/min.

Test method:

After the system communication is established, the GA and VA start the test at a distance of 5 m. Compare the difference between the test indication value and the actual test.

Reference Requirements:

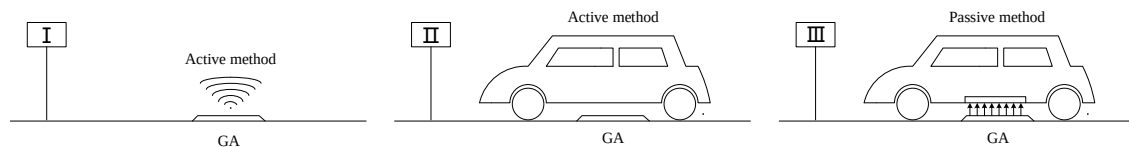
The system should have the alignment function.

Test item	System alignment function (PD) test		
Test Result	1-5m Precision/cm	0.5-1m Precision/mm	Final break Position Precision/mm

7.10 Foreign object detection (FOD) test

Test condition:

1. System input: 220 Vac/50 Hz;
2. The FOD test is performed separately in the three working modes of the WPT system, as shown in the figure below.



- 2.1 Mode I: Communication is completed waiting for the alignment phase,
- 2.2 Mode II: During the alignment process or the alignment is completed, waiting for the charging phase,
- 2.3 Mode III: Power transmission phase.VA
3. Water temperature of VA inlet: 45 °C, flow rate: 8 L/min.

Test method:

Do the ignition test and temperature rise test of the foreign object on the GA surface under rated input and rated output condition. The test foreign objects are listed as follows:

Test Objects	Temp. Rise Test	Ignition Test	Test Result	Remarks
Paper Clip	x	x		Length 29mm
Coins	x			¥1, ¥0.5, ¥0.1
Nail	x			Ref: M5
Beverage Can	x			Vol. 350mL or equiv.
Aluminum Sheet	x			50x75x1mm
Steel Sheet	x			50x75x1mm
Steel Wool	x	x		Standard Steel wool (e.g. for pan washing)
Foil	x	x		Foil with Paper Backing from Food Wrappers and Cigarette (25x25mm; 300x300mm)
Extension Cord	x			30m, 16 Gauge, coiled in >25loops

Reference Requirement:

For the defined test object, the temperature shall not exceed the maximum temperature limit specified in GB/T 16895.2-2005 when accessible, as follows:

- a) exposed surface of the metal part: 80 °C.
- b) Non-metallic part surface: 90 °C.

The system can detect foreign objects and stop charging before the test object reaches the upper temperature limit.

7.11 Living object detection (LOD) test**Test condition:**

1. System input: 220 Vac/50 Hz.
2. At nominal ground clearance, Z = 175 mm.
3. Water temperature of VA inlet: 45 °C, flow rate: 8L/min.

Test method:

TBD

Reference Requirement:

TBD

8 System Safety Test

8.1 Insulation resistance test

Test conditions:

1. Test without power.
2. Input to ground (chassis), output to ground (chassis) insulation resistance test:
 - Test voltage DC 1000V, test time 30s, insulation resistance not less than 10 MΩ.

Reference Requirement:

The insulation resistance between the non-electrical connection and the live circuits of the wireless charging system and between the individual live circuits and the ground (metal casing) shall not be less than 10 MΩ.

Test Item	Insulation resistance test					
Test objects	WB		GA		VA	
	Input to GND	Output to GND	Input to GND	Output to GND	Input to GND	Output to GND
Test Result						
Reference Standard	QC/T 895 · GB/T 18487.1-2015					

8.2 Dielectric strength test

Test conditions:

(A) For ground side equipment (WB, GA)

1. Test without power.
2. Input to GND (Casing), Output to GND (Casing) withstanding voltage test: Based on the test voltage specified in the table below, test for 1 min, no insulation breakdown and arcing during the test.
3. Input to communication port, output to communication port withstanding voltage test: Based on the test voltage specified in the table below, test for 1 min, no insulation breakdown and arcing during the test.

(B) Repeat the same conditions for vehicle side equipment (VA)

Rated Insulation Voltage U_i /V	Withstand voltage test voltage
≤ 60	DC 1.4kV / AC 1.0kV
$60 < U_i \leq 300$	DC 2.8kV / AC 2.0kV
$300 < U_i \leq 700$	DC 3.360kV / AC 2.4kV
$700 < U_i \leq 950$	DC $(2.8 \cdot U_i + 1400)$ V / AC $(2 \cdot U_i + 1000)$ V

Reference Requirements:

The wireless power transfer system should be able to withstand the test voltage specified in the above table for 1 min. There should be no insulation breakdown and arcing during test.

Test Item	Dielectric Strength Test		
Test Result	WB	Input to Ground	
		Output to Ground	
	GA	Input to Ground	
	VA	Output to Ground	
		Input to Com. Port	
		Output to Com. Port	
Reference Standard	QC/T 895 、GB/T 18487.1-2015		

8.3 Impulse withstand voltage test

Test condition:

1. Test without power.
2. Apply 3 positive and 3 negative lightning strokes specified in the table below with not more than 5 s interval. The lightning strikes are characterized by a 1.2/50 μ s voltage waves with 500 Ω source impedance.
3. Other circuit and exposed conductive parts are grounded during test.
4. During the test, there should be no breakdown discharge at the test site, allowing flashover without damage to the insulation. If flashover occurs, the dielectric strength should be reviewed.
5. The dielectric strength test voltage is 75 % of the specified value.

Rated Insulation Voltage U_i /V	Impulse Voltage test voltage
≤ 60	1 kV
$60 < U_i \leq 300$	± 5 kV
$300 < U_i \leq 700$	± 6 kV
$700 < U_i \leq 950$	± 6 kV

Reference Requirement:

The short-term impulse voltage test of the standard lightning wave specified in the table above shall be able to withstand the working voltage between the live circuits of the wireless charging system and between the live circuits and the ground (metal case). There should be no breakdown discharge during the test.

Test Item	Impulse withstand voltage test		
Test Result	WB	Input to Ground	
		Output to Ground	
		Input to Output	
	GA	Input to Ground	
Reference Standard	GB/T 18487.1-2015		

8.4 Ground Resistance Test

Test condition:

Test without power

Test Method:

Select the input terminal of the wireless charging system as a test point, another test point away from the input ground terminal; pass the twice input rated current between the two test points, and measure the voltage drop between the two points with a voltmeter. Calculate the resistance between two points.

1. If the rated current of the circuit to be tested is less than or equal to 16A, the test current is 2 times the rated current of the circuit under test and the test time is 2 min.
2. If the rated current of the circuit to be tested is greater than 16A, the test current is 2 times the rated current of the circuit under test, and the test time refers to the requirements of Table 2E of GB4943.1-2011.

Reference Requirement:

The ground resistance should below 0.1 Ω .

Test Item	Ground Resistance Test	
Test Result	2 times input rated current (A)	Ground Resistance (Ω)
Reference standard	GB/T 7251.1-2013	

8.5 Contact Current Test

Test conditions:

1. Input Voltage: 220 Vac/50 Hz.
2. At nominal ground clearance, Z = 175 mm.

Remarks: Before test, use an isolation transformer to isolate the grid to the wireless power transfer system. The power rating of the isolation transformer should over the rated power of the wireless power system

Test Method:

Contact current tester (simulated human impedance) is used to test the leakage current of each input line including live, neutral and protective ground line.

Reference Requirement:

Leakage current below 17 mA (Effective value).

Test Item	Contact Current Test	
Test Result	Leakage current between Live and ground	Leakage current between neutral and ground
Reference Standard	GB/T 18487.1-2015, GB/T 12113-2003, GB 4943.1-2011 & GB/T 2423.4-2008	

8.6 The discharge of the capacitor device

Test condition:

1. System input: 220 Vac/50 Hz.
2. At nominal ground clearance, $Z = 175$ mm. No offset.
3. Power output: $P_{out} = 6.6$ kW.
4. Water temperature of VA inlet: 45 °C, flow rate: 8 L/min.

Test Method:

Wireless power transfer system is operating normally. Disconnect the plug and start measuring the discharge time. Measure the duration when the voltage between the conductive parts or between any conductive part and the protective conductor less than 60 Vdc or the residual energy is less than 0.2 J. Alternatively, check the disconnected terminal at the external connection of the device. If the specified ingress protection can be met, no discharge time test is required.

Reference Requirement:

1s after the plug is disconnected from the socket, the voltage between the conductive parts or between any conductive part and the protective conductor shall be less than or equal to 60 Vdc or the residual energy shall be less than 0.2J.

Test Item	The discharge of the capacitor device	
Test Result	Voltage between the conductive parts or between any conductive part and the protective conductor (V)	Time (s)
Reference Standard		

8.7 Electromagnetic environment limit test

Test condition:

1. System input: 220 Vac/50 Hz;
2. At nominal ground clearance, $Z = 175$ mm. No offset.
3. Power output: $P_{out} = 6.6$ kW.
4. Water temperature of VA inlet: 45 °C, flow rate: 8 L/min.

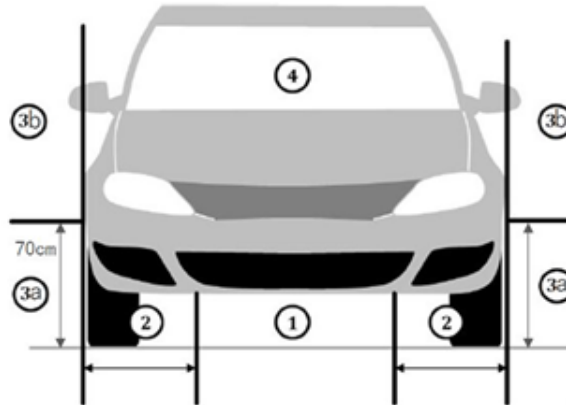
Test Method:

1. Ground side equipment WB

Under the condition of rated input and rated power output, using Narda ELT-400 to test the ground equipment WB and 20 cm directly above the surface of the connecting cable.

2. Vehicle protection area (3a, 3b & 4)

The magnetic field strength of the protected area is tested using a Narda ELT-400 device at rated input and rated power output.



Reference Requirement:

Electromagnetic field environment limit $<6.25\mu\text{T}$

Test Item	Electromagnetic environment limit test				
Test area	WB	WB to GA cable	3a	3b	4
Test result (no offset)					
Test result (maximum offset)					

9 EMC Test for VA

9.1 Overview

9.1.1 Classification

The EMC test project is divided into two parts: the vehicle component test and the off-vehicle component test. The tests of 7.2-7.x were used to verify that the external harassment of the test sample meets certain limits and has a certain degree of immunity to external disturbances.

The vehicle components include a coil portion at the vehicle end and a coil control portion at the vehicle end. The off-vehicle components include a coil portion at the ground end and a coil control portion at the ground end. For on-board EMC testing, the test setup may or may not include ground end components depending on the specific requirements of the test project. For non-vehicle end EMC testing, the test setup includes ground end components and vehicle end components.

9.1.2 EMC test criteria for VA

Criterion I: The design function can be completed during and after the test.

Criterion II: The design function cannot be completed during the test, but it can automatically return to the normal state after the test.

Criterion III: The design function cannot be completed during the test. After the test, it cannot be restored to the normal state without the simple operation of the driver/passenger, for example, by turning the DUT off/on or restarting the ignition switch.

Criterion IV: The design function cannot be completed during the test. After the test, more complicated operations are required to return to the normal state, and the function of the DUT should not cause any permanent damage. For example, disconnect the battery or power supply before connecting.

9.2 Radiation emission of electrical and electronic components – ALSE method

9.2.1 Overview

Through this test, the acceptable radio reception effect on the whole vehicle is guaranteed, mainly including the broadcasting band, the mobile service band, and the positioning system band.

Radiation field strength measurements should be made within the ALSE to eliminate the effects of additional electromagnetic disturbances from electrical equipment and broadcast stations. Since the test arrangement itself generates radiation, it is recommended to test the environmental level before measuring the EUT's radiation, *i.e.*, the DUT is not connected and energizes the load, housing, and simulator. The ambient index should be below the limit of 6 dB and the data should be recorded.

Test equipment should meet the requirements of IEC CISPR 25. Refer to CISPR 25: 2016 6.5 for test layout and test methods.

9.2.2 Test conditions

9.2.2.1 Antenna selection

A linearly polarized electric field antenna with a nominal 50 Ω output impedance is used. Depending on the measurement frequency, select the different types of antennas in the table below.

150kHz-30MHz	1m long vertical monopole antenna
30MHz-300MHz	biconical antenna
200MHz-1GHz	Logarithmic period antenna
1GHz-2.5GHz	Horn or log periodic antenna

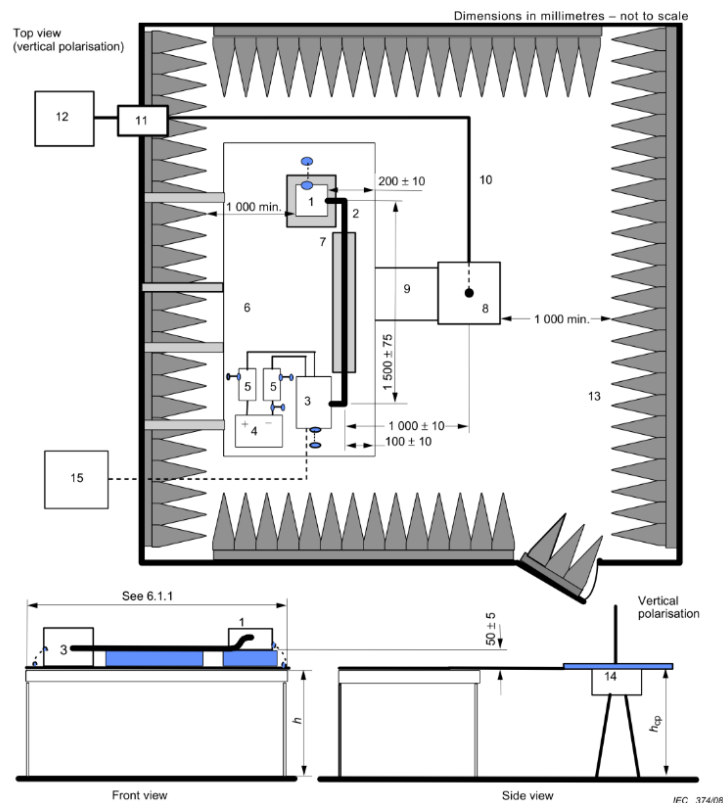
9.2.2.2 Test arrangement

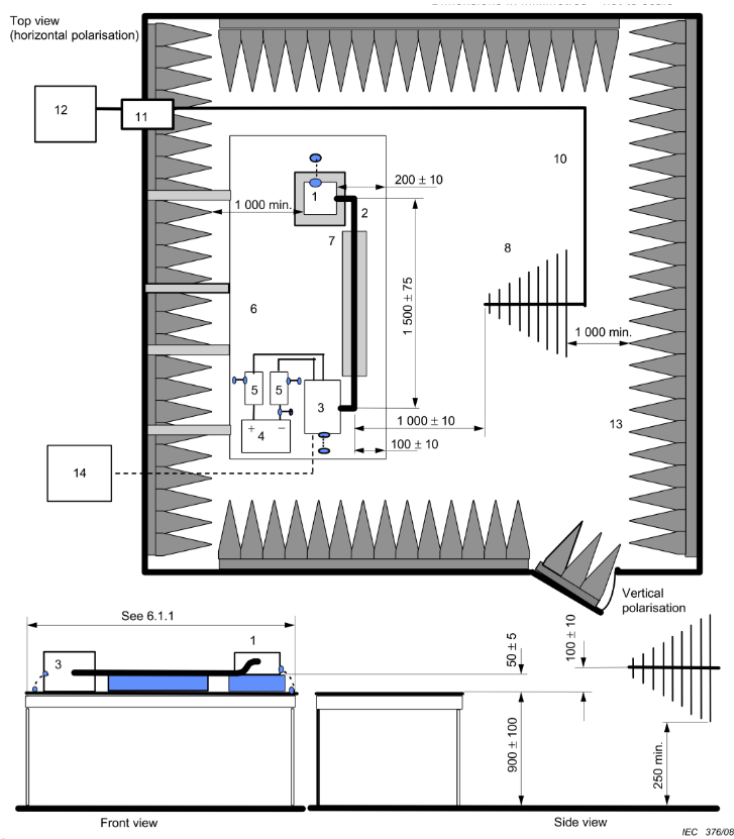
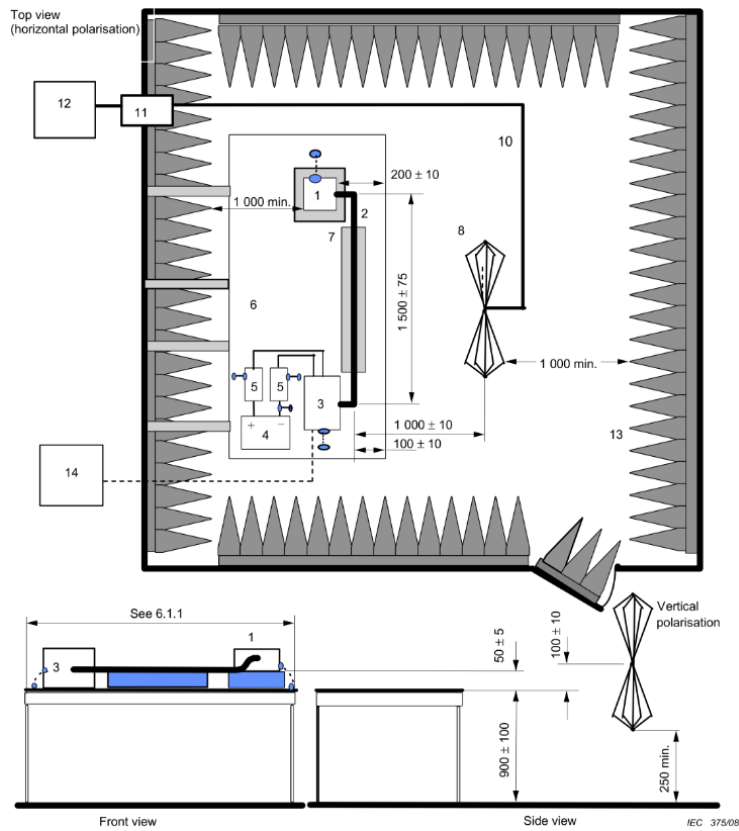
The EUT should be placed on a non-conductive, low relative dielectric constant material. The EUT is $50 \text{ mm} \pm 5 \text{ mm}$ above the ground plane. The EUT is at least $200 \text{ mm} \pm 10 \text{ mm}$ from the edge of the ground plane. The ground plane is $900 \text{ mm} \pm 100 \text{ mm}$ from the ground.

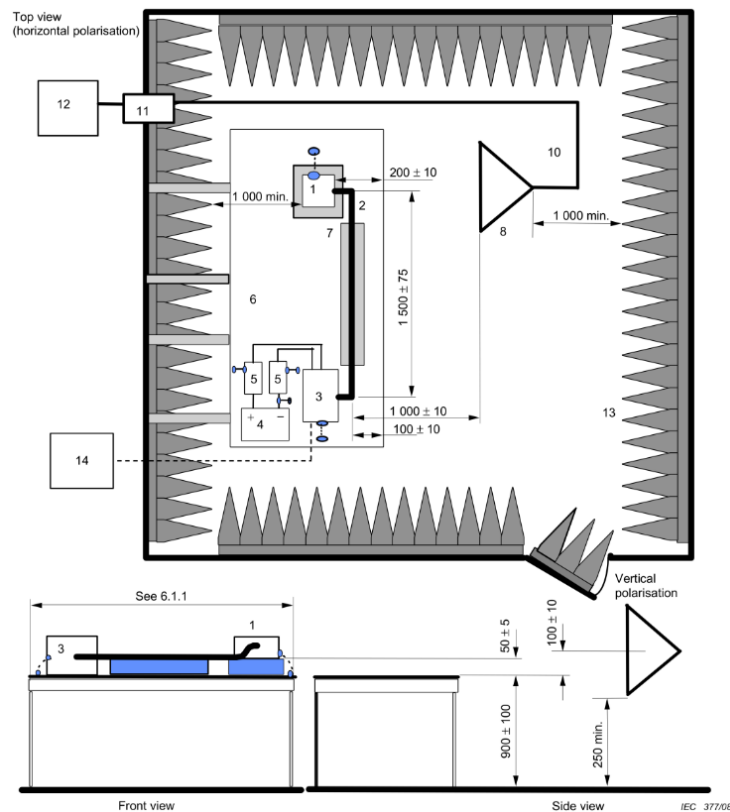
The total length of the test harness between the EUT and the load simulator or RF boundary should not exceed 2000 mm . The test harness shall be placed on a non-conducting, low relative dielectric constant material. The test harness is $50 \text{ mm} \pm 5 \text{ mm}$ above the ground plane. The long side of the test harness shall be placed parallel to the edge of the ground plane, the length shall be $1500 \text{ mm} \pm 75 \text{ mm}$, and the edge of the antenna shall be $100 \text{ mm} \pm 10 \text{ mm}$. The wire bending angle is $90^\circ \pm 45^\circ$.

The height of the balance plate of the rod antenna relative to the ground plane is $+10/-20 \text{ mm}$. The phase center of the biconical antenna, the logarithmic period antenna, and the horn antenna is $100 \text{ mm} \pm 10 \text{ mm}$ above the ground plane. When the measurement frequency is below 1 GHz , the antenna is aligned with the center of the harness. When the measurement frequency is above 1 GHz , the antenna is aligned with the center of the EUT.

The experimental arrangement of a monopole antenna, a biconical antenna, a logarithmic period antenna, a horn or a logarithmic period antenna corresponds to the following figures 1 to 4, respectively.







9.2.3 Test methods

9.2.3.1 EUT status

The EUT is set to the following four working modes:

1. In charging mode with maximum wireless transmission power with the coils are aligned. Set the load voltage to 350V and simulate battery SOC between 20%-80%. The ground side coil and the vehicle side coil are aligned (0,0).
2. In charging mode with maximum wireless transmission power with the coils are in maximum offset. Set the load voltage to 350V and simulate battery SOC between 20%-80%. The ground side coil and the vehicle side coil are in maximum offset (75,100).
3. The communication has been established and the wireless transmission has not started.
4. The communication was not established.

9.2.3.2 Test procedure

The EUT shall be made to operate under typical loading and other conditions as in the vehicle. The typical loading when the system is in wireless transmission charging mode, while the other conditions as in the vehicle when it is in non-wireless power transmission.

From 150 kHz to 30 MHz measurements shall be performed in vertical polarization only. From 30 MHz to 2.5 GHz measurements shall be performed in vertical and horizontal and polarizations.

9.2.3.3 Measurement frequency band

The measurement frequency bands are mainly divided into broadcasting, mobile services, and positioning systems according to services, and each of them includes several sub-bands. When there is a band overlap, the limits used are to be specified in the test plan.

9.2.4 Test criteria

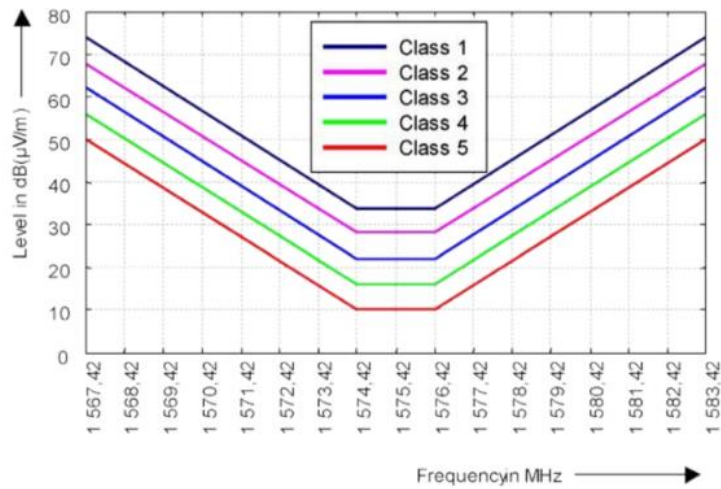
9.2.4.1 Broadcast and mobile services

The radiated noise of the broadcasting and mobile service bands should not exceed the limits in the table below.

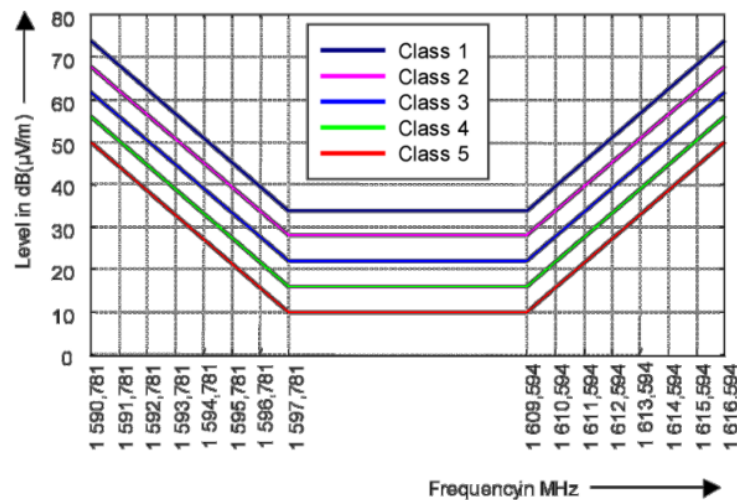
Service / Band	Frequency MHz	Levels in dB(μV/m)														
		Class 5			Class 4			Class 3			Class 2			Class 1		
		Peak	Quasipeak	Average	Peak	Quasipeak	Average	Peak	Quasipeak	Average	Peak	Quasipeak	Average	Peak	Quasipeak	Average
BROADCAST																
LW	0,15 to 0,30	46	33	26	56	43	36	66	53	46	76	63	56	86	73	66
MW	0,53 to 1,8	40	27	20	48	35	28	56	43	36	64	51	44	72	59	52
SW	5,9 to 6,2	40	27	20	46	33	26	52	39	32	58	45	38	64	51	44
FM	76 to 108	38	25	18	44	31	24	50	37	30	56	43	36	62	49	42
TV Band I	41 to 88	28	–	18	34	–	24	40	–	30	46	–	36	52	–	42
TV Band III	174 to 230	32	–	22	38	–	28	44	–	34	50	–	40	56	–	46
DAB III	171 to 245	26	–	16	32	–	22	38	–	28	44	–	34	50	–	40
TV Band IV	468 to 944	41	–	31	47	–	37	53	–	43	59	–	49	65	–	55
DTTV	470 to 770	45	–	35	51	–	41	57	–	47	63	–	53	69	–	59
DAB L Band	1 447 to 1 494	28	–	18	34	–	24	40	–	30	46	–	36	52	–	42
SDARS	2 320 to 2 345	34	–	24	40	–	30	46	–	36	52	–	42	58	–	48
MOBILE SERVICES																
CB	26 to 28	40	27	20	46	33	26	52	39	32	58	45	38	64	51	44
VHF	30 to 54	40	27	20	46	33	26	52	39	32	58	45	38	64	51	44
VHF	68 to 87	35	22	15	41	28	21	47	34	27	53	40	33	59	46	39
VHF	142 to 175	35	22	15	41	28	21	47	34	27	53	40	33	59	46	39
Analogue UHF	380 to 512	38	25	18	44	31	24	50	37	30	56	43	36	62	49	42
RKE	300 to 330	32	–	18	38	–	24	44	–	30	50	–	36	56	–	42
RKE	420 to 450	32	–	18	38	–	24	44	–	30	50	–	36	56	–	42
Analogue UHF	820 to 960	44	31	24	50	37	30	56	43	36	62	49	42	68	55	48
GSM 800	860 to 895	44	–	24	50	–	30	56	–	36	62	–	42	68	–	48
EGSM/GSM 900	925 to 960	44	–	24	50	–	30	56	–	36	62	–	42	68	–	48
GPS L1 civil	1 567 to 1 583	–	–	10	–	–	16	–	–	22	–	–	28	–	–	34
GLONASS L1	1 591 to 1 616	–	–	10	–	–	16	–	–	22	–	–	28	–	–	34
GSM 1800 (PCN)	1 803 to 1 882	44	–	24	50	–	30	56	–	36	62	–	42	68	–	48
GSM 1900	1 850 to 1 990	44	–	24	50	–	30	56	–	36	62	–	42	68	–	48
3G / IMT 2000	1 900 to 1 992	44	–	24	50	–	30	56	–	36	62	–	42	68	–	48
3G / IMT 2000	2 010 to 2 025	44	–	24	50	–	30	56	–	36	62	–	42	68	–	48
3G / IMT 2000	2 180 to 2 172	44	–	24	50	–	30	56	–	36	62	–	42	68	–	48
Bluetooth/802.11	2 400 to 2 500	44	–	24	50	–	30	56	–	36	62	–	42	68	–	48

9.2.4.2 Positioning system

GPS System



GLONASS System



9.3 CE Conducted emission of electrical and electronic components - Voltage method

9.3.1 Overview

Through this experiment, to ensure acceptable vehicle radio receiving effects, including broadcast band, mobile traffic bands, band positioning system. Voltage measurements are able to characterize the emissions on single leads only. The test method is not usable to characterize the radiated emission transmitted or to characterize the efficiency of shielding. Therefore, voltage measurements are not able to characterize the complete EUT emission. At lower frequencies (e.g. in the AM-bands) voltage measurements usually ensure more dynamic range than radiated measurements.

Test equipment should meet the requirements of IEC CISPR 25. Refer to CISPR 25: 2016 6.3 for test layout and test methods.

9.3.2 Test conditions

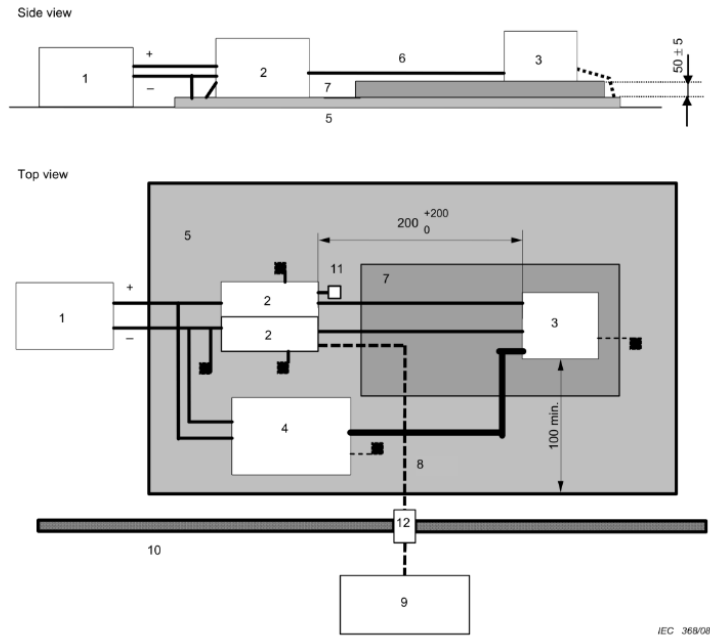
9.3.3 Test arrangement

The EUT shall be placed on a non-conductive, low relative permittivity material, at 50mm±5mm above the ground plane. The EUT shall be at least 100mm from the edge of the ground plane.

Cable length between the EUT and Artificial connector network connector to be 200mm-400mm. Cable should be placed on a straight line along a non-conductive, low relative permittivity material. The cable is 50mm ± 5mm above the ground plane. The cable is at least 100 mm from the edge of the ground plane.

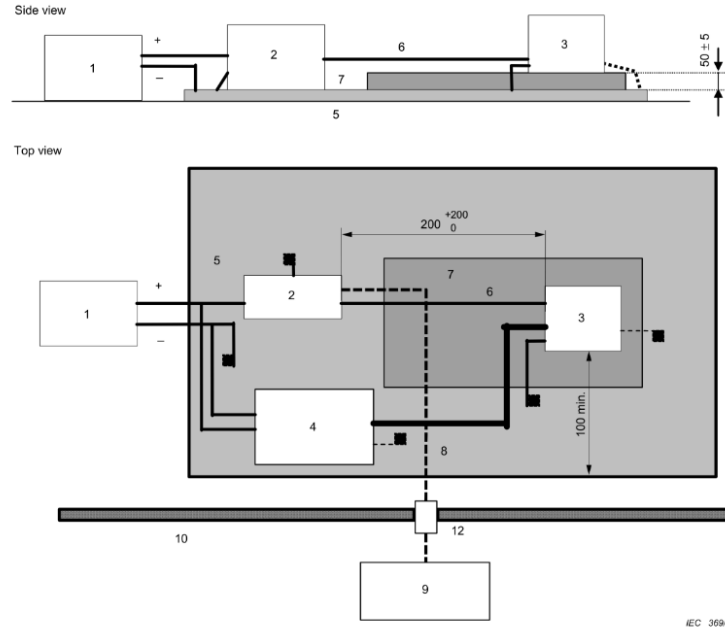
The vertical distance of the input/output wires from the power line connected between the artificial network and the EUT is ≥ 200 mm. The total length of the test harness (excluding cables) shall not exceed 2 m. All wires are at least 100 mm from the edge of the ground plane.

Power return line remotely grounded and power return line locally grounded are shown below in figure 1 and figure 2 respectively.

**Key**

- | | |
|--|--|
| 1 Power supply (may be placed on the ground plane) | 7 Low relative permittivity support ($\epsilon_r \leq 1,4$) |
| 2 Artificial network | 8 High-quality coaxial cable e.g. double-shielded (50 Ω) |
| 3 EUT (housing grounded if required in test plan) | 9 Measuring instrument |
| 4 Load simulator (metallic casing grounded if required in test plan) | 10 Shielded enclosure |
| 5 Ground plane | 11 50 Ω load |
| 6 Power supply lines | 12 Bulkhead connector |

NOTE The EUT housing ground lead, when required in the test plan, should not be longer than 150 mm.

**Key**

- | | |
|--|--|
| 1 Power supply (may be placed on the ground plane) | 7 Low relative permittivity support ($\epsilon_r \leq 1,4$) |
| 2 Artificial network | 8 High-quality coaxial cable e.g. double-shielded (50 Ω) |
| 3 EUT (housing grounded if required in test plan) | 9 Measuring instrument |
| 4 Load simulator (metallic casing grounded if required in test plan) | 10 Shielded enclosure |
| 5 Ground plane | 12 Bulkhead connector |
| 6 Power supply line | |

NOTE The EUT housing ground lead, when required in the test plan, should not be longer than 150 mm.

9.3.4 Test Methods

9.3.4.1 EUT Status

Same as 7.2.3.1.

9.3.4.2 Test procedure

The EUT shall be made to operate under typical loading and conditions that produce maximum electromagnetic emissions in the vehicle. The typical loading when the system is in wireless transmission charging mode, while the conditions as in the vehicle when it is in non-wireless power transmission.

Voltage measurements can be measured relative to ground on both the supply line and its return line or only measure the power line relative to ground. When the EUT is remotely grounded, i.e. the vehicle power return line is greater than 200mm, the voltage measurement is measured on the power supply line and the return line. When the EUT is grounded locally, i.e. the vehicle power return line is less than or equal to 200mm, the voltage measurement is measured on the power supply line only.

Measuring frequency is 150kHz-88MHz °

9.3.4.3 Measurement frequency band

The measurement frequency bands are mainly divided into broadcast and mobile services according to services, and each of them includes several sub-bands. When there is a band overlap, the limits used are to be specified in the test plan.

9.3.5 Test criteria

The radiated noise of the broadcasting and mobile service bands should not exceed the limits in the table below.

Service / Band	Frequency MHz	Levels in dB(μV)														
		Class 5			Class 4			Class 3			Class 2			Class 1		
		Peak	Quasipeak	Average	Peak	Quasipeak	Average	Peak	Quasipeak	Average	Peak	Quasipeak	Average	Peak	Quasipeak	Average
BROADCAST		70	57	50	80	67	60	90	77	70	100	87	80	110	97	90
LW	0,15 to 0,30	54	41	34	62	49	42	70	57	50	78	65	58	86	73	66
MW	0,53 to 1,8	53	40	33	59	46	39	65	52	45	71	58	51	77	64	57
SW	5,9 to 6,2	38	25	18	44	31	24	50	37	30	56	43	36	62	49	42
FM	76 to 108	34	–	24	40	–	30	46	–	36	52	–	42	58	–	48
TV Band I	41 to 88	Conducted emission – Voltage method Not applicable														
TV Band III	174 to 230															
DAB III	171 to 245															
TV Band IV	468 to 944															
DTTV	470 to 770															
DAB L Band	1 447 to 1 494															
SDARS	2 320 to 2 345															
MOBILE SERVICES		44	31	24	50	37	30	56	43	36	62	49	42	68	55	48
CB	26 to 28	44	31	24	50	37	30	56	43	36	62	49	42	68	55	48
VHF	30 to 54	38	25	18	44	31	24	50	37	30	56	43	36	62	49	42
VHF	68 to 87	Conducted emission – Voltage method Not applicable														
VHF	142 to 175															
Analogue UHF	380 to 512															
RKE	300 to 330															
RKE	420 to 450															
Analogue UHF	820 to 960															
GSM 800	860 to 895															
EGSM/GSM 900	925 to 960															
GPS L1 civil	1 567 to 1 583															
GLONASS L1	1 591 to 1 613															
GSM 1800 (PCN)	1 803 to 1 882															
GSM 1900	1 850 to 1 990															
3G / IMT 2000	1 900 to 1 992															
3G / IMT 2000	2 010 to 2 025															
3G / IMT 2000	2 180 to 2 172															
Bluetooth/802.11	2 400 to 2 500															

9.4 CTE Electrical transient conduction test of electrical and electronic components along the power line

9.4.1 Overview

Through this test, the EUT's power line transient emissions are evaluated. Artificial networks are used in the tests and the measurement arrangements are not subject to interference from the surrounding electromagnetic environment.

During the test, the ambient temperature should be $23\text{ °C} \pm 5\text{ °C}$. The test voltage is $13.5 \pm 0.5\text{V}$ (when the generator is working) or $12 \pm 0.2\text{V}$ (when the battery is powered)

Test arrangements and test methods refer to ISO 7637-2: 2011 4.3.

9.4.2 Test Conditions

9.4.2.1 Artificial network and shunt resistor

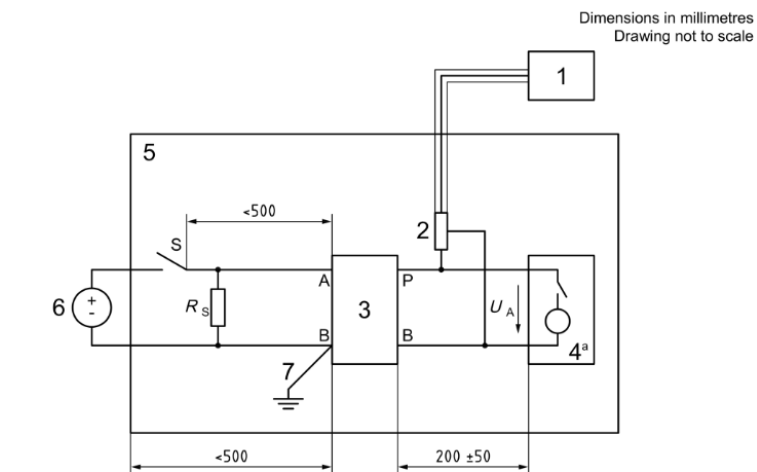
The artificial network is also called the power impedance stabilization network. The artificial network was used in the test to provide a stable impedance to the EUT instead of the impedance of the vehicle wiring harness. If the artificial network has a metal housing, it should be placed flat on the ground plane.

The shunt resistor is used to simulate the DC resistance of other electrical devices of the vehicle in parallel with the EUT. The connection of these electrical devices to the EUT is not controlled by the ignition switch. The shunt resistance value R_s can be measured from the wire harness between the open ignition switch terminal and ground. When not explicitly defined, $R_s = 40\ \Omega$ is selected. In order to simulate the worst case, the shunt resistor can be disconnected

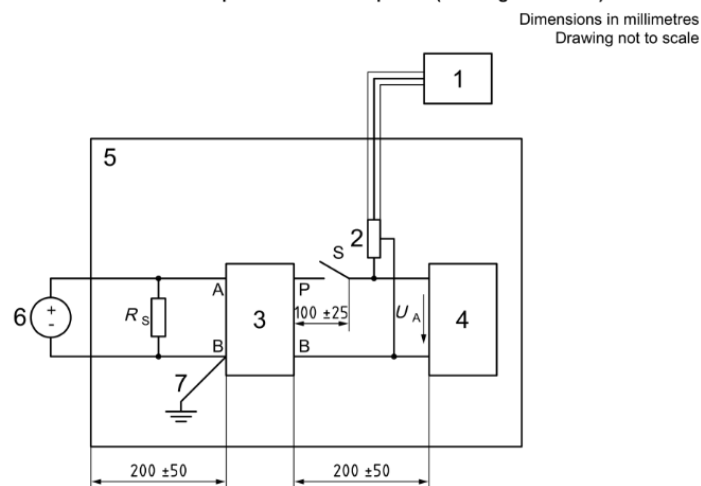
9.4.2.2 Test arrangement

The artificial network, switch, EUT and all connection wiring between them should be placed 50mm-60mm above the point grounding plate. The EUT should be placed on a non-conductive material above the ground plane, and it should place 50mm-60mm above the ground plane. When measuring the disturbance voltage, the measuring device should be as close as possible to the terminals of the EUT.

The test arrangement of the slow pulse transient emission test and the fast pulse transient emission tests are shown in figure 1 and 2 below.



a) Transient emission test set-up to measure slow pulses (ms-range or slower)



b) Transient emission test set-up to measure fast pulses (ns to μ s range)
for DUT without internal switch

9.4.3 Test Methods

9.4.3.1 EUT status

Same as 7.2.3.1.

9.4.3.2 Test procedure

Repeated transients should be measured when switch S is closed. If the transient is caused by a disconnection of the power cord, the measurement should be made when switch S is open.

The test is divided into two types: slow pulse and fast pulse. The measurement is first performed using a slow pulse test setup to determine the maximum amplitude of the slow pulse (in millisecond range) and then measured using the test arrangement of the fast pulse to determine the maximum amplitude of the transient (nanosecond range).

Collect 10 waveforms and record parameters such as voltage amplitude, rise time, fall time, and transient width.

9.4.4 Test criteria

A single transient waveform and burst transient waveform are shown below.

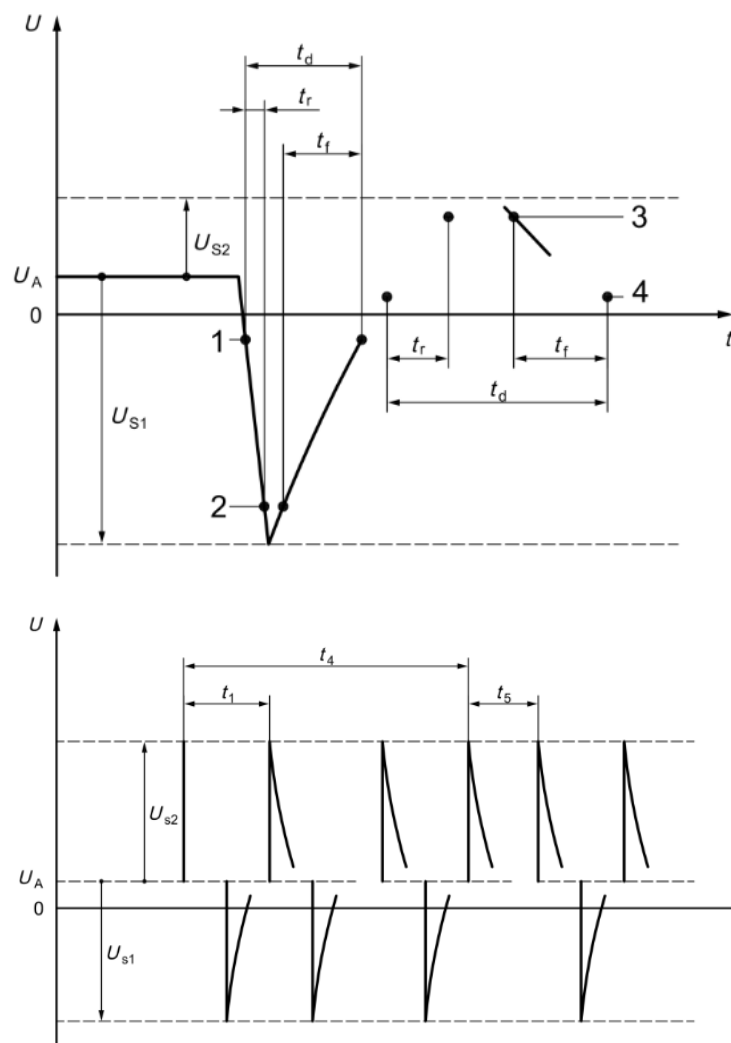


Figure B.2 — Burst transient waveform

The positive and negative pulse voltage amplitudes of the transient waveform should not exceed the limits in the table below.

Pulse amplitude (U_s)	Suggested limit for U_s for severity levels I to V			
	I / II	III	IV	V ^a
Positive slow pulses (ms range or slower)	+25 V	+37 V	+75 V	
Negative slow pulses (ms range or slower)	-50 V	-75 V	-100 V	
Positive fast pulses (μ s to ns range)	+50 V	+75 V	+100 V	
Negative fast pulses (μ s to ns range)	-75 V	-112 V	-150 V	

^a Values to be determined by vehicle manufacturer and equipment supplier.

9.5 RI Immunity test of electrical and electronic components for narrowband radiation-ALSE method

9.5.1 Overview

Through this test, simulate the electromagnetic environment of the car in which the EUT is located. EUT's immunity to continuous narrow band radiated electrical disturbances is assessed. The test uses an antenna and an RF energy source that produces the required field strength to form a radiated electromagnetic field. The test shall be carried out in an anechoic chamber (simulated open field) with or without absorbing material on the ground.

During the test, the ambient temperature should be $23\text{ }^{\circ}\text{C} \pm 5\text{ }^{\circ}\text{C}$. The test voltage was $13.5 \pm 0.5\text{V}$.

Test arrangements and test methods refer to ISO 11452-2: 2004.

9.5.2 Test conditions

9.5.2.1 Power and Artificial Networks

The power cord of each EUT should be connected to the power supply via an artificial network. When EUT is remotely grounded (vehicle power return line is greater than 200mm), two artificial networks are required, one connected to the positive pole of the power supply and one connected to the power return line. When the EUT is grounded locally, an artificial network is required to connect to the positive terminal of the power supply.

The parameters of the artificial network are $5\mu\text{H}$, 50Ω . The measurement port of each artificial network should be connected to a load of 50Ω .

9.5.2.2 Test arrangement

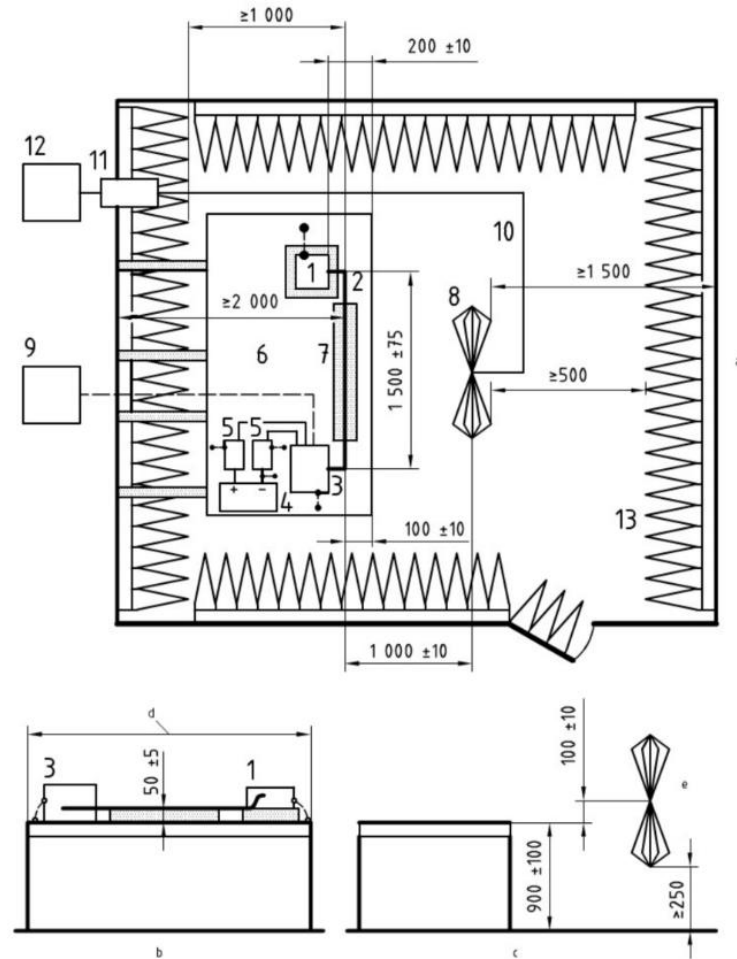
The EUT should be placed on a non-conducting, low relative dielectric constant material located $50\text{ mm} \pm 5\text{ mm}$ above the ground plane. The front surface of the EUT is $200\text{ mm} \pm 10\text{ mm}$ from the edge of the ground plane.

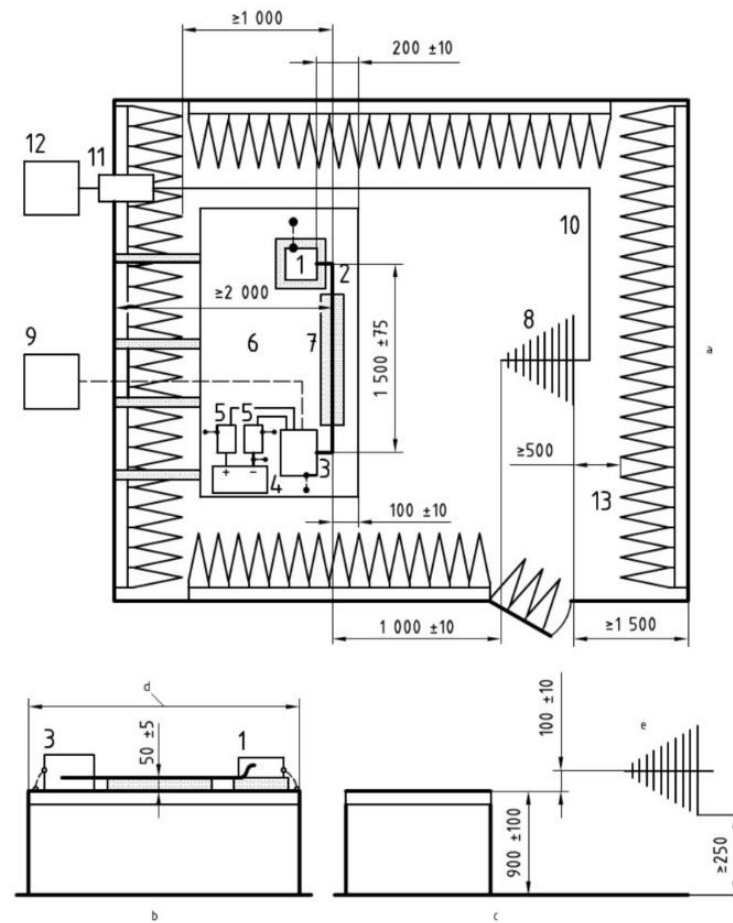
The test harness shall be placed on a non-conductive, low relative dielectric constant material located $50\text{ mm} \pm 5\text{ mm}$ above the ground plane. The length of the parallel portion of the test harness to the front edge of the ground plane shall be $1500\text{ mm} \pm 75\text{ mm}$ and the distance shall be $100\text{ mm} \pm 10\text{ mm}$. The total length of the test harness between the EUT and the load simulator or RF boundary should not exceed 2000 mm.

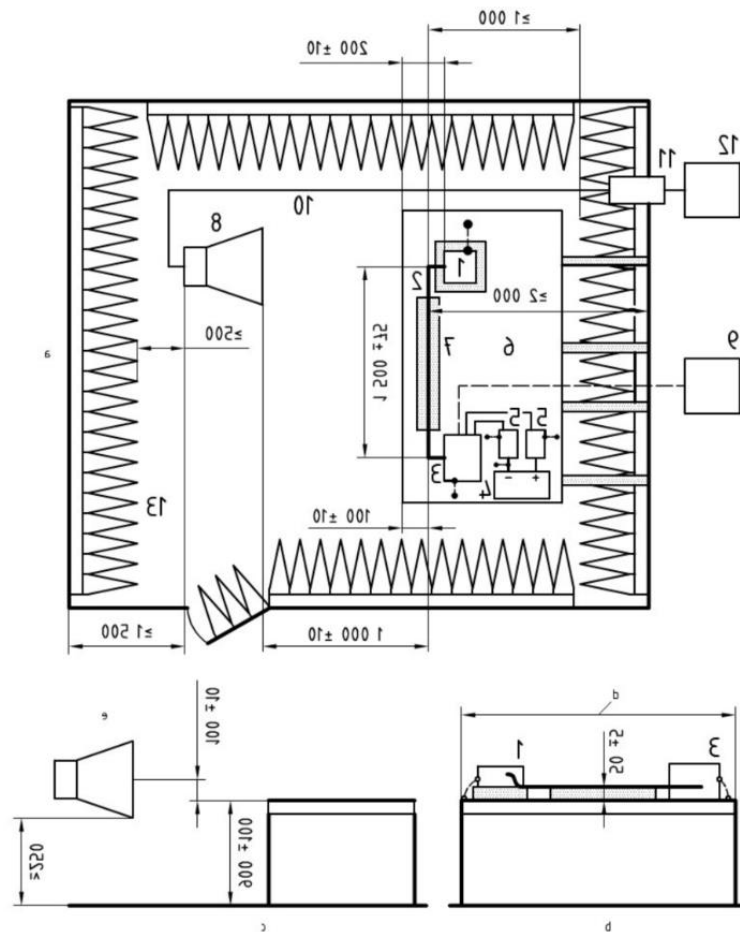
The height of the antenna phase center should be $100\text{mm} \pm 10\text{mm}$ above the ground plane. The antenna distance test harness should be $1000\text{mm} \pm 10\text{mm}$. When the measurement frequency is below 1 GHz, the antenna is aligned with the center of the wire harness (15000 mm portion).

When the measurement frequency is above 1 GHz, the antenna is aligned with the center of the EUT.

The test arrangement of a biconical antenna, a logarithmic period antenna, a horn or a logarithmic period antenna corresponds to figures 1 to 3, respectively.







9.5.3 Test methods

9.5.3.1 EUT status

Same as 7.2.3.1.

9.5.3.2 Test procedure

The field strength of the test level shall be calibrated using an unmodulated sine wave prior to testing. The field strength probe is used to measure the field strength and the forward power required to produce the specified field strength at each test frequency is recorded.

The test was carried out according to a predetermined calibration value. When the measurement frequency is between 80MHz and 400MHz, only the horizontal polarization direction is measured. When the measurement frequency is above 400 MHz, it is necessary to measure both the horizontal polarization and the vertical polarization.

The type of modulation signal can be selected from CW (15Hz-18GHz) unmodulated sine wave; or an AM (10 kHz-800 MHz) modulation frequency of 1 kHz, an amplitude modulated sine wave with a modulation depth of 0.8; or a PM-type1 (800MHz-1.2GHz, 1.4GHz-2.7GHz) pulse-width modulated sine wave with a pulse width of 577μs and a period of 4600μs; or PM-type 2 (1.2 GHz-1.4 GHz, 2.7 GHz-18 GHz) pulse-modulated sine wave with a pulse width of 3 μs and a period of 3333 μs.

The dwell time of each frequency point during the test sweep is at least 1 s, and the step size should be less than the specified step size in the table below.

Frequency band	Linear steps	Logarithmic steps
15 Hz to 100 Hz	10 Hz	10 %
> 100 Hz to 1 kHz	100 Hz	10 %
> 1 kHz to 10 kHz	1 kHz	10 %
> 10 kHz to 100 kHz	10 kHz	10 %
> 100 kHz to 1 MHz	100 kHz	10 %
> 1 MHz to 10 MHz	1 MHz	10 %
> 10 MHz to 200 MHz	5 MHz	5 %
> 200 MHz to 400 MHz	10 MHz	5 %
> 400 MHz to 1 GHz	20 MHz	2 %
> 1 GHz to 18 GHz	40 MHz	2 %

9.5.4 Test criteria

The test severity level and test frequency band shall comply with Table 1 - Table 2 below.

Test severity level	Value V/m
I	25
II	50
III	75
IV	100
V	Specific value agreed between the users of this part of ISO 11452, if necessary

Frequency band	Frequency range MHz
F1	> 80 to ≤ 400
F2	> 400 to ≤ 1 000
F3	> 1 000 to ≤ 10 000
F4	> 10 000 to ≤ 18 000

9.6 BCI Immunity test of electrical and electronic components to narrow-band radiated electromagnetic energy-BCI method

9.6.1 Overview

Through this test, simulate the electromagnetic environment of the car in which the EUT is located. EUT's immunity to continuous narrow band radiated electrical disturbances is assessed. The high current injection method is a method of performing an immunity test by directly coupling a disturbance signal to a wire harness passing through the transformer using a current injection probe.

The test includes both replacement methods and closed loop methods. The replacement method refers to that the pre-calibrated RF dry-wound signal is amplified by the RF power amplifier and injected into the cable by the current probe, and the current measurement probe is used for real-time measurement, which is an open-loop test. The closed-loop method of limiting power refers to measuring the real-time current value in the cable by using the current measuring probe while injecting the current, comparing the real-time value with the target value, and achieving the preset target value by modulating the level of the signal, which is a closed-loop test. Since the

replacement method is more adopted by the car manufacturer and more accurate, this section only covers the replacement method.

During the test, the ambient temperature should be $23\text{ }^{\circ}\text{C} \pm 5\text{ }^{\circ}\text{C}$. The test voltage was $13.5 \pm 0.5\text{V}$.

Test arrangements and test methods refer to ISO 11452-4: 2005.

9.6.2 Test conditions

9.6.2.1 Power and Artificial Networks

The power cord of each EUT should be connected to the power supply via an artificial network. When the EUT is remotely grounded (vehicle power return line is greater than 200mm), two artificial networks are required, one connected to the positive pole of the power supply and one connected to the power return line. When the EUT is locally grounded, an artificial network is required to connect to the positive terminal of the power supply.

The parameters of the artificial network are $5\mu\text{H}$, 50Ω . The measurement port of each artificial network should be connected to a load of 50Ω .

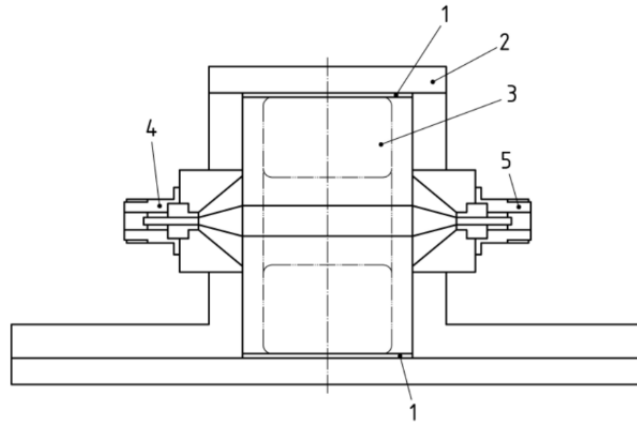
9.6.2.2 Test arrangement

The grounding plate is located on the test bench with a height of $900\text{mm} \pm 100\text{mm}$ from the ground. The minimum size of the ground plane is $2000\text{mm} \times 1000\text{mm} \times 0.5\text{mm}$ and is at least 200mm larger than each side of the EUT.

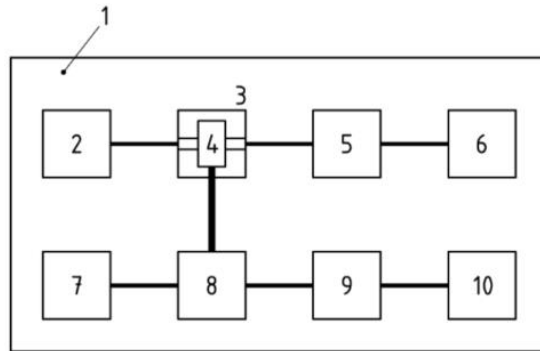
The EUT should be placed on an insulated bracket, $50\text{mm} \pm 5\text{mm}$ above the ground plane. The EUT surface is at least 100 mm from the edge of the ground plane.

The total length of the test harness between the EUT and the load simulator or RF boundary shall be $1000\text{ mm} \pm 10\text{ mm}$. The harness should be placed on the insulating bracket at a distance of $50\text{ mm} \pm 5\text{ mm}$ above the ground plane. The test harness should be placed in a straight line, and the harness should pass through the current injection probe and the current measurement probe. Both the test harness and the wires connected to the load simulator should be fixed and the wires connected to the load simulator should be shorter than the test harness.

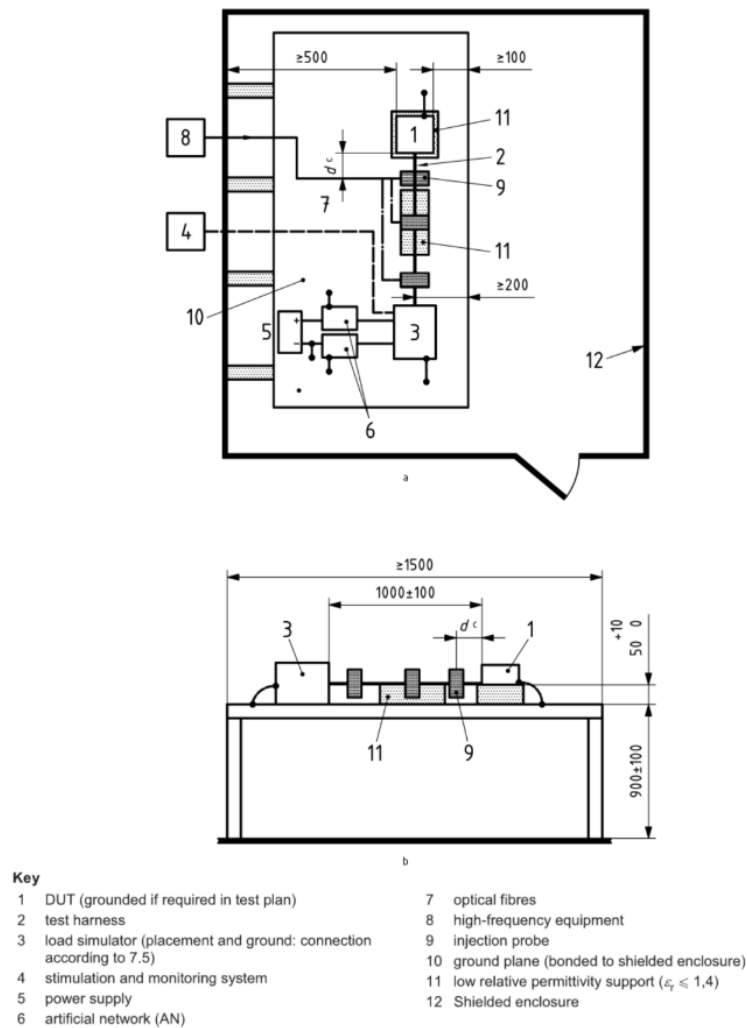
The distance between the injection probe and the EUT connector is $150\text{ mm} \pm 10\text{ mm}$, $450\text{ mm} \pm 10\text{ mm}$, and $750\text{ mm} \pm 10\text{ mm}$, respectively. The calibration fixture, calibration arrangement and test arrangement are shown in figure 1 - 3, respectively. One end of the calibration fixture is connected to a 50Ω load, the other end is connected to a 50Ω RF power meter, and a 50Ω attenuator of corresponding power is connected in series to protect the power meter.

**Key**

- 1 insulation
- 2 removable metal cover
- 3 current injection probe
- 4 direct connection to 50 Ω measurement equipment
- 5 direct connection to 50 Ω load

**Key**

- 1 shielded enclosure
- 2 50 Ω coaxial load, VSWR 1,2:1 max
- 3 calibration fixture
- 4 injection probe
- 5 50 Ω attenuator
- 6 spectrum analyser or equivalent
- 7 RF power level measuring device (two are required)
- 8 RF 50 Ω dual directional coupler (with 30 dB minimum decoupling coefficient)
- 9 broadband amplifier with 50 Ω output impedance
- 10 RF signal generator



9.6.3 Test methods

9.6.3.1 EUT status

Same as 7.2.3.1.

9.6.3.2 Test procedure

The replacement method uses forward power as a reference for calibration and testing. It is divided into two phases, calibration and testing.

Before the test, the calibration test should be carried out with a $50\ \Omega$ calibration fixture and a non-modulated sine wave. The forward power required to produce the specified current on a 50-ohm calibration fixture at each test frequency was recorded.

The EUT is applied with a test signal based on a predetermined calibration value. A current measuring probe can be added between the current injection probe and the EUT. The distance between the current measuring probe and the EUT connector should be $50\text{ mm} \pm 10\text{ mm}$, but the measuring probe may affect the injection current.

9.6.4 Test criteria

The test severity level and test frequency band shall comply with Table 1 - Table 2 below.

Test severity level	Value mA
I	25
II	50
III	75
IV	100
V	Specific value agreed between the users of this part of ISO 11452, if necessary.

Frequency band	Frequency range MHz
F1	≥ 1 to ≤ 10
F2	> 10 to ≤ 30
F3	> 30 to ≤ 80
F4	> 80 to ≤ 200
F5	> 200 to ≤ 400

9.7 CI Transient Immunity Test of Electrical Transient Conduction of Electrical and Electronic Components along Power Lines

9.7.1 Overview

Through this test, the electromagnetic environment of the vehicle in which the EUT is located is simulated, and the EUT's immunity to transients transmitted through the power line is evaluated. Artificial networks are used in the tests and the measurement arrangements are not subject to interference from the surrounding electromagnetic environment. The test should use an artificial network and a parallel resistor.

During the test, the ambient temperature should be $23\text{ }^{\circ}\text{C} \pm 5\text{ }^{\circ}\text{C}$. The test voltage was $13.5 \pm 0.5\text{V}$.

Test arrangements and test methods refer to ISO 7637-2: 2004. Latest for 2011

9.7.2 Test conditions

9.7.2.1 Artificial network and shunt resistor

The artificial network, also known as the power supply impedance stabilization network, uses artificial networks to provide a stable impedance to the EUT instead of the impedance of the vehicle wiring harness. If the artificial network has a metal housing, it should be placed flat on the ground plane.

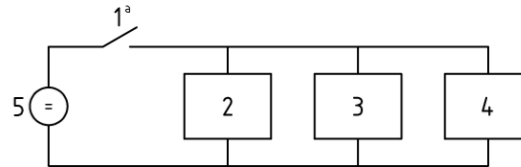
The shunt resistor is used to simulate the DC resistance of other electrical devices of the vehicle in parallel with the EUT. The connection of these electrical devices to the EUT is not controlled by the ignition switch. The shunt resistance value R_s can be measured from the wire harness between the open ignition switch terminal and ground. When not explicitly defined, $R_s = 40\text{ }\Omega$ is selected. In order to simulate the worst case, the shunt resistor can be disconnected.

9.7.2.2 Test pulse

The test pulse is derived from the pulses measured in the vehicle electrical system under certain conditions, as well as the pulses generated by the basic configuration of the electrical circuit. There are several types of pulses: pulse 1, pulse 2a, 2b, pulse 3, pulse 4, pulse 5.

Pulse 1 simulates a transient that occurs when the power supply and the inductive load are disconnected.

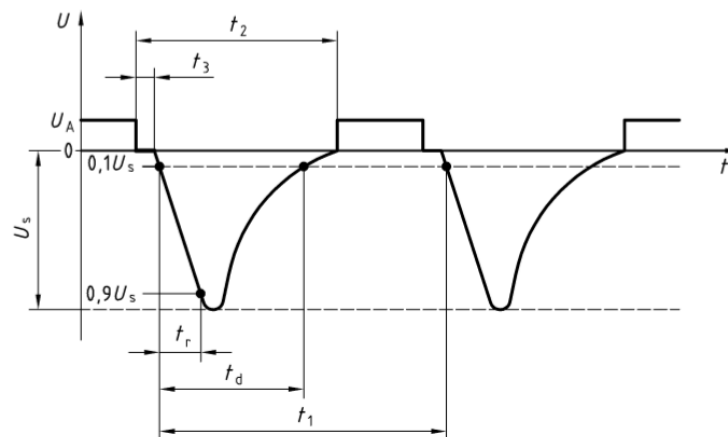
As shown in Figure 1 below, pulse 1 occurs when the switch is open. It affects devices that remain in direct parallel with the inductive load. The test pulse is shown in Figure 2 below, and the test pulse parameters are shown in Figure 3 below.



Key

- 1 ignition switch
- 2 inductive load
- 3 load R_s (see 4.2)
- 4 DUT
- 5 power supply

^a Pulse 1 occurs when the switch is opened.

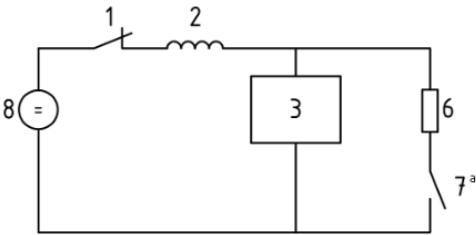


Parameter	12 V system	24 V system
U_s	-75 V to -100 V	-450 V to -600 V
R_i	10 Ω	50 Ω
t_d	2 ms	1 ms
t_r	$\begin{pmatrix} 1 \\ -0.5 \end{pmatrix} \mu\text{s}$	$\begin{pmatrix} 3 \\ -1.5 \end{pmatrix} \mu\text{s}$
t_1^a	0.5 s to 5 s	
t_2	200 ms	
t_3^b	< 100 μs	

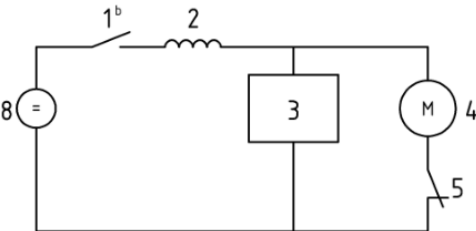
^a t_1 shall be chosen such that the DUT is correctly initialized before the application of the next pulse.

^b t_3 is the smallest possible time necessary between the disconnection of the supply source and the application of the pulse.

Pulse 2a simulates a transient phenomenon caused by a sudden interruption of current in the device in parallel with the DUT due to the inductance of the wire harness. The pulse 2b simulates a DC motor acting as a generator, and the transient phenomenon when the ignition switch is turned off is shown in Figure 1 below. The test pulse and test pulse parameters are as shown in Figure 2 and Figure 3 below.



a) Pulse 2a



b) Pulse 2b

Key

- 1 ignition switch
- 2 wiring harness (inductive)
- 3 DUT
- 4 d.c. motor with internal inductance
- 5 motor switch
- 6 load
- 7 load switch
- 8 power supply

- a Pulse 2a occurs when the load switch (7) is opened while the ignition switch (1) is closed.
- b Pulse 2b occurs when the motor is running and the ignition switch (1) is opened.

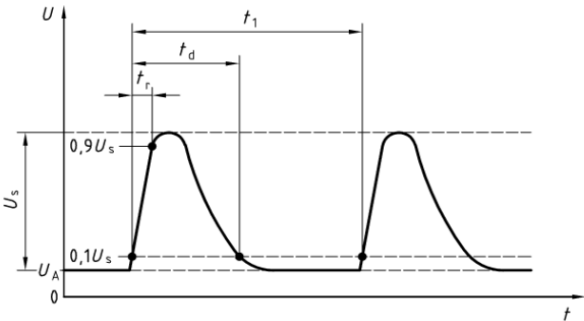


Figure 6 — Test pulse 2a

Table 4 — Parameters for test pulse 2a

Parameter	12 V system	24 V system
U_s	+ 37 V to + 50 V	
R_l	2 Ω	
t_d	0,05 ms	
t_r	$\left(1_{-0,5}^0\right) \mu s$	
t_1^a	0,2 s to 5 s	

^a The repetition time t_1 can be short, depending on the switching. The use of a short repetition time reduces the test time.

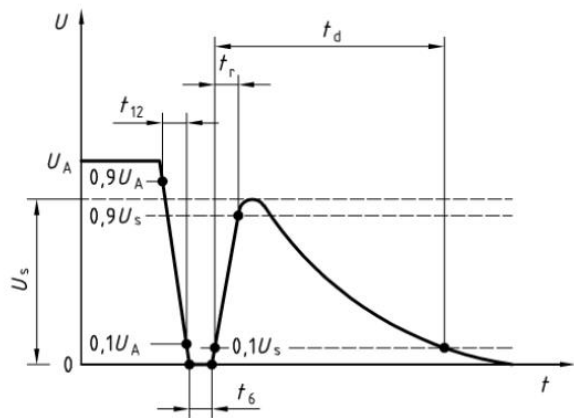
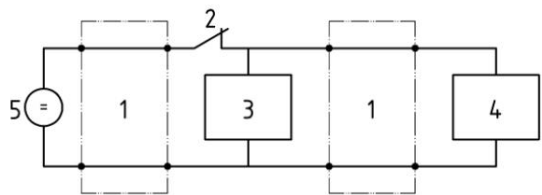


Figure 7 — Test pulse 2b

Table 5 — Parameters for test pulse 2b

Parameter	12 V system	24 V system
U_s	10 V	20 V
R_i	0 Ω to 0,05 Ω	
t_d	0,2 s to 2 s	
t_{12}	1 ms $\pm$ 0,5 ms	
t_r	1 ms $\pm$ 0,5 ms	
t_6	1 ms $\pm$ 0,5 ms	

Pulses 3a and 3b simulate transient phenomena caused by the switching process. The characteristics of these transient phenomena are affected by the distributed capacitance and the distributed inductance, as shown in Figure 1 below. The test pulse is shown in Figure 2 and Figure 3 below. The test pulse parameters are shown in Figures 4 and 5 below.



- Key**
- 1 wiring harness with distributed inductance and capacitance
 - 2 switch
 - 3 DUT
 - 4 inductive load
 - 5 power supply

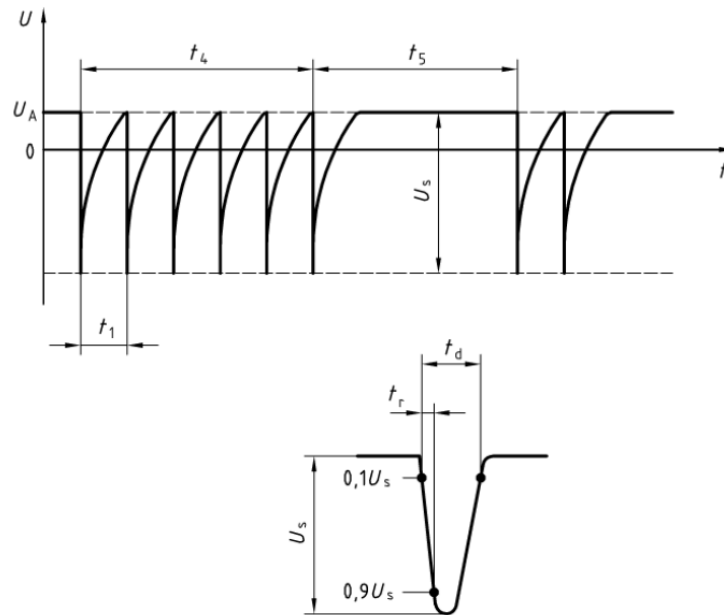


Figure 8 — Test pulse 3a

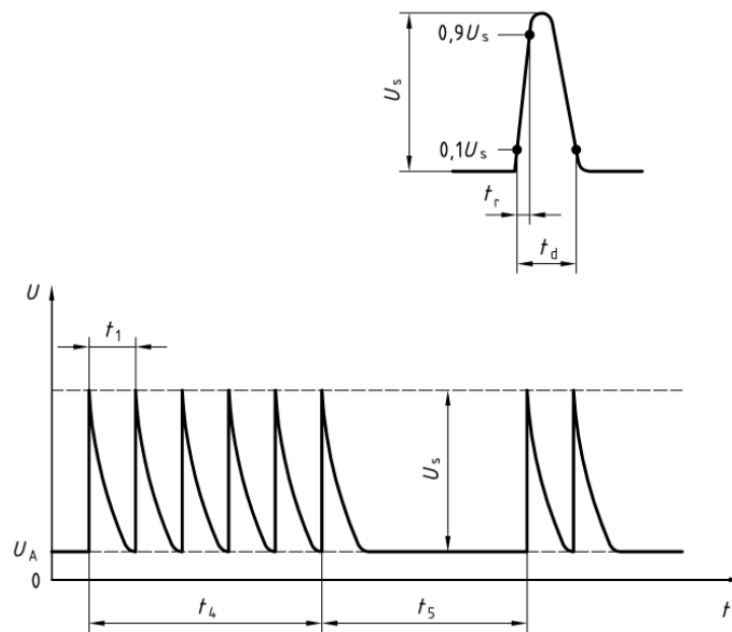


Figure 9 — Test pulse 3b

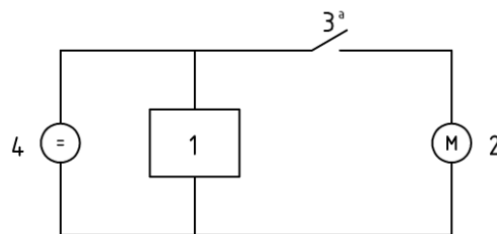
Table 6 — Parameters for test pulse 3a

Parameter	12 V system	24 V system
U_s	- 112 V to - 150 V	- 150 V to - 200 V
R_l	50 Ω	
t_d	$(0,1^{+0,1}_0)$ μ s	
t_r	5 ns $\pm$ 1,5 ns	
t_1	100 μ s	
t_4	10 ms	
t_5	90 ms	

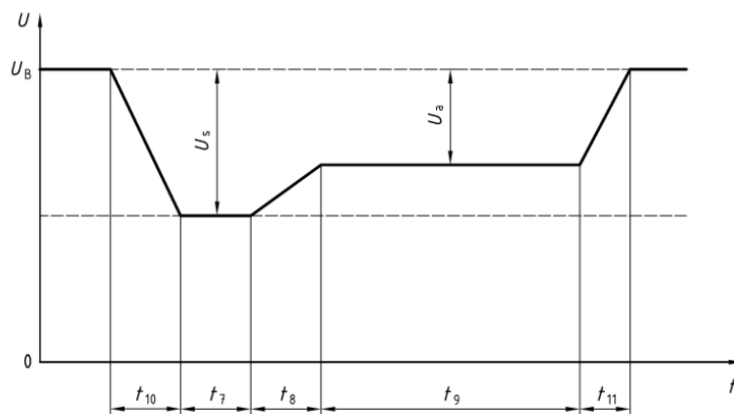
Table 7 — Parameters for test pulse 3b

Parameter	12 V system	24 V system
U_s	+ 75 V to + 100 V	+ 150 V to + 200 V
R_i	50 Ω	
t_d	$(0,1^{+0,1}_0) \mu s$	
t_r	5 ns $\pm$ 1,5 ns	
t_1	100 μs	
t_4	10 ms	
t_5	90 ms	

Pulse 4 simulates a transient phenomenon that causes a drop in the supply voltage when the internal combustion engine is started and energized, as shown in Figure 1 below. The test pulse is shown in Figure 2 below. The test pulse parameters are as shown in Figure 3 below.

**Key**

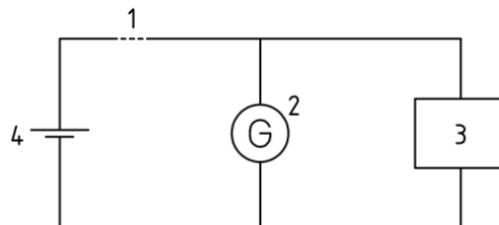
- 1 DUT
- 2 starter motor
- 3 switch
- 4 power supply
- a Pulse 4 occurs when the starter motor is energised.



Parameter	12 V system	24 V system
U_s	- 6 V to - 7 V	- 12 V to - 16 V
U_a	- 2,5 V to - 6 V with $ U_a \leq U_s $	- 5 V to - 12 V with $ U_a \leq U_s $
R_i	0 Ω to 0,02 Ω	
t_7	15 ms to 40 ms ^a	50 ms to 100 ms ^a
t_8	≤ 50 ms	
t_9	0,5 s to 20 s ^a	
t_{10}	5 ms	10 ms
t_{11}	5 ms to 100 ms ^b	10 ms to 100 ms ^c
^a The value used should be agreed between the vehicle manufacturer and the equipment supplier to suit the proposed application. ^b $t_{11} = 5$ ms is typical of the case when engine starts at the end of the cranking period, while $t_{11} = 100$ ms is typical of the case when the engine does not start. ^c $t_{11} = 10$ ms is typical of the case when engine starts at the end of the cranking period, while $t_{11} = 100$ ms is typical of the case when the engine does not start.		

Pulse 5 simulates a load dump transient, i.e. when the alternator generates a charging current, disconnects from the battery (debt state), and the other load remains connected to the alternator, as shown in Figure 1 below. The degree of load dump depends on the speed of the generator and the strength of the excitation field when the battery is disconnected. The throwing pulse width depends mainly on the time constant of the excitation circuit and the pulse amplitude. The test pulse and test pulse parameters are shown in Figures 2 and 3 below.

The load can be disconnected from the battery if the cable is corroded, the contact is poor, or the engine is running.


Key

- 1 defective connection
- 2 alternator
- 3 DUT
- 4 battery

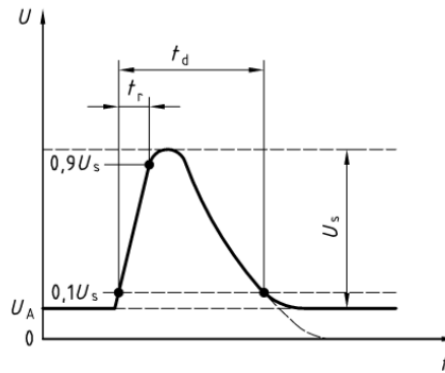
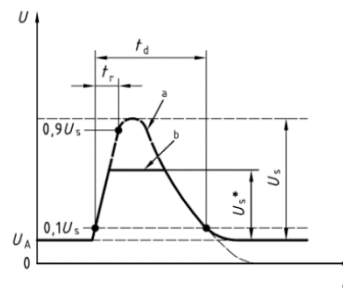


Figure 11 — Test pulse 5a

Table 9 — Parameters for test pulse 5a

Parameter	12 V system	24 V system
U_s	65 V to 87 V	123 V to 174 V
R_i	0,5 Ω to 4 Ω	1 Ω to 8 Ω
t_d	40 ms to 400 ms	100 ms to 350 ms
t_r	$\begin{pmatrix} 10 & 0 \\ 0 & -5 \end{pmatrix}$ ms	



- a Unsuppressed.
b Suppressed.

Figure 12 — Test pulse 5b

Table 10 — Parameters for test pulse 5b

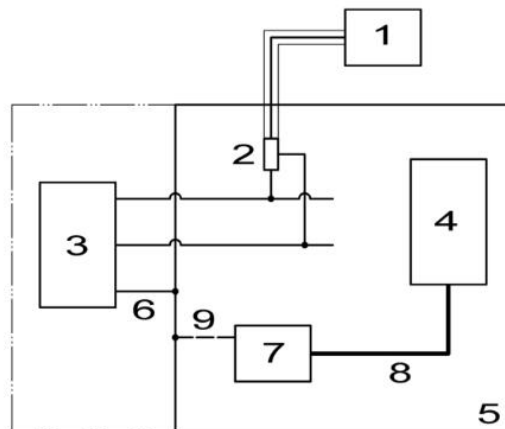
Parameter	12 V system	24 V system
U_s	65 V to 87 V	123 V to 174 V
U_s^*	As specified by customer	
t_d	Same as unsuppressed value	

9.7.2.3 Test arrangement

The EUT should be placed on a material with a low relative permittivity and the EUT is 50 mm $\pm$ 5 mm above the ground plane.

For the test items of the pulses 3a, 3b, the connecting lines between the terminals should be placed in parallel on the material with a low relative dielectric constant, the cable is 50 mm $\pm$ 5 mm above the ground plane, and the length is 500 mm $\pm$ 100 mm.

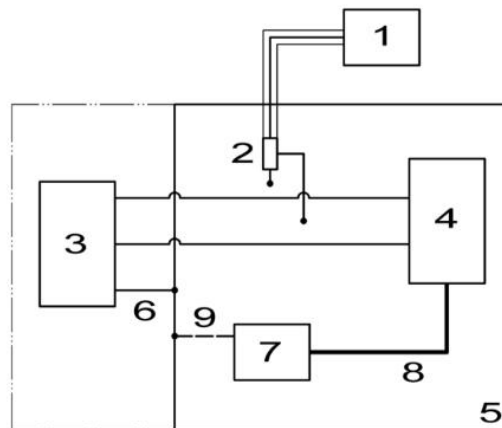
Prior to testing, the pulse generator was used to generate a specific pulse of polarity, amplitude, duration and impedance without an EUT and was measured using an oscilloscope. The peak voltage U_s is modulated to a specified test level with an error of 0%-10%. The waveform adjustment layout is shown in Figure 1. Then, connect the EUT to the pulse generator and disconnect the oscilloscope to inject the test pulse into the EUT. The pulse injection arrangement is shown in Figure 2 below. Suppression diodes and clamping voltages are non-standard and are required information from automotive manufacturers.



a) Pulse adjustment

Key

- 1 oscilloscope or equivalent
- 2 voltage probe
- 3 test pulse generator
- 4 DUT disconnected
- 5 ground plane
- 6 DC power ground connection; maximum length for test pulse 3 is 100 mm
- 7 load simulator (connected to ground plane if required)
- 8 interconnect cable routed away from DUT power leads under test to avoid coupling
- 9 load simulator ground (if required)



b) Pulse injection

Key

- 1 oscilloscope or equivalent
- 2 voltage probe disconnected
- 3 test pulse generator
- 4 DUT
- 5 ground plane
- 6 DC power ground connection; maximum length for test pulse 3 is 100 mm
- 7 load simulator (connected to ground plane if required)
- 8 interconnect cable routed away from DUT power leads under test to avoid coupling
- 9 load simulator ground (if required)

9.7.3 Test method

9.7.3.1 EUT Status

Same as 7.2.3.1.

9.7.3.2 Test procedure

First use the oscilloscope to adjust the peak voltage of the test pulse to the specified test level and inject the test pulse into the EUT. The function of the EUT was evaluated during and after the application of the test pulse.

The test level of the 12V system is shown in the figure below.

Test pulse ^a	Selected test level ^b	Test level, U_s^c V				Min. number of pulses or test time ^f	Burst cycle/pulse repetition time	
		I	II	III min.	IV max.		min.	max.
1		g	g	- 75	- 100	5 000 pulses	0,5 s	5 s
2a		g	g	+ 37	+ 50	5 000 pulses	0,2 s	5 s
2b		g	g	+ 10	+ 10	10 pulses	0,5 s	5 s
3a		g	g	- 112	- 150	1 h	90 ms	100 ms
3b		g	g	+ 75	+ 100	1 h	90 ms	100 ms
4		g	g	- 6	- 7	1 pulse	d	d
5 ^e		g	g	+ 65	+ 87	1 pulse	d	d

9.7.4 Test criteria

Meet the test requirements.

9.8 CI Transient Immunity Test of Electrical Transient Coupling of Electrical and Electronic Components along Non-Power Lines

9.8.1 Overview

Through this test, the electromagnetic environment of the vehicle in which the EUT is located is simulated, and the EUT's immunity to transients transmitted through the power line is evaluated. Artificial networks are used in the tests and the measurement arrangements are not subject to interference from the surrounding electromagnetic environment. The test should use an artificial network and a parallel resistor.

During the test, the ambient temperature should be $23\text{ }^{\circ}\text{C} \pm 5\text{ }^{\circ}\text{C}$. The test voltage was $13.5 \pm 0.5\text{V}$.

Test arrangements and test methods refer to ISO 7637-2: 2004. Latest for 2011

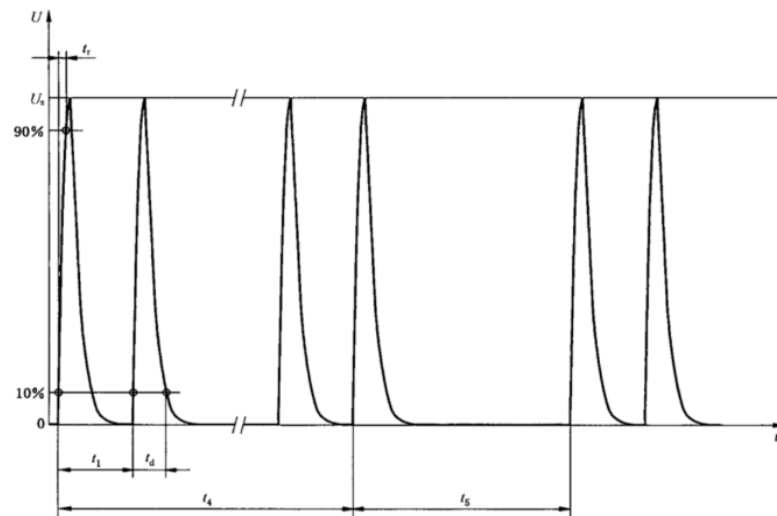
9.8.2 Test conditions

9.8.2.1 Pulse generator

9.8.2.1.1 Fast electrical transient test pulse

The fast electrical transient test pulse simulates the electrical transients generated by the switching process. The characteristics of electrical transients are affected by the distributed capacitance and inductance of the harness.

The pulse shapes and parameters of the fast electrical transient pulses a and b are as shown in Figures 1 and 2.



说明:

t ——时间;

U ——电压,单位为 V。

参数	12 V 电气系统	24 V 电气系统	42 V 电气系统
U_s/V	见表 B.1	见表 B.2	见表 B.3
t_r/ns	5	5	5
$t_d/\mu s$	0.1	0.1	0.1
$t_1/\mu s$	100	100	100
t_4/ms	10	10	10
t_5/ms	90	90	90
R_i/Ω	50	50	50

图 8 快速电瞬态试验脉冲 b

9.8.2.1.2 Slow electrical transient test pulse

The slow electrical transient test pulse simulates electrical transients that occur when a large inductive load circuit is interrupted, such as a radiator fan motor, an air conditioner compressor clutch.

The pulse shapes and parameters of the slow electrical transient positive and negative pulses are shown in Figures 1 and 2.

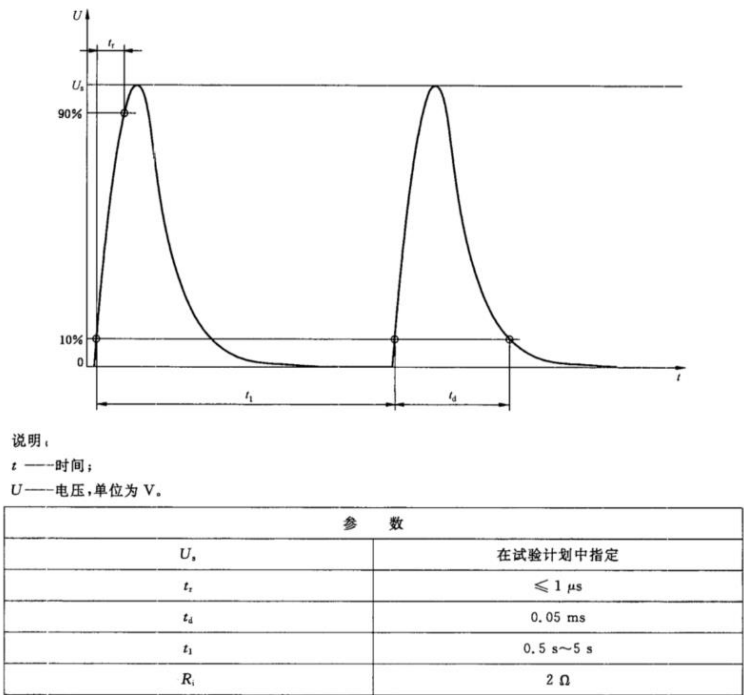


图 9 慢速电瞬态试验脉冲——正脉冲

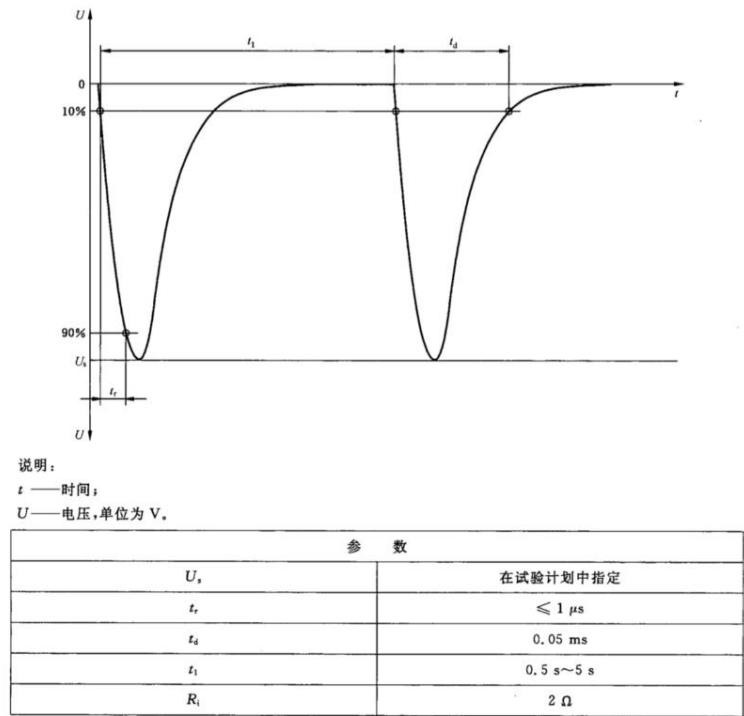


图 10 慢速电瞬态试验脉冲——负脉冲

9.8.2.2 Test arrangement

The minimum size of the grounding plate should be 2m*1m*1mm. The EUT should be placed on an insulated support plate with the EUT 50mm-100mm above the ground plane. The distance between the EUT and all other conductive structures (except the ground plane) should be greater than 0.5 m.

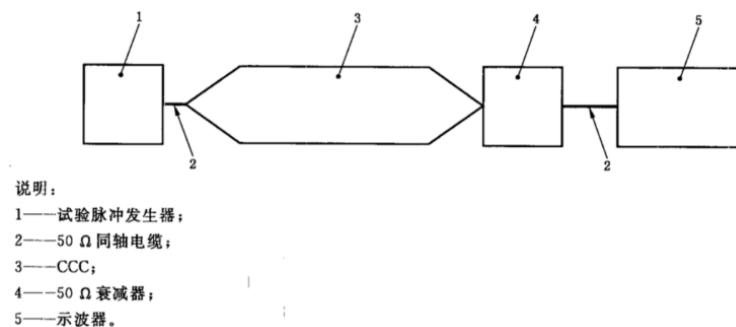
There are three methods for coupling the disturbance signal to the conduction line, which are the capacitive coupling clamp CCC method, the direct capacitor coupling DCC method, and the inductive coupling clamp ICC method.

瞬态类型	CCC 方法	DCC 方法	ICC 方法
4.3.3 的慢速脉冲	不适用	适用	适用
4.3.2 的快速脉冲 a 和 b	适用	适用	不适用

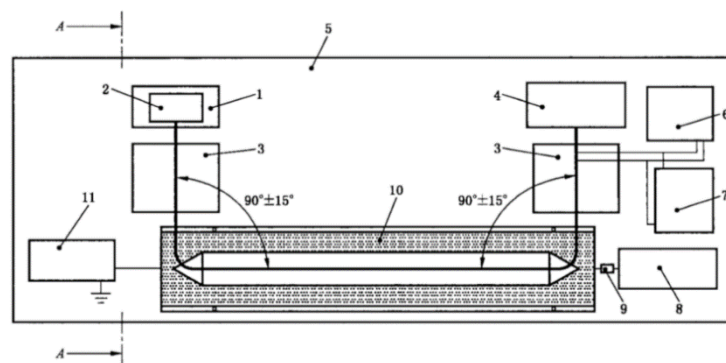
9.8.2.2.1 Capacitive coupling clamp CCC method

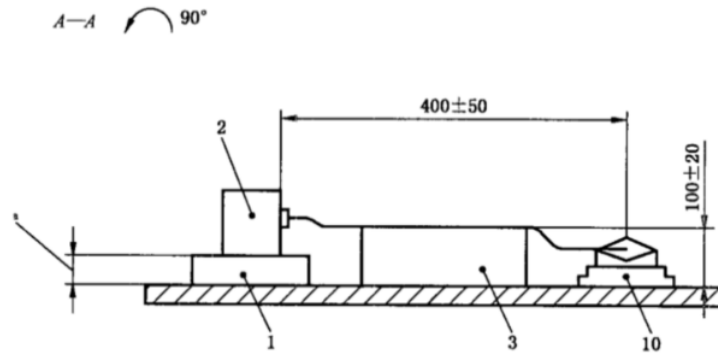
The capacitive coupling clamp method is suitable for coupling fast electrical transient test pulses, especially for EUTs with medium or large numbers of wires to be tested. The capacitive coupling clamp method is not suitable for coupling slow electrical transient test pulses.

The test pulse generator should be calibrated before the test and the pulse amplitude is calibrated using an oscilloscope. The calibration arrangement is shown below. 50Ω coaxial cable is connected to a 50Ω oscilloscope and a 50Ω attenuator for voltage measurement. Wires are not allowed to pass through the coupling clamp during calibration.



The arrangement of the test pulse injection is shown in the figure below. The coupling circuit is formed by a capacitive coupling clamp, and the wire of the EUT passes through the capacitive coupling clamp with a coupling length of 1 m. The power supply line outside the coupling clamp should be 1m long. The distance between the EUT and the CCC, the distance between the auxiliary device and the CCC should be 400 mm ± 50 mm. The part of the conductor to be tested outside the CCC shall be placed at a height of 100 mm ± 20 mm above the ground plane and at an angle of $90^\circ \pm 15^\circ$ with the longitudinal axis of the CCC. The EUT and pulse generator should be placed on the same side of the CCC, and the length of the EUT harness should not exceed 2 m.





说明：

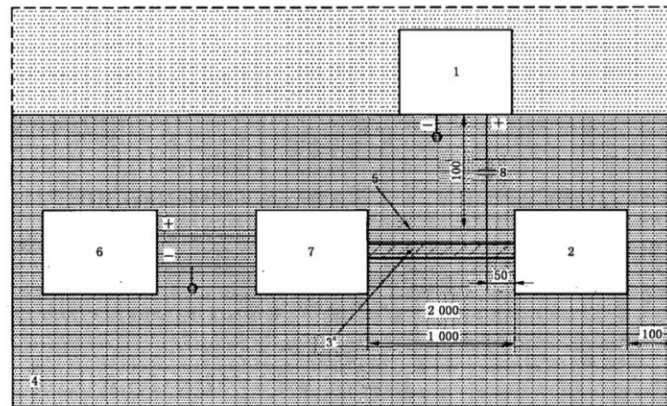
- 1——绝缘支撑板(如果 DUT 没有与车辆接地连接)；
- 2——DUT；
- 3——试验线束的绝缘支撑板；
- 4——辅助装置(如传感器,负载,配件),与在车辆上安装相同；
- 5——接地平板；
- 6——电源；
- 7——蓄电池；
- 8——示波器；
- 9——50 Ω 衰减器；
- 10——CCC；
- 11——试验脉冲发生器。

* 所选尺寸应在试验计划中规定并记录在试验报告中。

9.8.2.2.2 Direct capacitive coupling DCC method

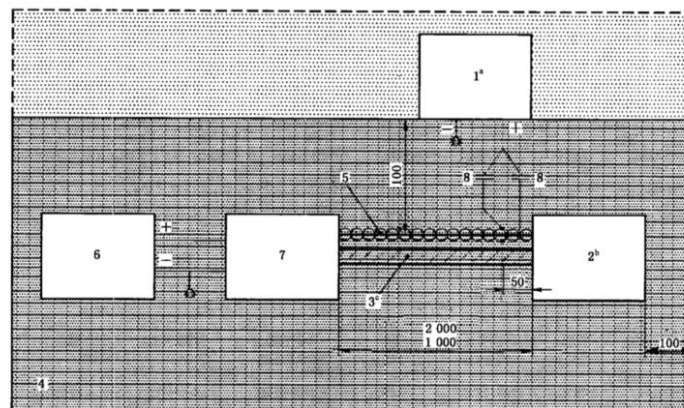
The direct capacitive coupling method uses the recommended capacitance values for EUT wire-coupled fast electrical transient test pulses and couples the same voltage. When the direct capacitance coupling method uses the recommended capacitance value, it is effective to couple the slow electrical transient test pulse. For fast electrical transient tests, each wire needs to be tested separately. The test pulse generator should be calibrated before the test. The output of the pulse generator is placed in series with the coupling capacitor, and a high-impedance oscilloscope is used to measure the peak voltage at the open end of the capacitor output. Adjust the output of the pulse generator so that the peak voltage at the open end of the capacitor reaches the level required by the test.

The arrangement of the test pulse injection is shown in the figure below. The length of the harness should be between 1000mm and 2000mm, and all harnesses should be 50mm $\pm$ 5mm above the ground plane. When using a fast electrical transient test pulse, the capacitor value is 100 pF. When using a slow electrical transient test pulse, the capacitor value is 0.1 μ F.



- 说明：
- 1——试验脉冲发生器；
 - 2——DUT；
 - 3——线束；
 - 4——接地平板；
 - 5——被测 I/O 线；
 - 6——电源；
 - 7——DUT 辅助装置；
 - 8——高电压(最小 200 V)陶瓷电容器。

The arrangement of the test pulse injection on the CAN bus is shown in the figure below. The length of the harness should be between 1000mm and 2000mm, and all harnesses should be $50\text{mm} \pm 5\text{mm}$ above the ground plane. When using a fast electrical transient test pulse, the capacitor value is 100 pF. When using a slow electrical transient test pulse, the capacitor value uses 470pF.

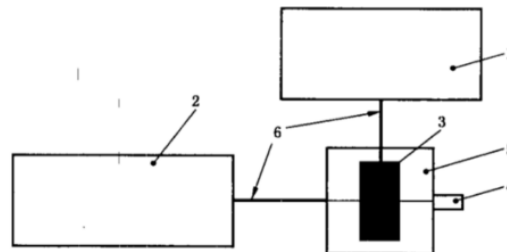


- 说明：
- 1——试验脉冲发生器；
 - 2——DUT；
 - 3——线束；
 - 4——接地平板；
 - 5——CAN 总线；
 - 6——电源；
 - 7——DUT 辅助装置；
 - 8——额定电压不小于 200 V 的陶瓷电容。

9.8.2.2.3 Inductive coupling clamp ICC method

The inductive coupling clamp method is suitable for coupling slow electrical transient test pulses, especially for EUTs with a medium or large number of wires to be tested.

The test pulse generator should be calibrated before the test. The test pulse generator shall be connected to the inductive coupling clamp by a single cable or multiple cables, and the cable length shall not exceed 0.5 m. CCC and DCC use the open circuit voltage of the pulse generator to specify the test level. The ICC uses the voltage measured by the calibration test arrangement as shown below to specify the test level. The pulse should meet the requirements of the following table



说明:

- 1——试验脉冲发生器;
- 2——示波器(1 MΩ 输入);
- 3——ICC;
- 4——短路;
- 5——校准夹具;
- 6——50 Ω 同轴电缆。

用于 ICC 试验方法的校准夹具的例子见附录 A。

参数	12V 电气系统
$t_d / \mu s$	7 ± 2.1
$t_r / \mu s$	≤ 1.2

9.8.3 Test methods

9.8.3.1 EUT status

Same as 7.2.3.1.

9.8.3.2 Test procedure

The test pulse generator is first calibrated so that the test level meets the test requirements. Inject the calibrated test level into the wires of the EUT (except the power line). Check whether the functions of the EUT are immune to interference during or after the test.

The test level of the EUT is as shown in the table below.

试验脉冲 ^a	协议试验电平 ^b	试验等级, U_i , ^c				试验时间 min
		I 最低	II	III	IV 最高	
快 a(DCC 和 CCC)		-10	-20	-40	-60	10
快 b(DCC 和 CCC)		+10	+20	+30	+40	10
DCC 慢+		+8	+15	+23	+30	5
DCC 慢-		-8	-15	-23	-30	5
ICC 慢+		+3	+4	+5	+6	5
ICC 慢-		-3	-4	-5	-6	5

9.8.4 Test criteria

Fulfil requirements.

9.9 ESD Immunity test of electrical and electronic components against electrostatic discharge

9.9.1 Overview

By using the human body ESD model, the electrostatic discharge caused by the EUT in the passenger compartment and the passenger entering the passenger compartment from the outside is simulated. Verify the EUT's immunity to statics in the vehicle environment.

During the test, the test environment is recommended to be 20 ° C and 30% RH, or to maintain the ambient temperature of 23 ° C $\pm$ 5 ° C, relative humidity between 30% -60%. The test voltage was 13.5 $\pm$ 0.5V.

Test arrangements and test methods refer to ISO 10605: 2001. Latest 2008

9.9.2 Test conditions

9.9.2.1 Grounding plane and Insulating pad

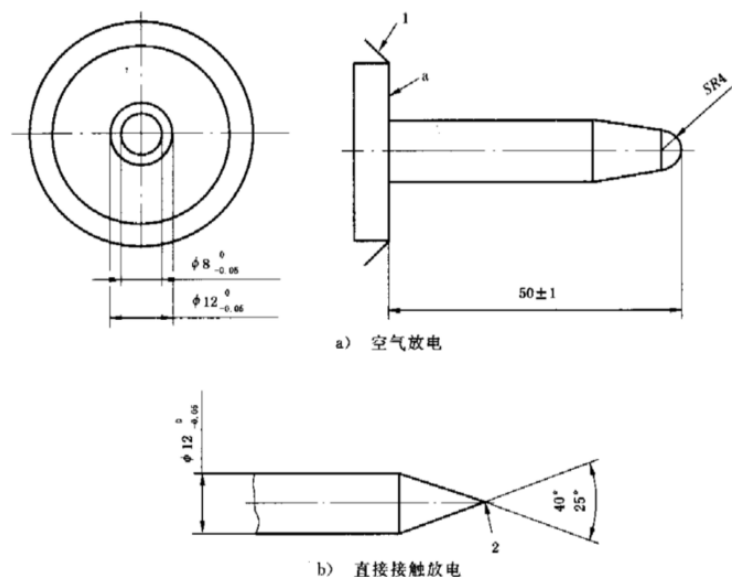
The ground plane should be a piece of foil. The ground plane has a thickness of at least 1 mm and an area of at least 1 m². The outer edge of the ground plane is at least 100 mm beyond the EUT projection profile.

The wire connecting the grounding plane to the EUT ground should have a length of less than 2m and width of at least 50mm.

The insulating spacer should be a clean, non-hygroscopic material. The thickness of the insulating spacer is 25mm $\pm$ 2.5mm, and the distance of the outer edge beyond the projection contour of the EUT is at least 20mm.

9.9.2.2 ESD Simulator discharge terminals

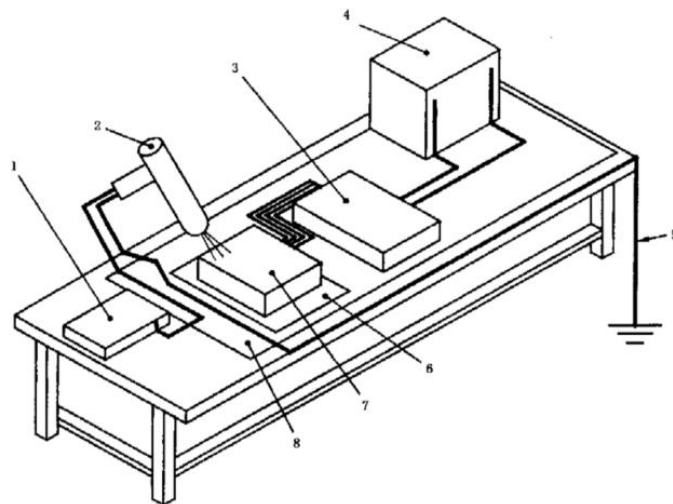
The discharge terminals of the insulating pad air discharge and direct contact discharge is as shown in the figure below.



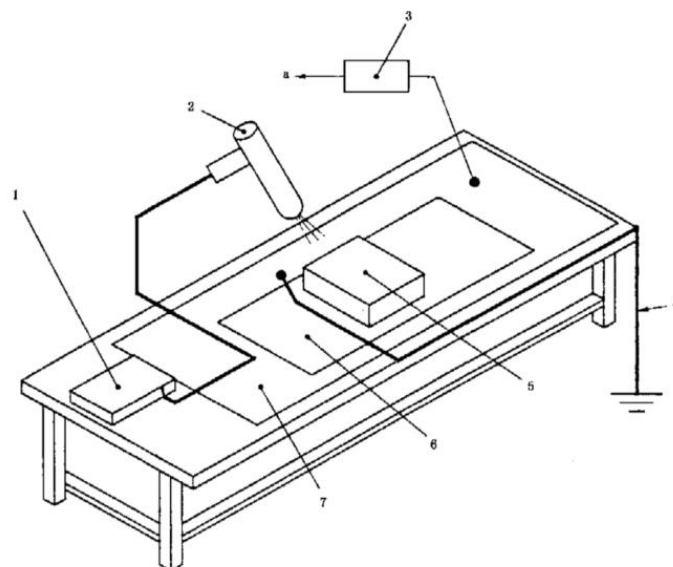
9.9.2.3 Test arrangement

The EUT is placed on a ground plane. Since the vehicle-end coil and coil controls are mounted directly on the chassis, the EUTs are placed directly on the ground plane and connected. The test arrangement is shown in Figure 1 below.

The package handling test arrangement is shown in Figure 2 below. The EUT is placed on the static electricity consuming material to release the charge accumulated in the EUT rack.



- 1—ESD 电源；
- 2—ESD 模拟器；
- 3—动作执行器；
- 4—电池；
- 5—接地带；
- 6—绝缘块(如果需要)；
- 7—被试设备；
- 8—接地平板。



- 1—模拟器电源；
- 2—ESD 模拟器；
- 3—1 MΩ 静电消耗电阻；
- 4—接地带；
- 5—被试设备；
- 6—静电消耗材料；
- 7—接地平板(如果静电消耗材料的生产厂家要求)。

9.9.3 Test methods

9.9.3.1 EUT status

Same as 7.2.3.1.

9.9.3.2 Test Procedures

EUT in working mode, or at least in standby mode. Test the surface that is easily accessible.

Each discharge test point shall withstand at least 3 positive voltage discharges and 3 negative voltage discharges at each voltage level, and the discharge intervals shall be at least 5 s apart. At each voltage level, the discharge test point of the EUT should be subjected to a discharge test of one polarity and then subjected to a discharge test of reverse polarity.

The discharge modes include: direct contact discharge (contact discharge + coupling plate discharge) and air discharge (at least 15 mm from the EUT, $\pm 15^\circ$ perpendicular to the discharge end, aligned with the discharge position, and slowly approached at a speed of no more than 5 mm/s until discharge. If the contact still fails to discharge, stop the test at this voltage level.

9.9.3.2.1 ESD test with powered module

A discharge end with a capacitance of 330 pF was used. Use the ESD test arrangement of the electronic module. Direct contact discharge and air discharge tests were performed.

9.9.3.2.2 Vehicle ESD Test

All discharge points accessible in the vehicle were tested using 330pF, 2000 Ω discharge terminals, and discharged at 4kV, 8kV and 15kV voltages. Use 150pF, 2000 Ω discharge terminals, and discharged at 25kV voltage to test all discharge points that can be reached outside the vehicle and entering the vehicle.

9.9.3.2.3 ESD test when handling without power up

Use the ESD test arrangement for handling. Direct contact discharge and air discharge tests were performed. The test points are connecting pins, housings, buttons, switch displays, container nuts, openings and female connecting pins that are easily accessible during EUT handling. The residual charge of the EUT after each discharge should be shorted to the ground with a 1M Ω resistor. The shorting point is between the discharge position and ground, and between the EUT ground point and ground.

If the EUT case is grounded and the female pins are not easily accessible, this test is not necessary.

9.9.4 Test criteria

The ESD test for module, vehicle, and handling shall meet the test levels specified in Table 1, Table 2, and Table 3, respectively. In an ESD test in which the EUT is not powered, the EUT should be capable of withstanding at least ± 4 kV direct contact discharge.

表 B.1 电子模块(通电运行)试验严酷程度等级

放电类型	严酷程度等级/kV					最少放电次数 ^a
	自选等级	试验等级				
		I	II	III	IV	
接触放电	x ^b	±4	±6	±7	±8	3
空气放电	x ^b	±4	±8	±14	±15	
^a 最小放电间隔时间为 5 s。						
^b 车辆制造商和供应商协议值。						

表 B.2 整车试验严酷程度等级

试验放电点	严酷程度等级/kV					最少放电次数
	自选等级	试验等级				
		I	II	III	IV	
仅车内可触及	x ^b	±4	±8	±14	±15	3
从车外可触及	x ^b	±4	±8	±15	±25	
注：全部是空气放电。						
a 最小放电间隔时间为 5 s。						
b 整车制造商和供应商协议值。						

表 B.3 电子模块(不通电)分类试验等级

放电类型	试验等级 ^b /kV			最少放电次数 ^a
接触放电	±4	±6	±8	3
空气放电	±4	±15	±25	
^a 最小放电间隔时间为 5 s。 ^b 建议在进行 ±8 kV 接触放电之前先进行 ±8 kV 空气放电的试验。				

9.10 RE Radiated Emissions test

9.10.1 Overview

The radiation emission of EUT was measured by experiment, and the degree of radiation disturbance caused by EUT to external environment was evaluated. The test should be carried out in the test site, and the limit of ambient noise level in the test site should be guaranteed.

The measuring distance can be 3 m or 10 m. The measuring distance of 3 m is only applicable when the EUT (including cables) is in the range of 1.2 m in diameter and 1.5 m in the ground plane.

During the test, the ambient temperature should be 23°C±5°C.

Test arrangements, test methods and limits are referenced in CISPR 11: 2009. The test site shall meet the IEC/CISPR 16-1-4: 2012 alternative requirements, within ±4 dB of the normalized field attenuation theoretical value.

9.10.2 Test conditions

9.10.2.1 Ambient noise

The ambient noise level should be measured before the test and the EUT is not operating. The ambient noise level should be at least 6dB (9kHz-1GHz) or 10dB (above 1GHz) below the EUT radiation limit to ensure that the EUT's radiated emissions are distinguishable from the background of ambient noise and the readings are unaffected.

If the noise level cannot meet the requirement of less than 6 dB when the measurement distance is 10 m, the antenna can be placed at a closer distance but not less than 3 m, and the radiation limit at the new distance is converted accordingly.

9.10.2.2 Measuring antenna

In the test, different measuring antennas were selected according to different measuring frequencies. When the frequency band is lower than 30MHz, a loop antenna should be used, and the lowest point of the ring is 1m above the ground. When the frequency band is in the range of 30MHz-1GHz, the CISPR16-1-4 antenna should be used. The lowest point of the antenna is at least 0.2m above the ground. The horizontal and vertical polarization directions should be measured, and the antenna center should be in the height range of 1m-4m. When the frequency band is higher than 1 GHz, the CISPR16-1-4 antenna should be used.

9.10.2.3 Test arrangement

The EUT should be placed on the turntable when measuring at the test site. The measured distance is the horizontal distance between the measuring antenna and the nearest part when the EUT makes one revolution. Floor devices should be placed on the ground or separated by a thin insulating material. Portable or non-floor equipment should be placed on a non-metallic table 0.8 meters above the ground.

When the measuring distance is 3 meters, the EUT should be a small device, which satisfies the EUT (including the cable) in the range of 1.2 m in diameter and 1.5 m above the ground. When the measuring distance is between 3 and 10 meters, the corresponding limit shall be converted according to the limit of 3 meters or 10 meters. (Defining the limit level indicated by the logarithmic unit at x meters is L_x , the distance measured at x meters is D_x , then the D_x limit conversion formula is:

$$L_x = L_{10} - 20 * \lg \frac{D_x}{D_{10}}, \text{ this formula only applies to 30MHz-1GHz)}$$

9.10.3 Test methods

9.10.3.1 EUT Status

Same as 7.2.3.1.

9.10.3.2 Test procedure

The EUT is placed on the turntable. The measuring antenna is rotated in both horizontal and vertical polarizations. The turntable rotates at all angles and the antenna rises and falls at a height of 1m-4m. The highest level of radiated disturbance is recorded at each measurement frequency.

In radiation emissions exceeding $L - 10\text{dB}$, record at least 6 largest disturbances in each of the observed frequency bands and their corresponding frequency points. The record should include the antenna polarization direction, antenna height and turret angle of the recorded disturbance level.

According to CISPR 11, radiation emission measurements above 1 GHz are suitable for EUT operating frequencies above 400 MHz. According to CISPR 32, the radiated emission measurement above 1G is suitable for the maximum internal frequency of the EUT above 108 MHz. Therefore, the tentative test frequency band is 9 kHz-1 GHz.

9.10.4 Test criteria

The wireless charging system is a device that intentionally produces and operates at 85 kHz, in accordance with the classification of Group 2 in CISPR 11. Class A (industrial environment) or Class B (residential environment) is applicable depending on whether the equipment used in the residential low-voltage power supply network facility is connected.

There are no applicable limit requirements for 9kHz-150kHz. Refer to CISPR 11 Commercial Induction Cooker Limits and SAE J9254 recommendations as shown below.

Frequency range MHz	Limits at 3 m distance Quasi-peak dB(μA/m)
0,009 – 0,070	69
0,070 – 0,148 5	69 Decreasing linearly with logarithm of frequency to 39
0,148 5 – 4,0	39 Decreasing linearly with logarithm of frequency to 3
4,0 – 30	3

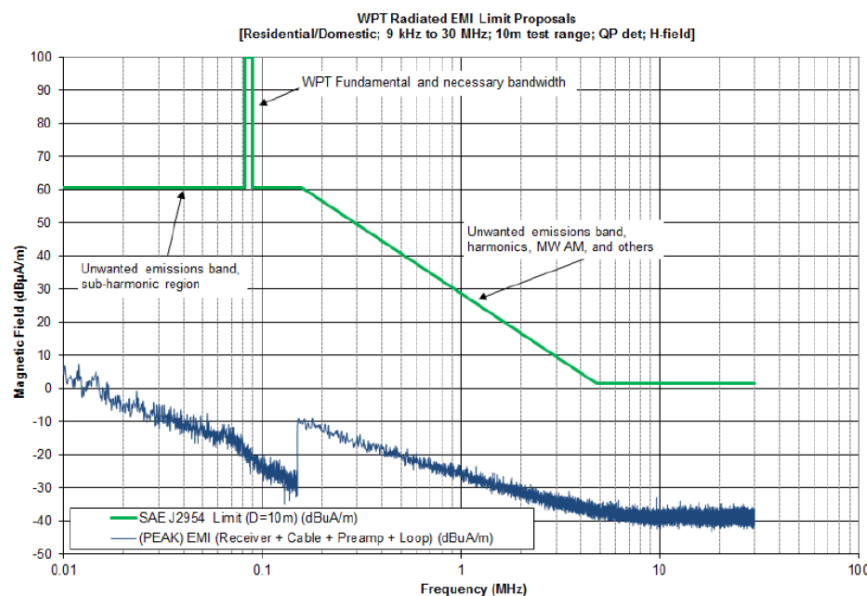


Figure 14 - Recommended limits (see Note 1)

For 150kHz – 30MHz, it should meet the limits of IEC 61980-1. Refer to the Class A or Class B limits of Group 2 in CISPR 11 as shown below.

Frequency range MHz	Magnetic field Quasi-peak Measurement distance $D = 3$ m dB(μ A/m)	Magnetic field Quasi-peak Measurement distance $D = 10$ m dB(μ A/m)
0,15 – 0,49	82	57,5
0,49 – 1,705	72	47,5
1,705 – 2,194	77	52,5
2,194 – 3,95	68	43,5
3,95 – 11	43,5 decreasing linearly with logarithm of frequency to 28,5	18,5
11 – 20	28,5	18,5
20 – 30	18,5	8,5

For 30MHz-1GHz, Class A or Class B limits for Group 2 of CISPR 11 should be met as shown in the following figure.

Frequency range MHz	Limits for a measuring distance of D in m			
	On a test site $D = 30$ m from the equipment		On a test site $D = 10$ m from the equipment	
	Electric field Quasi-peak dB(μ V/m)	Magnetic field Quasi-peak dB(μ A/m)	Electric field Quasi-peak dB(μ V/m)	Magnetic field Quasi-peak dB(μ A/m)
0,15 – 0,49	-	33,5	-	57,5
0,49 – 1,705	-	23,5	-	47,5
1,705 – 2,194	-	28,5	-	52,5
2,194 – 3,95	-	23,5	-	43,5
3,95 – 20	-	8,5	-	18,5
20 – 30	-	-1,5	-	8,5
30 – 47	58	-	68	-
47 – 53,91	40	-	50	-
53,91 – 54,56	40	-	50	-
54,56 – 68	40	-	50	-
68 – 80,872	53	-	63	-
80,872 – 81,848	68	-	78	-
81,848 – 87	53	-	63	-
87 – 134,786	50	-	60	-
134,786 – 136,414	60	-	70	-
136,414 – 156	50	-	60	-
156 – 174	64	-	74	-
174 – 188,7	40	-	50	-
188,7 – 190,979	50	-	60	-
190,979 – 230	40	-	50	-
230 – 400	50	-	60	-
400 – 470	53	-	63	-
470 – 1 000	50	-	60	-

Frequency range MHz	Limits for a measuring distance of D in m		
	Electric field		Magnetic field
	Measurement distance 10 m		Measurement distance 3 m
	Quasi-peak dB(μ V/m)	Average ^a dB(μ V/m)	Quasi-peak dB(μ A/m)
0,15 – 30	–	–	³⁹ Decreasing linearly with logarithm of frequency to 3
30 – 80,872	30	25	–
80,872 – 81,848	50	45	–
81,848 – 134,786	30	25	–
134,786 – 136,414	50	45	–
136,414 – 230	30	25	–
230 – 1 000	37	32	–

9.11 CE Conducted Emissions Test

9.11.1 Overview

The degree of radiated disturbance caused by the EUT to the external environment is evaluated by measuring the radiation emission value of the EUT.

During the test, the ambient temperature should be $23\text{ }^{\circ}\text{C} \pm 5\text{ }^{\circ}\text{C}$.

Test arrangements, test methods and limits are referenced in CISPR 11: 2009.

9.11.2 Test conditions

9.11.2.1 Ambient noise

The ambient noise level should be measured before the test, when the EUT is not operating. The ambient noise level should be at least 6dB (9kHz-1GHz) or 10dB (above 1GHz) below the EUT radiation limit to ensure that the EUT's radiated emissions are distinguishable from the background of ambient noise and the readings are unaffected.

An appropriate RF filter can be inserted between the artificial network and the power supply or measured in the shielded room. After the RF filter is connected, the impedance of the artificial power network should still meet the specified requirements at the measurement frequency.

9.11.2.2 Test arrangement

When measuring the disturbance of the power terminal, the extra length of the cable should be bundled into a wire harness of 0.3m-0.4m length near its midpoint. If increasing the number of cables in a place with multiple interfaces of the same type does not significantly affect the measurement results, just use one cable to cover one of the interfaces.

The test arrangement can be the same as the arrangement in radiation emission or use the following method.

A ground plane is used in the test site. The minimum size of the ground plane is 2m*2m (the ground or any wall of in shielded room can be used as the ground plane) and should be 0.5m larger than the boundary of the EUT. The floored EUT is placed on the ground, and the non-floored EUT should be placed on a non-metal table with a height of 0.4 m on the ground plane. The distance from all test equipment to other metal surfaces should be greater than 0.8m.

9.11.3 Test methods

9.11.3.1 EUT Status

Same as 7.2.3.1.

9.11.4 Test criteria

The wireless charging system is a device that intentionally produces and operates at 85 kHz. It complies with the classification of Group 2 in CISPR 11. Class A (industrial environment) or Class B (residential environment) is applicable depending on whether the equipment used in the residential low-voltage power supply network facility is connected.

There are no applicable limits for the 9 kHz-150 kHz band. Refer to CISPR 11 Induction Cooker Limits as shown below.

Frequency range MHz	Induction cooking appliance limits			
	All appliances except 100 V rated appliances without an earth connection		100 V rated appliances without an earth connection	
	Quasi-peak dB(μV)	Average dB(μV)	Quasi-peak dB(μV)	Average dB(μV)
0,009 – 0,050	110	–	122	–
0,050 – 0,148 5	90 Decreasing linearly with logarithm of frequency to 80	–	102 Decreasing linearly with logarithm of frequency to 92	–
0,148 5 – 0,5	66 Decreasing linearly with logarithm of frequency to 56	56 Decreasing linearly with logarithm of frequency to 46	72 Decreasing linearly with logarithm of frequency to 62	62 Decreasing linearly with logarithm of frequency to 52
0,5 – 5	56	46	56	46
5 – 30	60	50	60	50
At the transition frequency, the more stringent limit shall apply.				

Class A of the 150kHz-30MHz band shall meet the limits of the two sets of Class A equipment in CISPR 11 as shown in the following figure.

Frequency range MHz	Rated input power of ≤ 75 kVA		Rated input power of > 75 kVA ^a	
	Quasi-peak dB(μV)	Average dB(μV)	Quasi-peak dB(μV)	Average dB(μV)
0,15 – 0,50	100	90	130	120
0,50 – 5	86	76	125	115
5 – 30	90 decreasing linearly with logarithm of frequency to 73	80 decreasing linearly with logarithm of frequency to 60	115	105
At the transition frequency, the more stringent limit shall apply.				
NOTE 1 Limits only apply to Low Voltage (LV) a.c. mains input ports.				
NOTE 2 For class A equipment with a rated power ≤ 75 kVA intended to be connected solely to isolated neutral or high impedance earthed (IT) industrial power distribution networks (see IEC 60364-1), the limits defined for group 2 equipment with a rated input power > 75 kVA can be applied.				

Class B of the 150kHz-30MHz band shall meet the limits of the two sets of Class B equipment in CISPR 11 as shown in the following figure.

Frequency range MHz	Quasi-peak dB(μ V)	Average dB(μ V)
0,15 – 0,50	66 Decreasing linearly with logarithm of frequency to 56	56 Decreasing linearly with logarithm of frequency to 46
0,50 – 5	56	46
5 – 30	60	50

9.12 Harmonic Current Test

9.12.1 Overview

This test verifies the harmonic current injected into the utility power system during stable operation of the EUT.

9.12.2 Test Methods

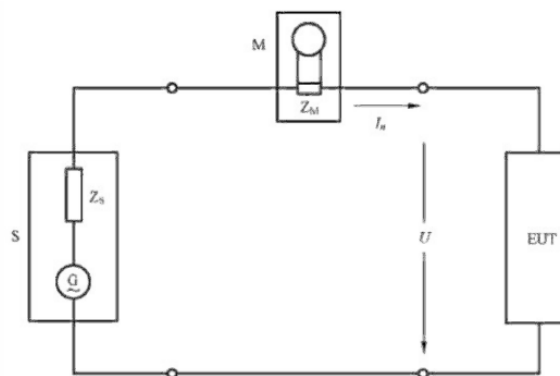
9.12.2.1 EUT Status

The EUT should be in a stable working process during the test. Harmonic current is not considered when the system is operated for the first 10s or from power up to steady state or during power termination. The EUT should not be in standby mode for more than 10% of the observation period.

The EUT should operate in the mode where the maximum THC is occurred (under normal operating conditions).

9.12.2.2 Test Procedures

The EUT should setup as below. The harmonic current of a single-phase device can be measured on the live or neutral.



- | | |
|---------------------|--------------------------|
| S —— 供电电源； | EUT —— 受试设备； |
| Z_M —— 测量设备的输入阻抗； | I_n —— 线电流的 n 次谐波分量； |
| M —— 测量设备； | U —— 试验电压； |
| Z_s —— 供电电源的内阻抗； | G —— 供电电源的开路电压。 |

9.12.3 Test criteria

The manufacturer must assume a short circuit ratio R_{sc} value. Equipment meeting the short circuit ratio $R_{sc}=33$ corresponding harmonic current limit can be connected to any point of the power supply system. The short circuit ratio is greater than 33, allowing higher harmonic emission value. Harmonic emission shall conform to the following table.

最小 R_{min}	可接受的单次谐波电流 I_n/I_1^* %						可接受的谐波电流畸变率 %	
	I_3	I_5	I_7	I_9	I_{11}	I_{13}	THD	PWHD
33	21.6	10.7	7.2	3.8	3.1	2	23	23
66	24	13	8	5	4	3	26	26
120	27	15	10	6	5	4	30	30
250	35	20	13	9	8	6	40	40
≥ 350	41	24	15	12	10	8	47	47
注 1: 12 次及以下偶次谐波的电流值不应超过 $(16/n)\%$ 。12 次以上偶次谐波与奇次谐波同样用 THD 和 PWHD 考虑。								
注 2: 允许在两个连续 R_{min} 间线性插值。参见附录 D。								
* I_1 为基波参考电流值, I_n 为谐波电流分量。								

9.13 ESD immunity test

9.13.1 Testing conditions

No power, standby (WB only), charging (WB only)

9.13.2 Test Methods

1. Air discharge: the electrode is continuously close to the gap of the product casing until it is discharged or contacted (10 times of discharge)
2. Contact discharge: the electrode should be directly in contact with the outside of the test product, and the discharge point is a metal part, such as a metal bolt, a metal heat sink, etc. (10 times of discharge)

Test Item	classification			Test Results	Reference Standard
Electrostatic discharge immunity test	WB	$\pm 6\text{kV}$ (Contact)	Criterion II		QC/T895 GBT18487.3-2001 GB/T 17626.2
		$\pm 8\text{kV}$ (Air)			
	VA	$\pm 8\text{kV}$ (Contact)	Criterion II		
		$\pm 8\text{kV}$ (Air)			
	System	$\pm 8\text{kV}$ (Contact)	Criterion II		
		$\pm 8\text{kV}$ (Air)			

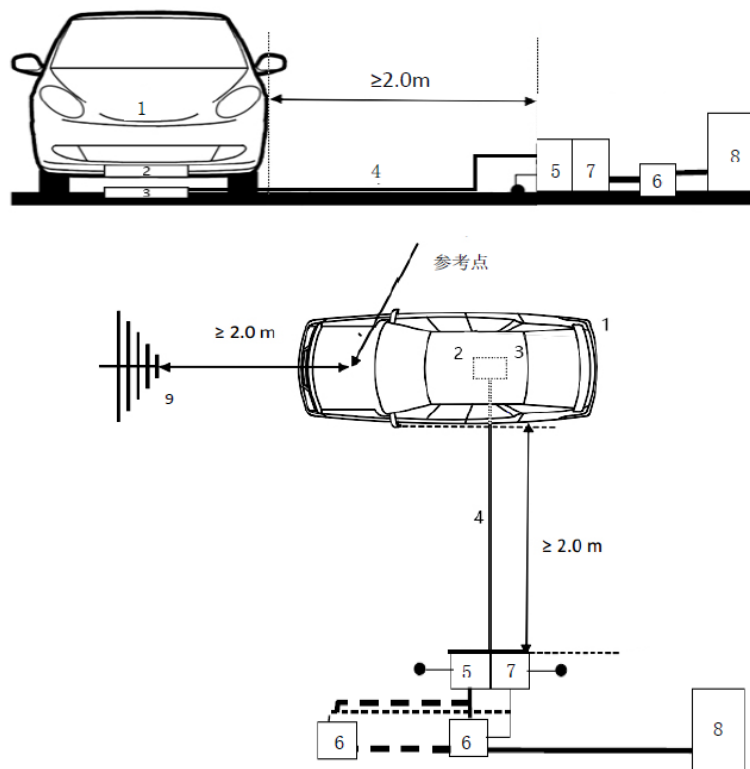
9.14 RF electromagnetic field radiation immunity test

9.14.1 Test Conditions

1. System input 220Vac/50Hz;
2. The ground clearance $Z=175\text{mm}$; no offset;
3. Power output $P_{\text{out}} = 6.6\text{KdW}$;
4. VA inlet water temperature: 45°C , flow: $8\text{L} / \text{min}$

9.14.2 Test Methods

Test arrangement (System)



说明:

1——被测车辆;

2——副边设备;

3——原边设备

4——充电线缆;

5——接地的人工电源网络;

6——电源插座 (图中示意了备选位置);

7——接地的不对称人工网络;

8——原边控制器;

9——发射天线。

Test Item	Classification			Test Results	Reference Standard
RF electromagnetic field radiation immunity test	WB	30 V/m (80MHz~1GHz)	Criterion I		GB/T17626.3 IEC 61000-4-3
		30 V/m (1.4GHz~2GHz)			
		30 V/m (2GHz~2.7GHz)			
	VA	30 V/m (20MHz~2GHz)	Criterion I		GB 34660-2017
	System	30 V/m (20MHz~2GHz)	Criterion I		GB 34660-2017

9.14.2.1 Test arrangement

The minimum size of the grounding plate should be 2m*1m*1mm. The EUT should be placed on an insulated support plate with the EUT 50mm-100mm above the ground plane. The distance between the EUT and all other conductive structures (except the ground plane) should be greater than 0.5 m.

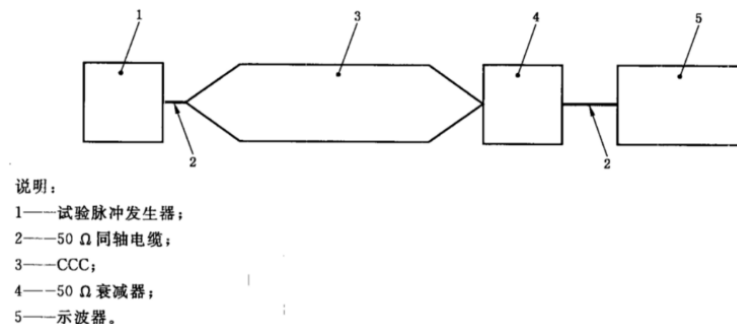
There are three methods for coupling the disturbance signal to the conduction line, which are the capacitive coupling clamp CCC method, the direct capacitor coupling DCC method, and the inductive coupling clamp ICC method.

瞬态类型	CCC 方法	DCC 方法	ICC 方法
4.3.3 的慢速脉冲	不适用	适用	适用
4.3.2 的快速脉冲 a 和 b	适用	适用	不适用

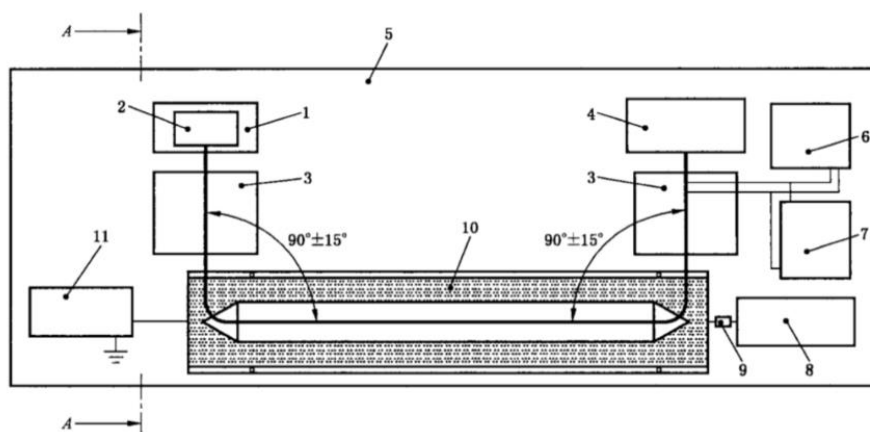
9.14.2.1.1 Capacitive coupling clamp CCC method

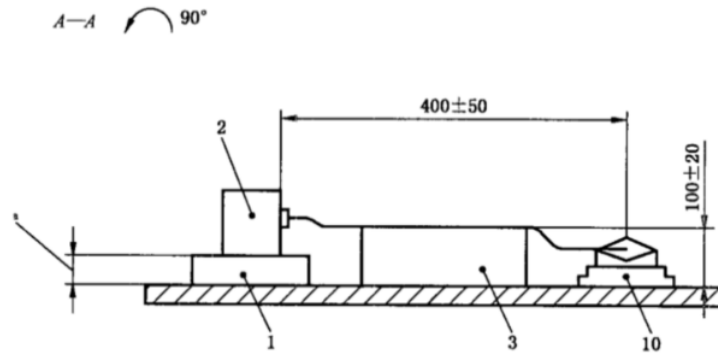
The capacitive coupling clamp method is suitable for coupling fast electrical transient test pulses, especially for EUTs with medium or large numbers of wires to be tested. The capacitive coupling clamp method is not suitable for coupling slow electrical transient test pulses.

The test pulse generator should be calibrated before the test and the pulse amplitude is calibrated using an oscilloscope. The calibration arrangement is shown below. 50Ω coaxial cable is connected to a 50Ω oscilloscope and a 50Ω attenuator for voltage measurement. Wires are not allowed to pass through the coupling clamp during calibration.



The arrangement of the test pulse injection is shown in the figure below. The coupling circuit is formed by a capacitive coupling clamp, and the wire of the EUT passes through the capacitive coupling clamp with a coupling length of 1 m. The power supply line outside the coupling clamp should be 1m long. The distance between the EUT and the CCC, the distance between the auxiliary device and the CCC should be 400 mm ± 50 mm. The part of the conductor to be tested outside the CCC shall be placed at a height of 100 mm ± 20 mm above the ground plane and at an angle of 90° ± 15° with the longitudinal axis of the CCC. The EUT and pulse generator should be placed on the same side of the CCC, and the length of the EUT harness should not exceed 2 m.





说明：

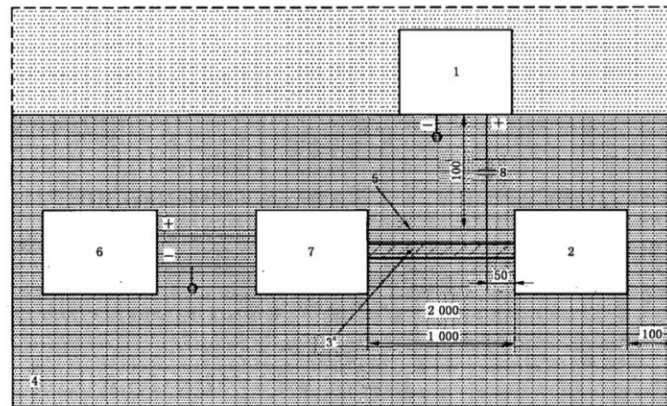
- 1——绝缘支撑板(如果 DUT 没有与车辆接地连接)；
- 2——DUT；
- 3——试验线束的绝缘支撑板；
- 4——辅助装置(如传感器,负载,配件),与在车辆上安装相同；
- 5——接地平板；
- 6——电源；
- 7——蓄电池；
- 8——示波器；
- 9——50 Ω 衰减器；
- 10——CCC；
- 11——试验脉冲发生器。

* 所选尺寸应在试验计划中规定并记录在试验报告中。

9.14.2.1.2 Direct capacitive coupling DCC method

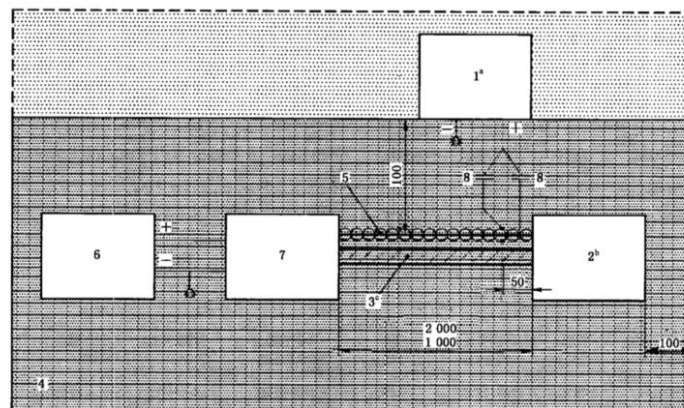
The direct capacitive coupling method uses the recommended capacitance values for EUT wire-coupled fast electrical transient test pulses and couples the same voltage. When the direct capacitance coupling method uses the recommended capacitance value, it is effective to couple the slow electrical transient test pulse. For fast electrical transient tests, each wire needs to be tested separately. The test pulse generator should be calibrated before the test. The output of the pulse generator is placed in series with the coupling capacitor, and a high-impedance oscilloscope is used to measure the peak voltage at the open of the capacitor output. Adjust the output of the pulse generator so that the peak voltage at the open end of the capacitor reaches the level required by the test.

The arrangement of the test pulse injection is shown in the figure below. The length of the harness should be between 1000mm and 2000mm, and all harnesses should be $50\text{mm} \pm 5\text{mm}$ above the ground plane. When using a fast electrical transient test pulse, the capacitor value is 100 pF. When using a slow electrical transient test pulse, the capacitor value is 0.1 μF .



- 说明：
- 1——试验脉冲发生器；
 - 2——DUT；
 - 3——线束；
 - 4——接地平板；
 - 5——被测 I/O 线；
 - 6——电源；
 - 7——DUT 辅助装置；
 - 8——高电压(最小 200 V)陶瓷电容器。

The arrangement of the test pulse injection on the CAN bus is shown in the figure below. The length of the harness should be between 1000mm and 2000mm, and all harnesses should be $50\text{mm} \pm 5\text{mm}$ above the ground plane. When using a fast electrical transient test pulse, the capacitor value is 100 pF. When using a slow electrical transient test pulse, the capacitor value uses 470pF.

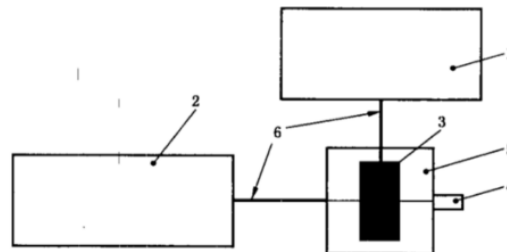


- 说明：
- 1——试验脉冲发生器；
 - 2——DUT；
 - 3——线束；
 - 4——接地平板；
 - 5——CAN 总线；
 - 6——电源；
 - 7——DUT 辅助装置；
 - 8——额定电压不小于 200 V 的陶瓷电容。

9.14.2.1.3 Inductive coupling clamp ICC method

The inductive coupling clamp method is suitable for coupling slow electrical transient test pulses, especially for EUTs with a medium or large number of wires to be tested.

The test pulse generator should be calibrated before the test. The test pulse generator shall be connected to the inductive coupling clamp by a single cable or multiple cables, and the cable length shall not exceed 0.5 m. CCC and DCC use the open circuit voltage of the pulse generator to specify the test level. The ICC uses the voltage measured by the calibration test arrangement as shown below to specify the test level. The pulse should meet the requirements of the following table



说明:

- 1——试验脉冲发生器;
- 2——示波器(1 MΩ 输入);
- 3——ICC;
- 4——短路;
- 5——校准夹具;
- 6——50 Ω 同轴电缆。

用于 ICC 试验方法的校准夹具的例子见附录 A。

参数	12V 电气系统
$t_d / \mu s$	7 ± 2.1
$t_r / \mu s$	≤ 1.2

9.14.3 Test methods

9.14.3.1 EUT status

Same as 7.2.3.1.

9.14.3.2 Test procedure

The test pulse generator is first calibrated so that the test level meets the test requirements. Inject the calibrated test level into the wires of the EUT (except the power line). Check whether the functions of the EUT are immune to interference during or after the test.

The test level of the EUT is as shown in the table below.

试验脉冲 ^a	协议试验电平 ^b	试验等级, U_i , ^c				试验时间 min
		I 最低	II	III	IV 最高	
快 a(DCC 和 CCC)		-10	-20	-40	-60	10
快 b(DCC 和 CCC)		+10	+20	+30	+40	10
DCC 慢+		+8	+15	+23	+30	5
DCC 慢-		-8	-15	-23	-30	5
ICC 慢+		+3	+4	+5	+6	5
ICC 慢-		-3	-4	-5	-6	5

9.14.4 Test criteria

Fulfil requirements.

9.15 ESD Immunity test of electrical and electronic components against electrostatic discharge

9.15.1 Overview

By using the human body ESD model, the electrostatic discharge caused by the EUT in the passenger compartment and the passenger entering the passenger compartment from the outside is simulated. Verify the EUT's immunity to statics in the vehicle environment.

During the test, the test environment is recommended to be 20 ° C and 30% RH, or to maintain the ambient temperature of 23 ° C $\pm$ 5 ° C, relative humidity between 30% -60%. The test voltage was 13.5 $\pm$ 0.5V.

Test arrangements and test methods refer to ISO 10605: 2001. Latest 2008

9.15.2 Test conditions

9.15.2.1 Grounding plane and Insulating pad

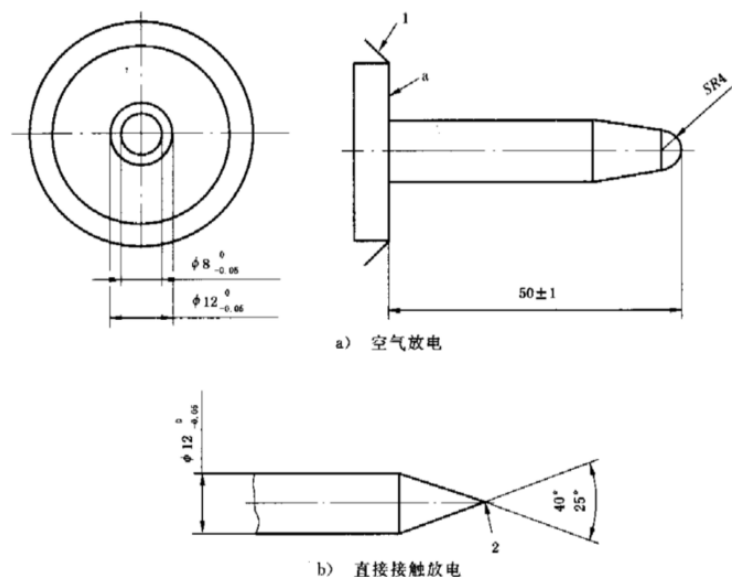
The ground plane should be a piece of foil. The ground plane has a thickness of at least 1 mm and an area of at least 1 m². The outer edge of the ground plane is at least 100 mm beyond the EUT projection profile.

The wire connecting the grounding plane to the EUT ground should have a length of less than 2m and width of at least 50mm.

The insulating spacer should be a clean, non-hygroscopic material. The thickness of the insulating spacer is 25mm $\pm$ 2.5mm, and the distance of the outer edge beyond the projection contour of the EUT is at least 20mm.

9.15.2.2 ESD Simulator discharge terminals

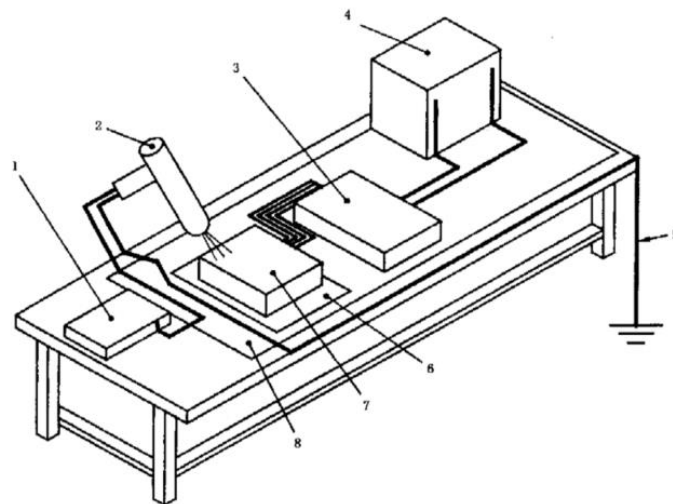
The discharge terminals of the insulating pad air discharge and direct contact discharge is as shown in the figure below.



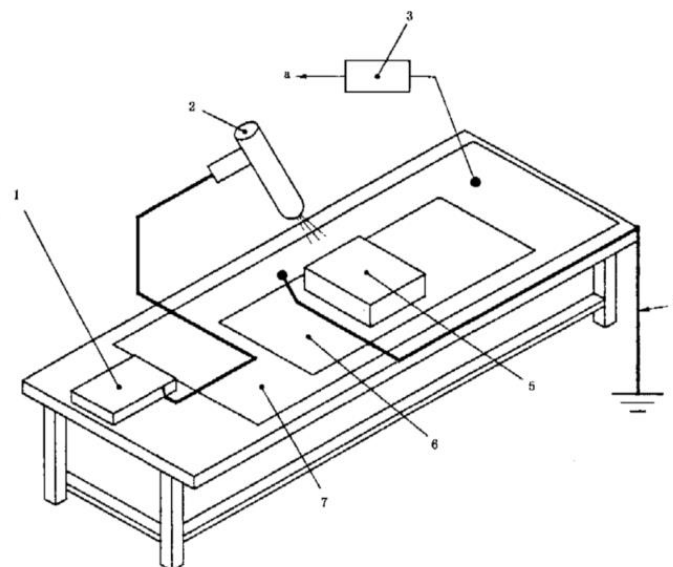
9.15.2.3 Test arrangement

The EUT is placed on a ground plane. Since the vehicle-end coil and coil controls are mounted directly on the chassis, the EUTs are placed directly on the ground plane and connected. The test arrangement is shown in Figure 1 below.

The package handling test arrangement is shown in Figure 2 below. The EUT is placed on the static electricity consuming material to release the charge accumulated in the EUT rack.



- 1—ESD 电源；
- 2—ESD 模拟器；
- 3—动作执行器；
- 4—电池；
- 5—接地带；
- 6—绝缘块(如果需要)；
- 7—被试设备；
- 8—接地平板。



- 1—模拟器电源；
- 2—ESD 模拟器；
- 3—1 MΩ 静电消耗电阻；
- 4—接地带；
- 5—被试设备；
- 6—静电消耗材料；
- 7—接地平板(如果静电消耗材料的生产厂家要求)。

9.15.3 Test methods

9.15.3.1 EUT status

Same as 7.2.3.1.

9.15.3.2 Test Procedures

EUT in working mode, or at least in standby mode. Test the surface that is easily accessible.

Each discharge test point shall withstand at least 3 positive voltage discharges and 3 negative voltage discharges at each voltage level, and the discharge intervals shall be at least 5 s apart. At each voltage level, the discharge test point of the EUT should be subjected to a discharge test of one polarity and then subjected to a discharge test of reverse polarity.

The discharge modes include: direct contact discharge (contact discharge + coupling plate discharge) and air discharge (at least 15 mm from the EUT, $\pm 15^\circ$ perpendicular to the discharge end, aligned with the discharge position, and slowly approached at a speed of no more than 5 mm/s until discharge. If the contact still fails to discharge, stop the test at this voltage level.

9.15.3.2.1 ESD test with powered module

A discharge end with a capacitance of 330 pF was used. Use the ESD test arrangement of the electronic module. Direct contact discharge and air discharge tests were performed.

9.15.3.2.2 Vehicle ESD Test

All discharge points accessible in the vehicle were tested using 330pF, 2000 Ω discharge terminals, and discharged at 4kV, 8kV and 15kV voltages. Use 150pF, 2000 Ω discharge terminals, and discharged at 25kV voltage to test all discharge points that can be reached outside the vehicle and entering the vehicle.

9.15.3.2.3 ESD test when handling without power up

Use the ESD test arrangement for handling. Direct contact discharge and air discharge tests were performed. The test points are connecting pins, housings, buttons, switch displays, container nuts, openings and female connecting pins that are easily accessible during EUT handling. The residual charge of the EUT after each discharge should be shorted to the ground with a 1M Ω resistor. The shorting point is between the discharge position and ground, and between the EUT ground point and ground.

If the EUT case is grounded and the female pins are not easily accessible, this test is not necessary.

9.15.4 Test criteria

The ESD test for module, vehicle, and handling shall meet the test levels specified in Table 1, Table 2, and Table 3, respectively. In an ESD test in which the EUT is not powered, the EUT should be capable of withstanding at least ± 4 kV direct contact discharge.

表 B.1 电子模块(通电运行)试验严酷程度等级

放电类型	严酷程度等级/kV					最少放电次数 ^a
	自选等级	试验等级				
		I	II	III	IV	
接触放电	x ^b	±4	±6	±7	±8	3
空气放电	x ^b	±4	±8	±14	±15	
^a 最小放电间隔时间为 5 s。						
^b 车辆制造商和供应商协议值。						

表 B.2 整车试验严酷程度等级

试验放电点	严酷程度等级/kV					最少放电次数
	自选等级	试验等级				
		I	II	III	IV	
仅车内可触及	x ^b	±4	±8	±14	±15	3
从车外可触及	x ^b	±4	±8	±15	±25	
注：全部是空气放电。						
^a 最小放电间隔时间为 5 s。 ^b 整车制造商和供应商协议值。						

表 B.3 电子模块(不通电)分类试验等级

放电类型	试验等级 ^b /kV			最少放电次数 ^a
接触放电	±4	±6	±8	
空气放电	±4	±15	±25	3
^a 最小放电间隔时间为 5 s。 ^b 建议在进行 ±8 kV 接触放电之前先进行 ±8 kV 空气放电的试验。				

9.16 RE Radiated Emissions test

9.16.1 Overview

The radiation emission of EUT was measured by experiment, and the degree of radiation disturbance caused by EUT to external environment was evaluated. The test should be carried out in the test site, and the limit of ambient noise level in the test site should be guaranteed.

The measuring distance can be 3 m or 10 m. The measuring distance of 3 m is only applicable when the EUT (including cables) is in the range of 1.2 m in diameter and 1.5 m in the ground plane.

During the test, the ambient temperature should be 23°C±5°C.

Test arrangements, test methods and limits are referenced in CISPR 11: 2009. The test site shall meet the IEC/CISPR 16-1-4: 2012 alternative requirements, within ±4 dB of the normalized field attenuation theoretical value.

9.16.2 Test conditions

9.16.2.1 Ambient noise

The ambient noise level should be measured before the test and the EUT is not operating. The ambient noise level should be at least 6dB (9kHz-1GHz) or 10dB (above 1GHz) below the EUT radiation limit to ensure that the EUT's radiated emissions are distinguishable from the background of ambient noise and the readings are unaffected.

If the noise level cannot meet the requirement of less than 6 dB when the measurement distance is 10 m, the antenna can be placed at a closer distance but not less than 3 m, and the radiation limit at the new distance is converted accordingly.

9.16.2.2 Measuring antenna

In the test, different measuring antennas were selected according to different measuring frequencies. When the frequency band is lower than 30MHz, a loop antenna should be used, and the lowest point of the ring is 1m above the ground. When the frequency band is in the range of 30MHz-1GHz, the CISPR16-1-4 antenna should be used. The lowest point of the antenna is at least 0.2m above the ground. The horizontal and vertical polarization directions should be measured, and the antenna center should be in the height range of 1m-4m. When the frequency band is higher than 1 GHz, the CISPR16-1-4 antenna should be used.

9.16.2.3 Test arrangement

The EUT should be placed on the turntable when measuring at the test site. The measured distance is the horizontal distance between the measuring antenna and the nearest part when the EUT makes one revolution. Floor devices should be placed on the ground or separated by a thin insulating material. Portable or non-floor equipment should be placed on a non-metallic table 0.8 meters above the ground.

When the measuring distance is 3 meters, the EUT should be a small device, which satisfies the EUT (including the cable) in the range of 1.2 m in diameter and 1.5 m above the ground. When the measuring distance is between 3 and 10 meters, the corresponding limit shall be converted according to the limit of 3 meters or 10 meters. (Defining the limit level indicated by the logarithmic unit at x meters is L_x , the distance measured at x meters is D_x , then the D_x limit conversion formula is:

$$L_x = L_{10} - 20 * \lg \frac{D_x}{D_{10}}, \text{ this formula only applies to 30MHz-1GHz)}$$

9.16.3 Test methods

9.16.3.1 EUT Status

Same as 7.2.3.1.

9.16.3.2 Test procedure

The EUT is placed on the turntable. The measuring antenna is rotated in both horizontal and vertical polarizations. The turntable rotates at all angles and the antenna rises and falls at a height of 1m-4m. The highest level of radiated disturbance is recorded at each measurement frequency.

In radiation emissions exceeding $L - 10\text{dB}$, record at least 6 largest disturbances in each of the observed frequency bands and their corresponding frequency points. The record should include the antenna polarization direction, antenna height and turret angle of the recorded disturbance level.

According to CISPR 11, radiation emission measurements above 1 GHz are suitable for EUT operating frequencies above 400 MHz. According to CISPR 32, the radiated emission measurement above 1G is suitable for the maximum internal frequency of the EUT above 108 MHz. Therefore, the tentative test frequency band is 9 kHz-1 GHz.

9.16.4 Test criteria

The wireless charging system is a device that intentionally produces and operates at 85 kHz, in accordance with the classification of Group 2 in CISPR 11. Class A (industrial environment) or Class B (residential environment) is applicable depending on whether the equipment used in the residential low-voltage power supply network facility is connected.

There are no applicable limit requirements for 9kHz-150kHz. Refer to CISPR 11 Commercial Induction Cooker Limits and SAE J9254 recommendations as shown below.

Frequency range MHz	Limits at 3 m distance Quasi-peak dB(μA/m)
0,009 – 0,070	69
0,070 – 0,148 5	69 Decreasing linearly with logarithm of frequency to 39
0,148 5 – 4,0	39 Decreasing linearly with logarithm of frequency to 3
4,0 – 30	3

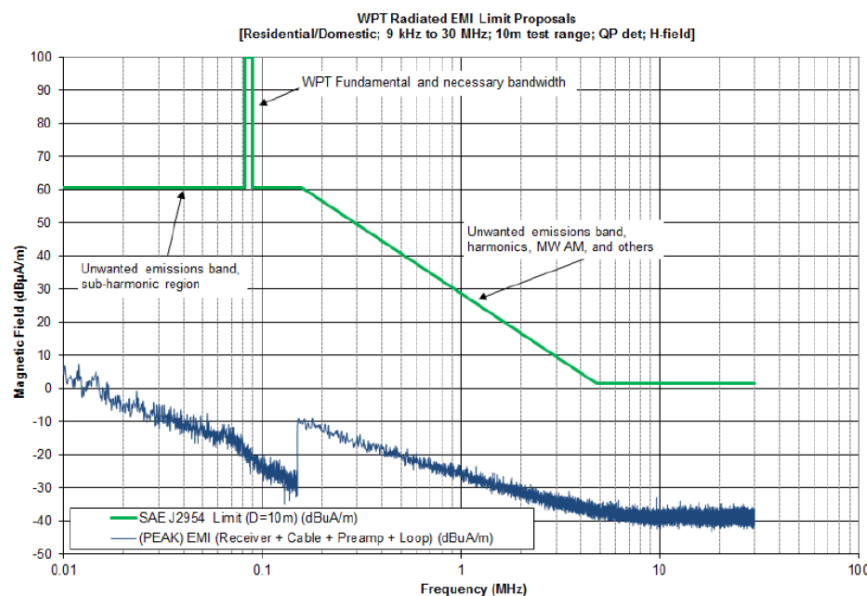


Figure 14 - Recommended limits (see Note 1)

For 150kHz – 30MHz, it should meet the limits of IEC 61980-1. Refer to the Class A or Class B limits of Group 2 in CISPR 11 as shown below.

Frequency range MHz	Magnetic field Quasi-peak Measurement distance $D = 3$ m dB(μ A/m)	Magnetic field Quasi-peak Measurement distance $D = 10$ m dB(μ A/m)
0,15 – 0,49	82	57,5
0,49 – 1,705	72	47,5
1,705 – 2,194	77	52,5
2,194 – 3,95	68	43,5
3,95 – 11	43,5 decreasing linearly with logarithm of frequency to 28,5	18,5
11 – 20	28,5	18,5
20 – 30	18,5	8,5

For 30MHz-1GHz, Class A or Class B limits for Group 2 of CISPR 11 should be met as shown in the following figure.

Frequency range MHz	Limits for a measuring distance of D in m			
	On a test site $D = 30$ m from the equipment		On a test site $D = 10$ m from the equipment	
	Electric field Quasi-peak dB(μ V/m)	Magnetic field Quasi-peak dB(μ A/m)	Electric field Quasi-peak dB(μ V/m)	Magnetic field Quasi-peak dB(μ A/m)
0,15 – 0,49	-	33,5	-	57,5
0,49 – 1,705	-	23,5	-	47,5
1,705 – 2,194	-	28,5	-	52,5
2,194 – 3,95	-	23,5	-	43,5
3,95 – 20	-	8,5	-	18,5
20 – 30	-	-1,5	-	8,5
30 – 47	58	-	68	-
47 – 53,91	40	-	50	-
53,91 – 54,56	40	-	50	-
54,56 – 68	40	-	50	-
68 – 80,872	53	-	63	-
80,872 – 81,848	68	-	78	-
81,848 – 87	53	-	63	-
87 – 134,786	50	-	60	-
134,786 – 136,414	60	-	70	-
136,414 – 156	50	-	60	-
156 – 174	64	-	74	-
174 – 188,7	40	-	50	-
188,7 – 190,979	50	-	60	-
190,979 – 230	40	-	50	-
230 – 400	50	-	60	-
400 – 470	53	-	63	-
470 – 1 000	50	-	60	-

Frequency range MHz	Limits for a measuring distance of D in m		
	Electric field		Magnetic field
	Measurement distance 10 m		Measurement distance 3 m
	Quasi-peak dB(μ V/m)	Average ^a dB(μ V/m)	Quasi-peak dB(μ A/m)
0,15 – 30	–	–	³⁹ Decreasing linearly with logarithm of frequency to ³
30 – 80,872	30	25	–
80,872 – 81,848	50	45	–
81,848 – 134,786	30	25	–
134,786 – 136,414	50	45	–
136,414 – 230	30	25	–
230 – 1 000	37	32	–

9.17 CE Conducted Emissions Test

9.17.1 Overview

The degree of radiated disturbance caused by the EUT to the external environment is evaluated by measuring the radiation emission value of the EUT.

During the test, the ambient temperature should be $23\text{ }^{\circ}\text{C} \pm 5\text{ }^{\circ}\text{C}$.

Test arrangements, test methods and limits are referenced in CISPR 11: 2009.

9.17.2 Test conditions

9.17.2.1 Ambient noise

The ambient noise level should be measured before the test, when the EUT is not operating. The ambient noise level should be at least 6dB (9kHz-1GHz) or 10dB (above 1GHz) below the EUT radiation limit to ensure that the EUT's radiated emissions are distinguishable from the background of ambient noise and the readings are unaffected.

An appropriate RF filter can be inserted between the artificial network and the power supply or measured in the shielded room. After the RF filter is connected, the impedance of the artificial power network should still meet the specified requirements at the measurement frequency.

9.17.2.2 Test arrangement

When measuring the disturbance of the power terminal, the extra length of the cable should be bundled into a wire harness of 0.3m-0.4m length near its midpoint. If increasing the number of cables in a place with multiple interfaces of the same type does not significantly affect the measurement results, just use one cable to cover one of the interfaces.

The test arrangement can be the same as the arrangement in radiation emission or use the following method.

A ground plane is used in the test site. The minimum size of the ground plane is 2m*2m (the ground or any wall of in shielded room can be used as the ground plane) and should be 0.5m larger than the boundary of the EUT. The floored EUT is placed on the ground, and the non-floored EUT should be placed on a non-metal table with a height of 0.4 m on the ground plane. The distance from all test equipment to other metal surfaces should be greater than 0.8m.

9.17.3 Test methods

9.17.3.1 EUT Status

Same as 7.2.3.1.

9.17.4 Test criteria

The wireless charging system is a device that intentionally produces and operates at 85 kHz. It complies with the classification of Group 2 in CISPR 11. Class A (industrial environment) or Class B (residential environment) is applicable depending on whether the equipment used in the residential low-voltage power supply network facility is connected.

There are no applicable limits for the 9 kHz-150 kHz band. Refer to CISPR 11 Induction Cooker Limits as shown below.

Frequency range MHz	Induction cooking appliance limits			
	All appliances except 100 V rated appliances without an earth connection		100 V rated appliances without an earth connection	
	Quasi-peak dB(μV)	Average dB(μV)	Quasi-peak dB(μV)	Average dB(μV)
0,009 – 0,050	110	–	122	–
0,050 – 0,148 5	90 Decreasing linearly with logarithm of frequency to 80	–	102 Decreasing linearly with logarithm of frequency to 92	–
0,148 5 – 0,5	66 Decreasing linearly with logarithm of frequency to 56	56 Decreasing linearly with logarithm of frequency to 46	72 Decreasing linearly with logarithm of frequency to 62	62 Decreasing linearly with logarithm of frequency to 52
0,5 – 5	56	46	56	46
5 – 30	60	50	60	50
At the transition frequency, the more stringent limit shall apply.				

Class A of the 150kHz-30MHz band shall meet the limits of the two sets of Class A equipment in CISPR 11 as shown in the following figure.

Frequency range MHz	Rated input power of ≤ 75 kVA		Rated input power of > 75 kVA ^a	
	Quasi-peak dB(μV)	Average dB(μV)	Quasi-peak dB(μV)	Average dB(μV)
0,15 – 0,50	100	90	130	120
0,50 – 5	86	76	125	115
5 – 30	90 decreasing linearly with logarithm of frequency to 73	80 decreasing linearly with logarithm of frequency to 60	115	105
At the transition frequency, the more stringent limit shall apply.				
NOTE 1 Limits only apply to Low Voltage (LV) a.c. mains input ports.				
NOTE 2 For class A equipment with a rated power ≤ 75 kVA intended to be connected solely to isolated neutral or high impedance earthed (IT) industrial power distribution networks (see IEC 60364-1), the limits defined for group 2 equipment with a rated input power > 75 kVA can be applied.				

Class B of the 150kHz-30MHz band shall meet the limits of the two sets of Class B equipment in CISPR 11 as shown in the following figure.

Frequency range MHz	Quasi-peak dB(μ V)	Average dB(μ V)
0,15 – 0,50	66 Decreasing linearly with logarithm of frequency to 56	56 Decreasing linearly with logarithm of frequency to 46
0,50 – 5	56	46
5 – 30	60	50

9.18 Harmonic Current Test

9.18.1 Overview

This test verifies the harmonic current injected into the utility power system during stable operation of the EUT.

9.18.2 Test Methods

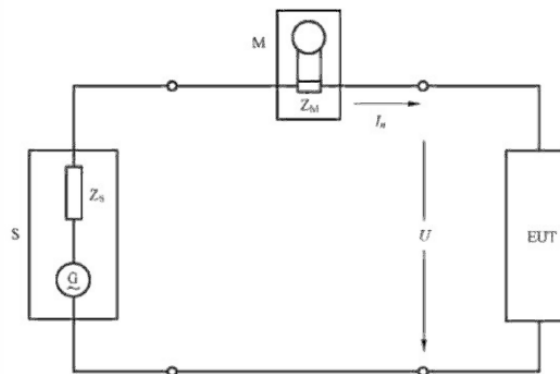
9.18.2.1 EUT Status

The EUT should be in a stable working process during the test. Harmonic current is not considered when the system is operated for the first 10s or from power up to steady state or during power termination. The EUT should not be in standby mode for more than 10% of the observation period.

The EUT should operate in the mode where the maximum THC is occurred (under normal operating conditions).

9.18.2.2 Test Procedures

The EUT should setup as below. The harmonic current of a single-phase device can be measured on the live or neutral.



- | | |
|--------------------|-------------------------|
| S — 供电电源； | EUT — 受试设备； |
| Z_M — 测量设备的输入阻抗； | I_n — 线电流的 n 次谐波分量； |
| M — 测量设备； | U — 试验电压； |
| Z_s — 供电电源的内阻抗； | G — 供电电源的开路电压。 |

9.18.3 Test criteria

The manufacturer must assume a short circuit ratio R_{sc} value. Equipment meeting the short circuit ratio $R_{sc}=33$ corresponding harmonic current limit can be connected to any point of the power supply system. The short circuit ratio is greater than 33, allowing higher harmonic emission value. Harmonic emission shall conform to the following table.

最小 R_{res}	可接受的单次谐波电流 I_n/I_1^* %						可接受的谐波电流畸变率 %	
	I_3	I_5	I_7	I_9	I_{11}	I_{13}	THD	PWHD
33	21.6	10.7	7.2	3.8	3.1	2	23	23
66	24	13	8	5	4	3	26	26
120	27	15	10	6	5	4	30	30
250	35	20	13	9	8	6	40	40
≥ 350	41	24	15	12	10	8	47	47
注 1: 12 次及以下偶次谐波的电流值不应超过 $(16/n)\%$ 。12 次以上偶次谐波与奇次谐波同样用 THD 和 PWHD 考虑。								
注 2: 允许在两个连续 R_{res} 间线性插值。参见附录 D。								
* I_1 为基波参考电流值, I_n 为谐波电流分量。								

9.19 ESD immunity test

9.19.1 Testing conditions

No power, standby (WB only), charging (WB only)

9.19.2 Test Methods

- Air discharge: the electrode is continuously close to the gap of the product casing until it is discharged or contacted (10 times of discharge)
- Contact discharge: the electrode should be directly in contact with the outside of the test product, and the discharge point is a metal part, such as a metal bolt, a metal heat sink, etc. (10 times of discharge)

Test Item	Classification			Test Results	Reference Standard
Electrostatic discharge immunity test	WB	$\pm 6\text{kV}$ (Contact)	Criterion		QC/T895 GBT18487.3-2001 GB/T 17626.2
		$\pm 8\text{kV}$ (Air)	II		
	VA	$\pm 8\text{kV}$ (Contact)	Criterion		
		$\pm 8\text{kV}$ (Air)	II		
	System	$\pm 8\text{kV}$ (Contact)	Criterion		
		$\pm 8\text{kV}$ (Air)	II		

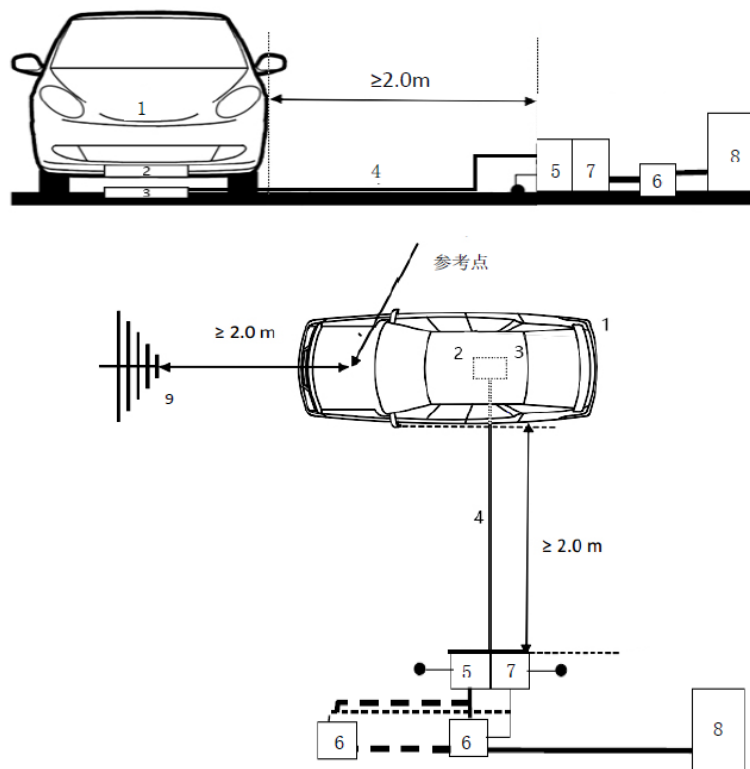
9.20 RF electromagnetic field radiation immunity test

9.20.1 Test Conditions

- System input 220Vac/50Hz;
- The ground clearance $Z=175\text{mm}$; no offset;
- Power output $P_{\text{out}} = 6.6\text{KdW}$;
- VA inlet water temperature: 45°C , flow: $8\text{L} / \text{min}$

9.20.2 Test Methods

Test arrangement (System)



说明:

1——被测车辆;

2——副边设备;

3——原边设备

4——充电线缆;

5——接地的人工电源网络;

6——电源插座 (图中示意了备选位置);

7——接地的不对称人工网络;

8——原边控制器;

9——发射天线。

Test Item	classification			Test Results	Reference Standard
RF electromagnetic field radiation immunity test	WB	30 V/m (80MHz~1GHz)	Criterion I		GB/T17626.3 IEC 61000-4-3
		30 V/m (1.4GHz~2GHz)			
		30 V/m (2GHz~2.7GHz)			
	VA	30 V/m (20MHz~2GHz)	Criterion I		GB 34660-2017
	System	30 V/m (20MHz~2GHz)	Criterion I		GB 34660-2017

9.21 Operating frequency magnetic field immunity test

9.21.1 Test Conditions

1. System input 220Vac/50Hz;
2. The ground clearance $Z=175\text{mm}$; no offset;
3. Power output $P_{out} = 6.6\text{KdW}$;
4. VA inlet water temperature: 45°C , flow: $8\text{L} / \text{min}$

9.21.2 Test Methods

The device is in a test magnetic field generated by an appropriately sized induction coil. The test should be repeated by moving the induction coil. The overall EUT was tested in each orthogonal direction. The test shall be carried out in steps of 50% of the shortest side of the coil, moving the coil to different positions along the side of the EUT. In order to expose the EUT to the test magnetic field in different directions, the induction coil should be rotated 90° and tested according to the same procedure.

Test Item	Classification			Test Results	Reference Standard
Operating frequency magnetic field immunity test	WB	30A/m (system below 32A) 100A/m (system above 32A)	Criterion I		GB/T 17626.8
	System	30A/m (System below 32A) 100A/m (system above 32A)	Criterion I		GB/T 17626.8

9.22 Electrical fast transient burst immunity test

9.22.1 Test Conditions

1. System input 220Vac/50Hz;
2. The ground clearance Z=175mm; no offset;
3. Power output Pout = 6.6KdW;
4. VA inlet water temperature: 45°C, flow: 8L / min

9.22.2 Test Methods

1. WB power terminal test: The test equipment coupling network replaces the power supply to the power supply terminal of the tested product. The pulse group interval is 300ms, and the test time is 60s.
2. VA signal terminal test: Test equipment coupling network using coupling clip. The signal line harness between VA and vehicle controller is inserted into coupling clip. The pulse group interval is 300ms, and test time is 60s.
3. VA power terminal test: The test equipment coupling network replaces the 12V DC power supply of the tested product. The pulse group duration is 10ms, the pulse group interval is 90ms, and the test time is 60s.
4. System test: Test equipment coupling network replaces the power supply to WB. The pulse group interval is 300ms, and test time is 60s

Test Item	Classification			Test Results	Reference Standard
Electrical fast transient	WB	±4kV 5/50 Tr/Th,ns 100kHz	Criterion II		QC/T895 GBT18487.3-2001 GB/T 17626.4

burst immunity test	VA signal terminal	±2kV (Coupling clip) 5/50 Tr/Th,ns 100kHz			
	VA power terminal	-150V/+100V 100 t1,us	Criterion B		ISO7637-2-3a ISO7637-2-3b
	System	±4kV 5/50 Tr/Th,ns 100kHz	Criterion II		QC/T895 GBT18487.3-2001 GB/T 17626.4

9.23 Surge (impact) immunity test

9.23.1 Test Condition

1. System input 220Vac/50Hz;
2. The ground clearance Z=175mm; no offset;
3. Power output Pout = 6.6KdW;
4. VA inlet water temperature: 45°C, flow: 8L / min

9.23.2 Test Methods

1. WB power terminal test: test equipment coupling network replaces the power supply terminal of the tested product. When L to N, the internal resistance is set to 2Ω. When L to PE, the internal resistance is set to 12Ω. When N to PE, the internal resistance is set to 12Ω. The three methods are done 5 times each with an interval of 60s.
2. VA power terminal test: The test equipment coupling network replaces the 12V DC power supply of the tested product and the pulse occurs once.
3. System test: test equipment coupling network replaces the power supply terminal of the WB power supply. When L to N, the internal resistance is set to 2Ω. When L to PE, the internal resistance is set to 12Ω. When N to PE, the internal resistance is set to 12Ω. The three methods are done 5 times each with an interval of 60s.

Test Item	Classification			Test Results	Reference Standard
Surge (impact) immunity test	WB	1.2/50 (8/20) μs ±4 kVa ±2 kVb	Criterion II		QC/T895 GBT18487.3-2001 GB/T 17626.5
	VA	87V 400 t1,ms 2 Re,Ω	Criterion B		ISO7637-5a
	System	1.2/50 (8/20) μs ±4 kVa ±2 kVb	Criterion II		QC/T895 GBT18487.3-2001 GB/T 17626.5

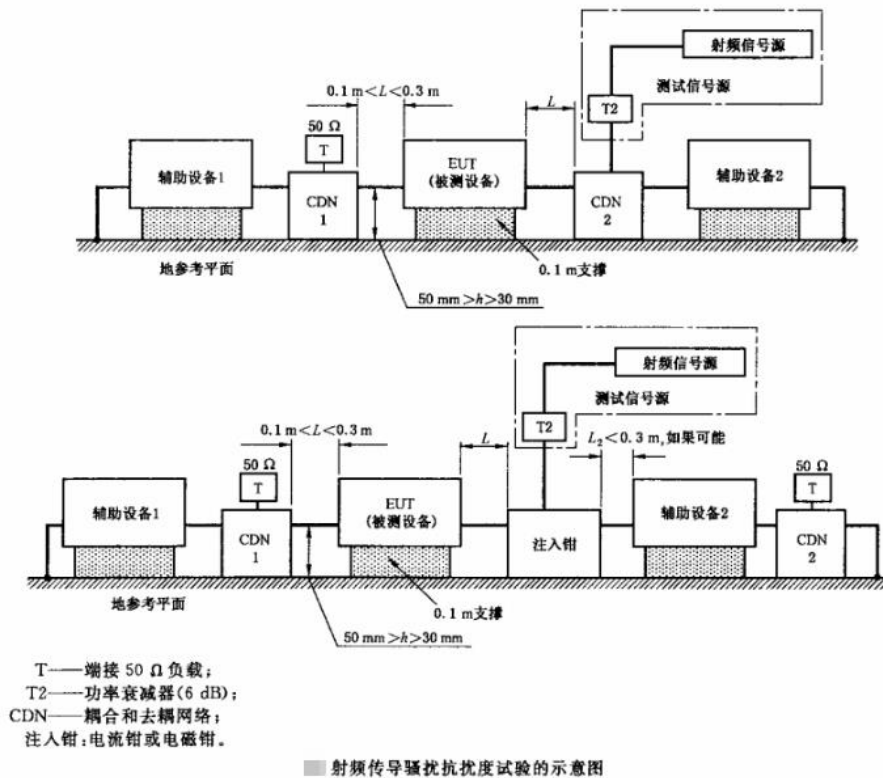
9.24 Conducted disturbance immunity test for RF field induction

9.24.1 Test Condition

1. System input 220Vac/50Hz;
2. The ground clearance Z=175mm; no offset;
3. Power output Pout = 6.6KdW;

4. VA inlet water temperature: 45°C, flow: 8L / min

9.24.2 Test Methods



As shown in the figures above, all the external cables of the unit under test are placed in the coupling device and scanned according to the standard.

Note: When sweeping, the dwell time of each frequency segment should not lower than the running and response time of the device under test and it is not less than 0.5s.

Test Item	Classification			Test Results	Reference Standard
Conducted disturbance immunity test for RF field induction	WB	10 V (rms) (0.15 ~80 MHz)	Criterion I		QC/T895 GBT18487.3-2001 GB/T 17626.6
	System	10 V (rms) (0.15 ~80 MHz)	Criterion I		

9.25 Voltage dip, short interruption and voltage variation immunity test

9.25.1 Test Condition

1. System input 220Vac/50Hz;
2. The ground clearance Z=175mm; no offset;
3. Power output Pout = 6.6KdW;
4. VA inlet water temperature: 45°C, flow: 8L / min

9.25.2 Test Methods

When the output voltage is 80% of the nominal value, the above requirements only need to be tested during the maximum duration of 5s.

When the output voltage is 70% and 40% of the nominal value, the above requirements only need to be tested during the maximum duration of 3s.

The Rise and fall times as well as overshoot and undershoot should be tested and verified at phase angles of 90° and 270° switch from 0% to 100%; 100% to 80%; 100% to 70%; 100% to 40% and 100% to 0%.

The accuracy of the phase angle should be in 9 phase angles from 0° to 360° with 45° increments. Switching tests are performed from 0% to 100% and 100% to 0%. And at phase angles of 90° and 180°, verified from 100% to 80% and 80% to 100%, 100% to 70% and 70% to 100%, 100% to 40% and 40% to 100%.

The EUT shall perform three voltage sag or interruption tests in the order of each selected test level and duration combination. A minimum discontinuity of 10 s (interval between trials) should be tested in each typical mode of operation.

For voltage sag, the change in supply voltage occurs at voltage zero crossings as determined by the relevant professional standardization technical committee or individual product specifications for additional testing. Each phase is preferably selected from 45°, 90°, 135°, 180°, 225°, 270°, and 315°. For short interruptions, the relevant professional standardization technical committees set the angle according to the worst case. If not specified, it is recommended to select one phase and test at a phase angle of 0°.

Test Item	Classification			Test Results	Reference Standard
Voltage dip, short interruption and voltage variation immunity test	WB	40% for 10 cycles	Criterion II		QC/T895 GBT18487.3-2001 GB/T 17626.34 (>16A)
		70% for 25 cycles	Criterion II		
		80% for 250 cycles	Criterion II		
		0% for 250 cycles	Criterion III		
	System	40% for 10 cycles	Criterion II		
		70% for 25 cycles	Criterion II		
		80% for 250 cycles	Criterion II		
		0% for 250 cycles	Criterion III		

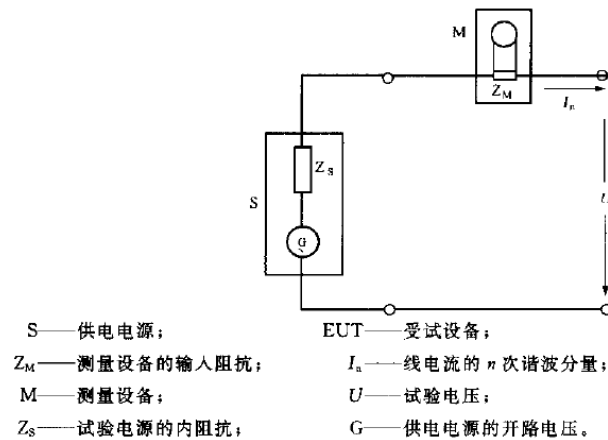
9.26 Harmonic current test

9.26.1 Test Condition

1. System input 220Vac/50Hz;
2. The ground clearance Z=175mm; no offset;
3. Power output Pout = 6.6KdW;
4. VA inlet water temperature: 45°C, flow: 8L / min

9.26.2 Test Methods

The measuring device is connected between the power supply and the system under test as follows:



When a device is manually or automatically put into operation or exit of operation, the harmonic current in first 10s after switching action is not considered.

The device under test should not exceed 10% of any observation period in standby mode.

Test Item	Classification		Test Results	Reference Standard
Harmonic current test	WB	GB/T 17625.8 (Rated current per phase >16A and ≤ 75A)		GB/T 17625.8
	System	GB/T 17625.8 (Rated current per phase >16A and ≤ 75A)		

9.27 Voltage fluctuation and flicker test

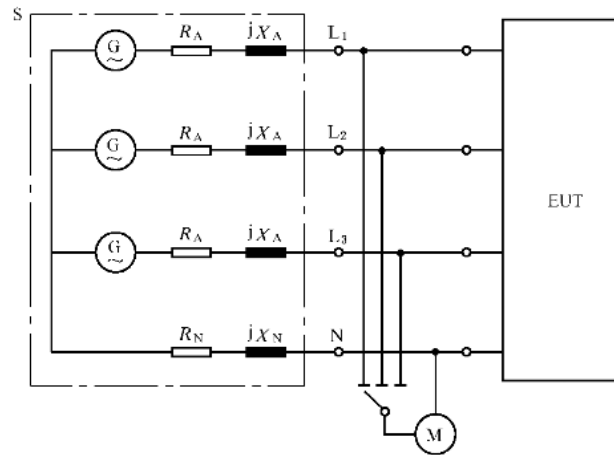
9.27.1 Test Condition

1. System input 220Vac/50Hz;
2. The ground clearance $Z=175\text{mm}$; no offset;
3. Power output $P_{out} = 6.6\text{KdW}$;
4. VA inlet water temperature: 45°C , flow: 8L / min

9.27.2 Test Methods

The test method is selected according to the system impedance.

The device is connected as shown below:



EUT——受试设备。

M——测量设备。

S——由电源电压发生器 G 和参考阻抗 Z 组成的供电电源,Z 由下列元件组成:

$$R_A = 0.24 \, \Omega; \quad jX_A = 0.15 \, \Omega \quad 50 \, \text{Hz}$$

$$R_N = 0.16 \, \Omega; \quad jX_N = 0.10 \, \Omega \quad 50 \, \text{Hz}$$

Test Item	Classification		Test Results	Reference Standard
Voltage fluctuation and flicker test	WB	GB/T 17625.7 (Rated current per phase >16A and ≤75A)		GB/T 17625.7
	System	GB/T 17625.7 (Rated current per phase >16A and ≤75A)		

9.28 Conducted disturbance test

9.28.1 Test Condition

1. System input 220Vac/50Hz;
2. The ground clearance Z=175mm; no offset;
3. Power output Pout = 6.6KdW;
4. VA inlet water temperature: 45°C, flow: 8L / min

9.28.2 Test Methods

The test was carried out in accordance with Group 2 Class B equipment.

表 7 在试验场地测量时,2 组 B 类设备的电源端子骚扰电压限值

频段/MHz	准峰值/dB(μV)	平均值/dB(μV)
0.15~0.50	66~56 随频率对数线性减小	56~46 随频率对数线性减小
0.50~5	56	46
5~30	60	50
在过渡频率上采用较严格的限值。		

Test Item	classification	Test Results	Reference Standard
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Conducted disturbance test	WB	Conducted disturbance (150kHz~30MHz)		QC/T895 GBT18487.2-2001 GB 4824 CISPR32
	System	Conducted disturbance (150kHz~30MHz)		QC/T895 GB 4824 CISPR32

9.29 Radiation disturbance test

9.29.1 Test Condition

1. System input 220Vac/50Hz;
2. The ground clearance Z=175mm; no offset;
3. Power output Pout = 6.6KdW;
4. VA inlet water temperature: 45°C, flow: 8L / min

9.29.2 Test Methods

Test Item	Classification		Test Results	Reference Standard
Radiation disturbance test	WB	Radiation disturbance (Requirements for protecting the vehicle's keyless entry system)		GB 4824
		Radiation disturbance		
	VA	Electromagnetic radiation emission test (protecting the receiver outside)		GB 14023/CISPR 12 GB/T 18655-201X
		Electromagnetic radiation emission test (protecting the car receiver)		
	System	Radiation disturbance (Requirements for protecting the vehicle's keyless entry system)		GB 14023/CISPR 12 GB/T 18655-201X GB/T 18387-2017
		Electromagnetic radiation emission test (protecting the receiver outside)		
		Electromagnetic radiation emission test (protecting the car receiver)		
		Low frequency electromagnetic field emission		

10 Environment Stress Test for VA

10.1 Constant low temperature test

Testing requirement:

1. Low temperature storage function should reach Class C as defined by ISO 16750-1:2006.

2. Low temperature operation function should reach Class A as defined in ISO 16750-1:2006.

Test Method:

1. Low temperature storage: Put the wireless charging system into the temperature box with the initial temperature of room temperature, adjust the temperature of the incubator to reach $-40^{\circ}\text{C} \pm 2^{\circ}\text{C}$ for 24 h, then remove it from the incubator and place it in the specified 1 h in environmental conditions. After the test is over, the wireless charging system should work properly.
2. Low temperature operation: Put the VA parts of the wireless charging system into the temperature box with the initial temperature of room temperature, in accordance with GB/T2423.1-2008, § 5.3 "Test Ad: Low Temperature Test for Gradual Temperature Change of Heat Dissipation Test Samples – Power up the test sample when sample temperature is stable". Test temperature: $-40^{\circ}\text{C} \pm 2^{\circ}\text{C}$. After the environmental test chamber reaches the test temperature, the system is powered, and it is kept in full load for 24 h.

Test Item	Classification		Test Results	Reference Standard
Low temperature test	Low temperature storage	-40°C , 24h		ISO 16750-1 : 2006 GB/T 2423.1-2008
	Low temperature operation	-40°C , 24h		ISO 16750-1 : 2006 GB/T 2423.1-2008

10.2 Constant high temperature test

Testing requirement:

1. High temperature storage function should reach Class C defined by ISO 16750-1: 2006.
2. High temperature operation function should reach Class A as defined in ISO 16750-1: 2006.

Test Method:

1. High temperature storage: Put the wireless charging system into the temperature box with the initial temperature of room temperature, adjust the temperature of the incubator to $105^{\circ}\text{C} \pm 2^{\circ}\text{C}$, and then keep it for 504 h, then remove the wireless charging system from the incubator and place it in 1 h in the specified environmental conditions. After the test is over, the wireless charging system should work properly.
2. High temperature operation: Put the wireless charging system into the temperature box with the initial temperature of room temperature, adjust the temperature of the incubator to $85^{\circ}\text{C} \pm 2^{\circ}\text{C}$, and let it be in full load for 96h.

Put the VA component of the wireless charging system into the environmental test chamber. According to GB/T2423.2-2008, § 5.3 "Test Bd: High Temperature Test for Gradual Temperature Change of Heat Dissipation Test Samples - Test Samples are not powered during temperature rise". Test temperature: 85°C . After the environmental test chamber reaches the test temperature, the system is powered, and it is kept at full load for 96 hours.

Test Item	classification	Test Results	Reference Standard
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High temperature test	High temperature storage	105°C, 504h		ISO 16750-1 : 2006 GB/T 2423.2-2008
	High temperature operation	85°C, 96h		ISO 16750-1 : 2006 GB/T 2423.2-2008

10.3 Damp heat cyclic test

Testing requirement:

The functional status shall be in accordance with Class A as defined in ISO 16750-1:2006.

Test Method:

The test is carried out in accordance with ISO 16750-4: 2010, Section 5.6.2.2 and IEC 60068-2-30, Test Db, Variant 1. A functional test (operating mode 3.2 in accordance with ISO 16750-1) when the maximum cycle is reached.

temperature is reached

Test Item	classification		Test Results	Reference Standard
Hot and humid cycle test	Hot and Humid Cycle	6 Cycles, High Temp. 55°C, Low Temp. 25°C		ISO 16750-4: 2010 GB-T2423.4

10.4 Composite temperature/humidity cyclic test

Testing requirement:

The functional status shall be in accordance with the Class A as defined in ISO 16750-1: 2006.

Test Method:

The test is carried out in accordance with ISO 16750-4: 2010, Section 5.6.2.3 and IEC 60068-2-38, Test Z/AD. A functional test (operating mode 3.2 in accordance with ISO 16750-1) when the maximum cycle is reached.

Test Item	Classification		Test Results	Reference Standard
Composite temperature/humidity cyclic test	Hot and Humid Cycle	24 h/Cycle, 10 Cycles, low temp. -10 °C		ISO 16750-4: 2010 GB-T2423.34

10.5 Temperature step test

Testing requirement:

At each temperature step between -40 °C and 85 °C, the wireless charging system should maintain normal function and the functional status should conform to Class A as defined in ISO 16750-1: 2006.

Test Method:

The test was carried out in accordance with ISO 16750-4-2010, Section 5.2. Decrease the temperature in steps of 5 °C from 20 °C to -40 °C, and then increase the temperature in steps of 5 °C from -40 °C to 85 °C. Wait at each step until the sample has reached the new temperature. Perform functional tests operating mode 3.2 in accordance with ISO 16750-1 at minimum supply voltage, U_{Smin} , and at maximum supply voltage, U_{Smax} , in accordance with the specified ISO 16750-2 code letter, at each new temperature step. Switch the DUT off during transition to the next temperature.

Test Item	Classification		Test Results	Reference Standard
Temperature step test	Temperature step			ISO 16750-1: 2006 ISO 16750-2: 2010 ISO 16750-4: 2010

10.6 Temperature cycle test (with specified change rate)**Testing requirement:**

The functional status shall be in accordance with the Class A as defined in ISO 16750-1:2006.

Test Method:

The test is carried out in accordance with ISO 16750-4: 2010, Section 5.3.1 and IEC 60068-2-14, Test Nb. The change rates in the temperature between -40 °C, 20 °C and 85 °C shall correspond to the specifications given in Table 2 of ISO 16750-4: 2010, Section 5.3.1. 30 test cycles shall be conducted on a sample.

Test Item	classification		Test Results	Reference Standard
Temperature cycle test	Temperature cycle	30 Cycles, 480 min for each cycle		ISO 16750-4: 2010 GB-T2423.22

10.7 Steady state damp heat test**Testing requirement:**

The functional status shall be in accordance with the Class A as defined in ISO 16750-1:2006.

Test Method:

The test is carried out in accordance with ISO 16750-4: 2010, Section 5.7.2 and IEC 60068-2-78 at temperature (40 ± 2) °C, relative humidity (85 ± 3) %, a test duration of 21 days and the following operation mode.

- Operation mode 2.1, in accordance with ISO 16750-1:2006, test duration: 20 days and 23 h

- Operating mode 3.2, in accordance with ISO 16750-1:2006, test duration: 1 h after operating mode 2.1.

Test Item	classification		Test Results	Reference Standard
Steady state damp heat test	Steady state damp heat	21 Days		ISO 16750-4: 2010 IEC 60068-2-78

10.8 Rapid Temperature Change with specified transition duration test

Testing requirement:

The functional status reaches Class C as defined by ISO 16750-1: 2006.

Test Method:

The test is carried out in accordance with ISO 16750-4: 2010, Section 5.3.2 and IEC 60068-2-14, Test Na. Raise the temperature from -40 °C to 85 °C within 30 s or less. Keep the sample at -40 °C or 85 °C for a given soak of 1 h after thermal stability is obtained. Use operating mode 1.1, as defined in ISO 16750-1. See Table 4 of ISO 16750-4: 2010 for the number of cycles.

Test Item	Classification		Test Results	Reference Standard
Thermal shock test	Thermal shock	30 Cycles		ISO 16750-4: 2010 GB-T2423.22

10.9 Ice water shock (submersion) test

Testing requirement:

The functional status shall be in accordance with Class A as defined in ISO 16750-1: 2006.

Test Method:

The test is carried out in accordance with ISO 16750-4: 2010, Section 5.4.3 using the test parameters as provided.

Test Item	Classification		Test Results	Reference Standard
Ice water shock (submersion) test	Ice water shock (submersion) test	Water Temp. ~4°C, 100 Cycles		ISO 16750-4: 2010 ISO 16750-1: 2006

10.10 Ingress Protection (IP) Test

Testing requirement:

According to ISO 6469-1:2019, Section 5.3.2, for the waterproof performance requirements of the rechargeable battery system, set the dust-proof and waterproof level of the vehicle-mounted end of the wireless charging system: IP67 specified in ISO 20963.

Test Method:

Tested in accordance with IP67 in ISO 20653.

Test Item	Classification		Test Results	Reference Standard
IP Test	IP Class	IP67: Completely dust-proof & protected against temporary flooding		ISO 20653 : 2013

10.11 Salt spray (corrosion) test

Testing requirement:

The functional status shall be in accordance with Class C as defined in ISO 16750-1:2006.

Test Method:

Tested in accordance with ISO 16750-4-2010 5.5.1.2.

Test Item	Classification		Test Results	Reference Standard
Salt spray test	Salt spray (corrosion)	5% sodium chloride solution, spray 2h per cycle		ISO 16750-4 : 2010 IEC 60068-2-52 GB-T2423.18

10.12 Salt spray (leakage) test

Testing requirement:

No saltwater may enter the casing. In the 3.2 mode of operation, the functional status shall comply with Class A as defined in ISO 16750-1:2006.

Test Method:

The test is carried out in accordance with ISO 16750-4: 2010, Section 5.5.2.2 and IEC 60068-2-11, Test Ka. Note that operate the sample with operating mode 3.2 in accordance with ISO 16750-1 between the fourth and fifth hour of each cycle.

Test Item	Classification		Test Results	Reference Standard
Salt spray test	Salt spray (leakage)	5% sodium chloride solution, spray for 8 h and stop for 16 h for a cycle to repeat a total of 6 cycles		ISO 16750-4: 2010 GB-T2423.17

10.13 Accelerated corrosion test

Testing requirement:

Meets the requirements of VCS 1027.1449.

Test Method:

According to the requirements of VCS 1027.1449, the on-board coil test period is 9 weeks, and the on-board controller is required for 6 weeks.

Test Item	Classification		Test Results	Reference Standard
Salt spray test	Circulating salt spray	Vehicle coil test cycle 9 weeks, on-board controller requires 6 weeks		VCS 1027.1449

10.14 Flow mixed gas corrosion test**Testing requirement:**

The functional status shall be in accordance with Class C as defined in ISO 16750-1: 2006.

Test Method:

Tested in accordance with ISO 16750-4-2010, Section 5.8.2 and IEC 60068-2-60, Test Ke, method 4.

Test Item	Classification		Test Results	Reference Standard
Flow mixed gas corrosion test	Flow mixed gas corrosion	21 Days		ISO 16750-4: 2010 GB-T2423.51

10.15 Chemical resistance performance test**Testing requirement:**

The functional status after the test should conform to Class C defined by ISO 16750-1:2006. There should be no change that impairs normal performance (e.g., sealing function), and the signs and labels should be clearly visible.

Test Method:

Tests are conducted according to ISO 16750-5: 2010.

Test Item	Classification		Test Results	Reference Standard
Chemical resistance performance test	Chemical resistance performance	At least 1 cycle		ISO 16750-5: 2010

10.16 Splash water test**Testing requirement:**

Functional status should conform to Class A defined by ISO 16750-1: 2006.

Test Method:

According to ISO 16750-4, Section 5.4.2, every 10 L of water plus 1 g of wetting agent, working mode 3.1.

Test Item	Classification		Test Results	Reference Standard
Splash water test	Splash water	Splash water 3s, transition time less than 20s, flow rate 1L/s~1.33L/s, cycle 100 times		ISO 16750-4: 2010

10.17 Noise test

This noise test of VA is to be tested with the WPT machine.

Testing purpose:

The overall working noise level of the charging equipment meets the requirements of relevant laws and regulations.

Testing requirement:

Working noise $\leq$ 55dB (A).

Test Method:

Under the condition that the rated load and the ambient noise are not more than 40dB, the working noise of the on-board charger is measured at a position 1m away from the noise source and 1m~1.5m from the ground.

10.18 Reliable and durable test**Testing requirement:**

The functional status shall be in accordance with Class A as defined in ISO 16750-1:2006.

Test Method:

Under the rated input conditions, the ambient temperature is 85 °C, the water temperature is 65 °C, and the wireless charging system is turned on to make it work at full load for 2000 hours.

Test Item	Classification		Test Results	Reference Standard
Reliable and durable test	Reliable and durable	2000h		ISO 16750-1

10.19 Restricted substances test

Testing requirement:

1. Meet GB/T 30512:2014 "Requirements for prohibited substances on automobiles".
2. The mass percentage of lead, mercury, hexavalent chromium, polybrominated biphenyls (PBBs) and polybrominated diphenyl ethers (PBDEs) shall not exceed 0.1%, and the mass percentage ratio of cadmium shall not exceed 0.01%.

Test Method:

According to GB/T 30512:2014 "Requirements for prohibited substances on automobiles", §5 detection method of prohibited substances,

- QC/T 941-2013 "Method for the Detection of Mercury in Automotive Materials"
- QC/T 942-2013 "Detection Method of Hexavalent Chromium in Automotive Materials"
- QC/T 943-2013 "Detection Method of Lead and Cadmium in Automotive Materials"
- QC/T 944-2013 Method for the Detection of Polybrominated Biphenyls (PBBs) and Polybrominated Diphenyl Ethers (PBDEs) in Automotive Materials.

11 Mechanical Stress Test for VA

11.1 Vibration test

Test Requirement:

Functional state A is reached in operating mode 3.2 defined by ISO 16750-1:2006, and functional state C is reached in other modes.

Test Method:

The vibration test is carried out in accordance with the requirements of 4.1 of the ISO 16750-3:2012 standard, where Tmax is 85 ° C, and the specific test method is carried out in accordance with 4.1.2.4.2.

Test criteria:

- After the vibration is completed, the IP protection level and insulation performance of each module should not be reduced;
- the live part should not be in contact with the outer casing due to permanent or temporary deformation;

After returning to normal state, the assembly function test, appearance, continuity, insulation, and withstand voltage check, all functions should be normal.

Test Item	Classification		Test Results	Reference Standard
Vibration test	Vibration mounted on elastic mass	Acceleration root mean square 27.127.8m/s ² Each face lasts 8 hours		ISO 16750-3 IEC60068-2-64

11.2 Sweep Vibration and Resonance Durability Testing

Test Requirement:

On the vibrating table, the excitation of the VA component is performed by exciting at an acceleration of 1 g and sweeping the XYZ in three directions from the low frequency to the high frequency. Resonance endurance testing was performed at the resonant frequency.

Test Method:

1. Mount the VA component on the fixture and fix it on the vibration test bench;
2. The sweep frequency range is 5~300Hz, and the scan rate is 1oct/min;
3. A three-way accelerometer for measuring the vibration response of the VA component;
4. Turn on the sweep program and record the vibration response acceleration and frequency curve;
5. Find the first-order resonant frequency in three directions.
6. Resonance endurance test is performed in the three directions of XYZ with the first-order resonance frequency in each direction. The test period is 24 hours.

Test criteria:

Assembly function test, appearance, continuity, insulation, pressure test, all functions should be normal. After the test, the component mounting bolt tightening torque is not more than 20%.

11.3 Gas-tightness test

Test Requirement:

Before and after the test, the sealing strip was not damaged, and the airtightness was good.

Test Method:

Air pressure 0.3Bar, inflation time 30S, pressure holding time 30S, test time 60S, leakage amount: 400Pa.

Test Item	classification	Test Results	Reference Standard
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Gas-tightness test	Gas-tightness	The seal is not damaged and has good air tightness		
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11.4 Coolant pressure test

Test Requirement:

Leakage $\leq$ 300Pa; resistant to 20 mbar vacuum pressure.

Test Method:

Resistance to air pressure 3.25Bar, inflation time 12S, holding time 30S, test time 60S, leakage: 300Pa.

Test Item	Classification		Test Results	Reference Standard
Coolant pressure test	Coolant pressure	Leakage $\leq$ 300Pa; vacuum pressure of 20mbar		

11.5 Metal cooling water pipe static abuse force

Test Requirement:

At the top of the water pipe, in the plane perpendicular to the nozzle, a load of 20 Nm is applied through the pipe diameter axis, and the water pipe and the VA are connected, and no liquid leakage is required according to 9.4.

Test Method:

Use a torque wrench to apply a load at the water pipe port for 5 s.

11.6 Cooling pressure drop test

Test Requirement:

The cooling pressure drop must not exceed 60 mBar at a flow rate of 10 L/min and +25°C.

Test Method:

The cooling system is normally connected and operated with the ODP waterway, and the pressure difference between the pressure and the back pressure of the cooling system is recorded.

Test Item	Classification		Test Results	Reference Standard
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Cooling pressure drop test	Cooling pressure drop	Cooling pressure drop cannot exceed 60 mBar		
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11.7 Mechanical shock test

Test Requirement:

After the test, its performance should not be reduced. The live part should not be in contact with the outer casing due to permanent or temporary deformation. The functional status is up to the C grade defined in ISO 16750-1:2006.

Test Method:

Working mode 3.2, with a half-sine pulse impact with a peak acceleration of 500m/s² and a nominal pulse duration of 18ms. Six directions are required, and each test direction is impacted 10 times.

Test Item	Classification		Test Results	Reference Standard
Mechanical shock test	Mechanical shock	After the test, its performance should not be reduced, and the live part and the outer casing should not be in contact due to permanent or temporary deformation, and the functional state reaches the C level.		ISO 16750-1 GB-T2423.5

11.8 Free fall, package drop test

Test Requirement:

Implicit damage is not allowed, and the casing is allowed to be slightly damaged without affecting the performance of the DUT. The functional status shall be Class C as defined by ISO 16750-1:2006.

Test Method:

The free fall test is carried out in accordance with the provisions of 4.3.2 of the ISO 16750-3:2012 standard.

Test Item	Classification		Test Results	Reference Standard
Free fall, package drop test	Free fall, package drop	1 meter, 2 drops		ISO 16750-3 GB-T2423.8

11.9 Testing of Bolt Axial Pull-out Force

Testing purpose:

1. Test bolt installation strength.
2. Scope of application: All bolt fasteners used for connection on each assembly; the bolts are made of non-ferrous alloy or other materials.

Test Method:

1. The test is performed on the assembly sample and each type and size of screw fastener is selected.
2. The assembly is fixed to the work surface before testing.
3. Each typical bolt and connection type are mounted with suitable bolts.
4. According to Table 9.12.7, select the appropriate axial pull-out force of each bolt and test one by one. The pull-out force time is 10s.

Bolt specification	Axial pull-out force (N)
M4	350
M5	350
M6	500
M8	500
M10	800
M12	800

Test criteria:

1. Visual inspection. The test bolt used is in its original installation position and in good condition.
2. The surrounding wrapping material in contact with the bolt has no cracks and no looseness.

Note: If there are already visible micro-cracks caused by air bubbles around the threaded hole before the test and the crack does not affect the axial pull-out force test, it can be ignored.

11.10 Elbow force test

Test purpose:

Verify that the outer casing of the part and the parts inside the outer casing are not affected by the elbow force on the part.

Scope of application:

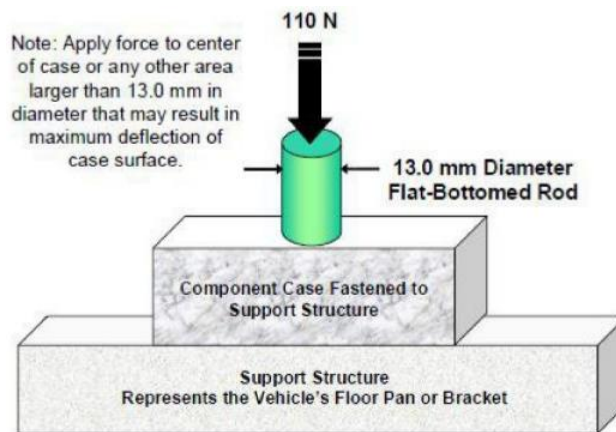
1. All parts.
2. Mode of operation: 1.1
3. Monitoring: When applying a force, measure the amount of deformation of the case to confirm the gap between the case and the parts on the board.

Test Method:

A uniform force of 110 N was applied to any area of the diameter of the outside of the part of 13.0 mm or more for a duration of 1 s. After all the parts have been tested, perform a 1-point function/parameter test.

Test criteria:

1. The component has no electrical degradation or permanent mechanical damage, and the functional status level must be C;
2. The deformation of the outer casing under stress shall not touch the parts of the circuit board other than the terminal block;
3. The deformation force does not cause the outer casing to separate or open.



11.11 Foot end force test

Test purpose:

Verify that the outer casing of the component and the components within the outer casing are not affected by the strength of the foot that is pressed against the component.

Scope of application:

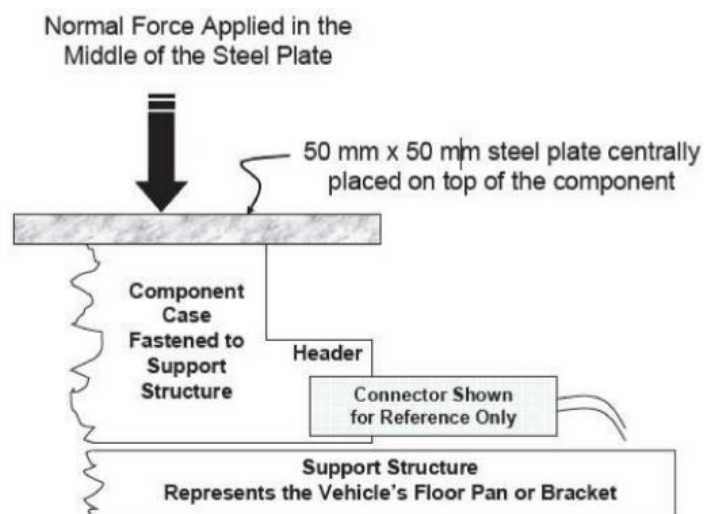
1. All parts that may be subjected to footing during vehicle assembly or after-sales service.
2. Mode of operation: 1.1
3. Monitoring: When applying a force, measure the amount of deformation of the case to confirm the gap between the case and the parts on the board.

Test Method:

Place a (50*50) mm hard steel plate on top of the part and apply a uniform force of 890 N for 1 minute. After all the parts have been tested, perform a 1-point function/parameter test.

Test criteria:

1. The component has no electrical degradation or permanent mechanical damage, and the functional status level must be C.
2. The deformation of the outer casing under stress shall not touch the parts of the circuit board other than the terminal block.
3. The deformation force does not cause the outer casing to separate or open.



11.12 Force on the socket test

11.12.1 Lateral abuse test

Test purpose:

Make sure that the component terminal block is not affected by side hand or elbow pressure during terminal connections or other assembly operations.

Scope of application:

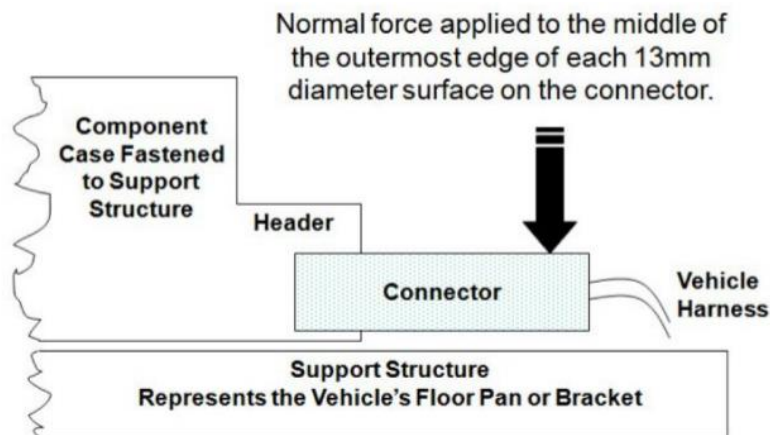
1. All parts with terminal blocks with at least 13.0 mm diameter area.
2. Operating mode: 3.2
3. Monitoring: Continuous monitoring.

Test Method:

Apply a uniform force of 110 N to any of the 13.0 mm or larger diameter areas outside the terminal block for a duration of 1 s.

Test criteria:

The functional status level is C. The terminal block, the housing, and the circuit board that connects the terminal block parts should withstand the mechanism pressure without shear, collapse, or insulation loss.

**11.12.2 Lateral abuse test****Test purpose:**

Verify that the component terminal block is not affected by the lateral foot force during terminal connections or other assembly operations.

Scope of application:

1. All parts that may be subjected to footing during vehicle assembly or after-sales service.
2. Operating mode: 3.2
3. Monitoring: continuous monitoring

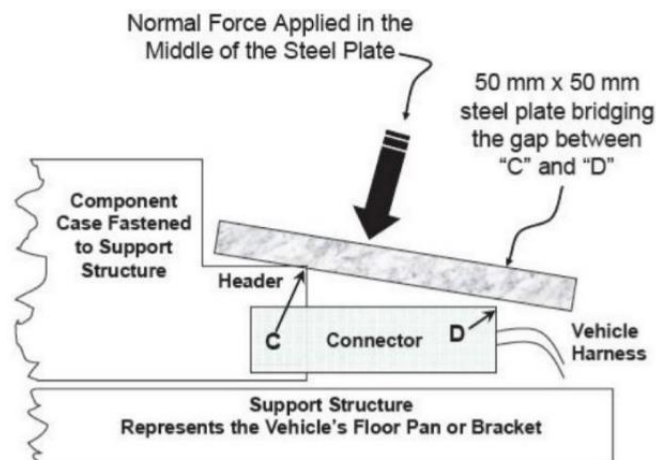
Test Method:

Place a (50*50) mm hard steel plate on the terminal block and terminal head and apply a uniform force of 890 N for 1 minute.

Test criteria:

1. Functional status level should be C.

- The head of the terminal seat shall withstand the pressure of the mechanism without shear, collapse or insulation loss. There is no degradation of PCB due to pressure transfer.



11.13 Cover crushed stone impact test

Test purpose:

- The simulated component is subjected to mechanical stress from the crushing stone.
- In order to ensure the stability of the components against defects, such as deformation and cracks.

Test Method:

According to ISO29567-1, the test is carried out with the following parameters.

Sample operation	Mode of operation: The sample is not powered but is connected to the plug and cable.
Shot peening dose	500g
Test pressure	2 Bar
Impact material	According to ISO 11124-2, the size of iron pellets is 4-5 mm.
Test surface on sample	Cover relative to the ground
Impact angle	54° against the impact direction
Test device	Multifunctional gravel impact testing machine in accordance with ISO 20567-1
Number of test cycles	2
Sample quantity	3

Test judgement:

- The sample function before and after the test must be in good condition and all parameters must be within the technical requirements. Verify by the following parameter detection.
- Temperature: room temperature;

3. The sample output voltage and output current are within the design specifications, the charging efficiency is within the allowable range, and the sample function is normal.
4. The sample must be visually inspected with the naked eye and shaken to see if the sample is loose or abnormal.
5. There must be no integral or partial inlay of the iron pellets into the specimen.
6. Prepare the changes and damages in the test report and conduct a final appraisal review with the customer.